


**Virtual University of Pakistan**

Real Analysis I (MTH621)

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**Virtual University**
**Learning Management System**

To my unknown students

## About the instructor

Dr. Malik did his MS and PhD (Mathematics) from University of La Rochelle, La Rochelle, France in 2009 and 2012, respectively. Prior to MS and PhD, Dr. Malik completed his MPhil and MSc (Mathematics) from Department of Mathematics, University of the Punjab, Lahore, Pakistan. He has been affiliated with several universities in Pakistan and abroad. He has the experience of teaching a wide range of mathematics courses at undergraduate and graduate level.

Dr. Malik has published several research articles in international journals and conferences. His area of research includes the study of differential equations with nonlocal operators and their applications to image processing. He is also interested in inverse problems related to reaction-diffusion equations with nonlocal integro-differential operators and boundary conditions. These models have numerous applications in anomalous diffusion/transport, biomedical imaging and non-destructive testing.

## About the handouts

The books followed during this course are: **W. Rudin**, Principles of Mathematical Analysis, Third Edition, McGraw-Hill, 1976. ISBN: 9780070542358. and **W. F. Trench**, Introduction to Real Analysis, Pearson Education, 2013. Consequently, the most of the examples considered in these notes are from the above mentioned books and their exercises, but not restricted to those books only. If you find any typing error in the text kindly report to me by writing an email to salman.amin.malik@gmail.com.

## Course Information

**Title and Course Code**: Real Analysis I (MTH621)

**Number of Credit Hours**: 3 credits

**Course Objective**: The Real Analysis I is the first course towards the rigorous (formal) treatment of the fundamental concepts of mathematical analysis. This course could be considered as the fundamental course in pursue of mathematical study at undergraduate or master level. Although, the topics of the Real Analysis I are self contained but someone having knowledge of Calculus (single and multivariable) and Differential Equations will be comfortable with the contents of the course. These subjects could be considered as prerequisite of the Real Analysis I. Learning Outcomes (for the whole course) Upon completion of this course students will be able to

- Understand the set theoretic statements, the real and complex number systems.
- Apply principle of mathematical induction, ordered sets.
- Decide about the convergent or divergent sequences and series.
- Define the limit of a function, prove various theorems about limits, sequences and functions.
- Check the continuity of real valued functions, prove various theorems about continuous functions with emphasize on the proofs.
- Understand the derivative of a function, proof of various theorems about differentiability of the function (LO6).
- Prove and apply Bolzano-Weierstrass theorem, Mean value theorem.
- Define Riemann integral, Riemann sums, proof of various results about the Riemann integrals.

**Prerequisites**: Calculus with Analytical Geometry

**The textbooks for this course**:

[1] W. Rudin, Principles of Mathematical Analysis, Third Edition, McGraw-Hill, 1976. ISBN: 9780070542358.

[2] W. F. Trench, Introduction to Real Analysis, Pearson Education, 2013.

**Reference books**:

[3] A. N. Kolmogorov and S. V. Fomin, Introductory Real Analysis, Revised English Edition Translated and Edited by R. A. Silverman, Dover Publication, Inc. New York.

[4] R. G. Bartle and D. R. Sherbert, Introduction to Real Analysis, Third Edition, 2000, John Wiley & Sons Inc.

- The real number system
- Sequences and Series
- Limits, Continuity and Differentiability
- The integration

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SET set is a collection of well-defined and distinct objects. A set is typically denoted using curly braces { } and its elements are separated by commas.
*** Uniqueness: Each element in a set is unique and cannot be repeated.
***Order doesn't matter: The order of elements in a set doesn't change the set itself.
e. g{1, 2, 3, 4, 5} is a set of five integers.

Real numbers can be defined as the union of both rational and irrational numbers. They can be both positive or negative and are denoted by the symbol "R".

LEC#1

CHAPTER 1

# The Real Number System

---

set:unordered and distint collection(order matter nahi krta duplicity nahi honi cheye)

ex:(1,2,3,4)

## 1.1 Basic Set Theory

In this section, we are going to provide some basic terminology required to understand the forthcoming concepts about the real number system.

It's like the big container that holds everything you're talking about.

### 1.1.1 Universal Set

sab sy larger set. jis ma koi b set ly us k all elements exist ho

The understanding that the members of all sets under consideration in any given context come from a specific collection of elements, called the **universal set**.

**For this chapter of the course our universal set will be the set of real numbers.**

e.gU = {1,2,3,a,b,c}, A = {1,2,3} B = {1,a,b,c}
U is the universal set of A and B

If an element $x$ is in $A$, we write $x \in A$ and say that $x$ is a member of $A$. If an element $x$ is not in $A$, we write $x \notin A$ and say that $x$ is not a member of $A$.

**How to define a set?** write about set AT start

$\{4, 5, 6, 7\}$.

Examples of Universal set

Ex:All Students in a School,All Employees in a Company,All Prime Numbers etc.

set builder notion mein eak specific property hold kry gi jo given ho gi.

### 1.1.2 Set Builder Notion

If $P$ is a property that is meaningful and unambiguous for elements of a set $S$, then we write

A notation used to "build" sets by classifying numbers according to their property.

$$\{x \in S : P(x)\},$$

for the set of all elements $x$ in $S$ for which the property $P$ is true.

If every element of a set $A$ also belongs to a set $B$, we say that $A$ is a subset of $B$ and write $A \subset B$.

subset ⊆. proper subset ⊂

**Proper subset**: We say that $A$ is proper subset of $B$ if there exist at least one element of $B$ which is not in $A$. A⊂B

**Equal sets**: Two sets are said to be equal if they contain the same number of elements. A=B

**Examples**: Consider the set of natural numbers $\mathbb{N} = \{1, 2, 3, ...\}$.

Examples of Set builder notion

It's like creating a set based on a rule or condition rather than listing out all its elements.

such as:Adults in a Population: {x | x is an adult},Even Numbers: {x | x is an even number}.

proper subset

A proper subset is a subset that contains some but not all of the elements of another set.

e. g : A= {12, 24} B= {12, 24, 36}

A is the proper subset of B because 36 is not present in the A.

mcqs

- $\{x \in \mathbb{N} : x^2 - 3x + 2 = 0\} = \{1, 2\}$.
- $\{x \in \mathbb{N} : x^2 - 4x + 2 = 0\}$.
- The set of even numbers $\{2k : k \in \mathbb{N}\}$.
- The set of odd numbers $\{2k - 1 : k \in \mathbb{N}\}$.

Let $S$ and $T$ be sets.

:)every set is subset of itself it is called trivial set.

- $S$ *contains* $T$, and we write $S \supset T$ or $T \subset S$, if every member of $T$ is also in $S$. In this case, $T$ is a *subset* of $S$. 1st set ky sab element 2nd mein hony cheye tu subset ho ga.(A<=B bhi kehty hn)
- $S - T$ is the set of elements that are in $S$ but not in $T$. proper subset

jab bhi 2 sets mein equilty show krni ho gi tu es ka mtlb hy ky 2nu sets mein na sirf same No. of elementds hn blky same members bhi hn

- $S$ *equals* $T$, and we write $S = T$, if $S$ contains $T$ and $T$ contains $S$; thus, $S = T$ if and only if $S$ and $T$ have the same members.

Let $S$ and $T$ be sets.

S mein kuch asy elemts hn ju T mein nhi hn,equilty hold nahi krti.

- $S$ *strictly contains* $T$ if $S$ contains $T$ but $T$ does not contain $S$; that is, if every member of $T$ is also in $S$, but at least one member of $S$ is not in $T$.

U-S

- The *complement* of $S$, denoted by $S^c$, is the set of elements in the universal set that are not in $S$.

jis ma dono k elements ko arrange krty h

- The *union* of $S$ and $T$, denoted by $S \cup T$, is the set of elements in at least one of $S$ and $T$

Let $S$ and $T$ be sets.

common

- The *intersection* of $S$ and $T$, denoted by $S \cap T$, is the set of elements in both $S$ and $T$ If $S \cap T = \emptyset$ (the empty set), then $S$ and $T$ are *disjoint sets*

Fira Neehson A singleton set is a set that contains exactly one element. EX;{5}

- A set with only one member $x_0$ is a *singleton set*, denoted by $\{x_0\}$.

Let $S$ and $T$ be sets.

- The *intersection* of $S$ and $T$, denoted by $S \cap T$, is the set of elements in both $S$ and $T$ If $S \cap T = \emptyset$ (the empty set), then $S$ and $T$ are *disjoint sets*

- A set with only one member $x_0$ is a *singleton set*, denoted by $\{x_0\}$.

Proper subset:any subset of A which is not trivial subset of A is called proper subset.
(jase subset mein A<=B hota hy wase hi proper subset mein A <B ho ga equal nahi hona cheye.)
ex:If A{1,2,3,4} then A'={1,2,3} which is proper subset of A.

Ex: Weekdays and Days in a Working Week.

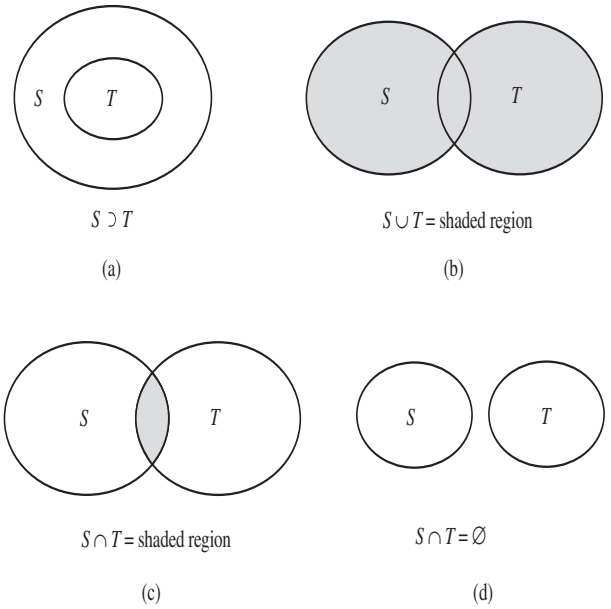



Figure 1.1: (a) Subset (b) Union of two sets (c) Intersection of two sets, (d) Disjoint sets

LEC#2

- **The set of natural numbers**
- **The set of prime numbers**
- **Fundamental Theorem of arithmetic**
- **Diophantines**
- **A little bit about number theory**

**The set of natural numbers**:

$$\mathbb{N} = \{1, 2, 3, ...\}.$$

## 1.2 Number Theory

Number theory is a branch of mathematics that studies the properties of, and the relationships between, particular types of numbers.

- The set of natural numbers $\mathbb{N}$.
- The prime numbers.

>number theory is all about the property of prime numbers such as:
if there"re infinite prime NO. then how many known prime numbers.
>ya phir eak prime No ko kis trha define kiya jye ky hmein pta chly ky given No prime hy or sab sy brha prime NO kon sa hy.

There are infinitely many prime numbers. This fact was first proven by the ancient Greek mathematician Euclid around 300 BCE. No matter how far you go in the sequence of prime numbers, there will always be more primes to be found.



The primes are the building blocks of the positive integers.

**Fundamental Theorem of Arithmetic**: Every positive integer can be uniquely written as the product of primes in nondecreasing order. prime factorization 12=2*2*3

How may prime numbers are there? (2500 years ago, Euclid provided the proof)

Many different approaches have been used to determine whether an integer is prime. **For example, in the nineteenth century, Pierre de Fermat showed that p divides $2^p - 2$ whenever $p$ is prime.**

**The search for integer solutions of equations is another important part of number theory.**

An equation with the added provision that only integer solutions are sought is called **diophantine**, after the ancient Greek mathematician Diophantus.

**Example**: kuch equations asi hoti hn jin mein humein sirf integer ya natural No. hi solution milta hy yeh euations diaphantine kehty hn.

$$a^n + b^n = c^n, \qquad n \in \mathbb{Z}.$$

**The set of nonnegative integers**:

$$\mathbb{Z}^+ = \{0, 1, 2, 3, ...\}.$$

Diophantine equations are equations that involve integer solutions.

**The set of nonpositive integers**:

$$\mathbb{Z}^- = \{0, -1, -2, -3, ...\}.$$

**The set of integers**:

$$\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, ...\}.$$

:) Zero is to be consider as non-negative and non-positive .

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**Some properties of the natural numbers**: commutative law

$$2 + 4 = 4 + 2$$

so naturals No. is a proper subset of Integer.

$$m + 3 = p + 3 \quad \text{then} \quad m = p.$$

$$4 + (2 + 7) = (4 + 2) + 7.$$

## 1.3 Principle of Mathematical Induction

asa tool ju hum mathematical statements ko true ya false prove krnty hn ky woh Natural No. hn ya phir Integer No.

Let $P(n)$be a mathematical statement, where $n \in \mathbb{N}$ (or $\mathbb{Z}^+$). If

- $P(1)$ is true.
- $P(n)$ is true implies $P(n+1)$ is true.

Then $P(n)$ is true for all $n \in \mathbb{N}$ (or $\mathbb{Z}^+$).

**Examples**:

technique used to prove statements about natural numbers.

it contains 2 steps:

Bases step p(1) is true

inductive step p (n+1) is true

MCQS

- $1^2 + 2^2 + 3^2 + ... + n^2 = \frac{n(n+1)(2n+1)}{6}$.
- $1^3 + 2^3 + 3^3 + ... + n^3 = \frac{n^2(n+1)^2}{4}$.
- Prove that if $(x + 1/x)$ is an integer then $(x^n + 1/x^n)$ is also an integer for any positive integer $n$.

**Theorem**: Suppose that $m, n, p \in \mathbb{Z}^+$. Then

under addition and substration it's hold.

under addition and multiplication it's holds.

- $m + n = n + m$, (commutativity)
- $(m + n) + p = n + (m + p)$, (associativity)
- if $m + n = p + n$, then $m = p$ (cancelation)
- if $m + n = 0$, then $m = n = 0$

**Proof**: Proof of first property.
**Step I**: Define $U = \{m \in \mathbb{Z}^+ : 0 + m = m + 0\}$.

**Step II**: Define $V = \{n \in \mathbb{Z}^+ : (m + 1) + n = (m + n) + 1, \text{ for all } m \in \mathbb{Z}^+\}$.

**Step III**: Define $W = \{n \in \mathbb{Z}^+ : m + n = n + m, \text{ for all } m \in \mathbb{Z}^+\}$.

$$m + (n + 1) = (m + n) + 1 \tag{1.1}$$

**Step I**: Define $U = \{m \in \mathbb{Z}^+ : 0 + m = m + 0\}$. Notice that $0 \in U$. Suppose $m \in U$ and consider $0 + (m + 1) = (m + 1)$ and we have $(0 + m) + 1 = m + 1$.

Thus $m + 1 \in U$, and so $U = \mathbb{Z}^+$.
**Step II**: Define $V = \{n \in \mathbb{Z}^+ : (m + 1) + n = (m + n) + 1, \text{ for all } m \in \mathbb{Z}^+\}$.
$0 \in V$, suppose $n \in V$, then

$$\begin{aligned} (m + 1) + (n + 1) &= ((m + 1) + n) + 1 \quad \text{by} \quad (1.1) \\ &= ((m + n) + 1) + 1, \quad \text{since} \quad n \in V \\ &= (m + (n + 1)) + 1 \quad \text{by} \quad (1.1). \end{aligned}$$

Hence we have $V = \mathbb{Z}^+$.
**Step III**: Define $W = \{n \in \mathbb{Z}^+ : m + n = n + m, \text{ for all } m \in \mathbb{Z}^+\}$.
Using the first step we have $0 \in W$. Suppose $n \in W$. Then

$$\begin{aligned} m + (n + 1) &= (m + n) + 1 \quad \text{by (1.1)} \\ &= (n + m) + 1, \quad \text{since} \quad n \in W \\ &= (n + 1) + m, \quad \text{from step II.} \end{aligned}$$

Hence $W = \mathbb{Z}^+$.

**Theorem**: Suppose that $m, n, p \in \mathbb{Z}^+$. Then

- $m.n = n.m$, (commutativity)
- $(m.n).p = n.(m.p)$, (associativity)
- if $m.n = p.n$ and $n \neq 0$, then $m = p$ (cancellation)
- if $m.n = 0$, then $m = 0$ or $n = 0$.
- $0.n = 0$ and $1.n = n$.

> e.g >agr koi set 20 elements contain krta hy tu us set mein 20 sy less sab elements hony cheye(without any gap).

### 1.3.1 An initial segment

LEC#3

An initial segment $I$ of $\mathbb{N}$ is a nonempty subset of $\mathbb{N}$ with the property that if $n \in I$ and $m \leq n$ then $m \in I$.

**Example**: kisi bhi set ka sb sy bara element initial segment ho ga agr os sy choty sab element set mein hon.

- Consider the set
$$I_{20} = \{1, 2, 3, ..., 20\}.$$
- Consider the set
$$I_n = \{1, 2, 3, ..., n\}.$$
- Consider the set $\{2, 4, 6, 8, 10\}$.

this is not an initial segment due to gap

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**Proposition**: If $I$ is an initial segment of $\mathbb{N}$ then either $I = \mathbb{N}$ or there exists $n \in \mathbb{N}$ such that $I = I_n = \{m \in \mathbb{N} : m \leq n\}$.

**Proof**: It follows immediately from the definition of an initial segment that if $m \notin I$ and $n \geq m$ then $n \notin I$. If $I \neq \mathbb{N}$, then $\mathbb{N} \backslash I$ is non-empty. Let $m_0$ be its least element. Suppose, if possible, that $m_0 = 1$. If $n \in \mathbb{N}$, then $n \geq 1$, so that $n \notin I$ and $I$ is the empty set. Thus $m_0 > 1$, and so there exists $n \in \mathbb{N}$ such that $m_0 = n + 1$. Then $n \in I$, and so $I_n \subseteq I$. But if $p > n$ then $p \geq n + 1 = m_0$, and so $p \notin I$. Thus $I \subseteq I_n$.

2 images lein agr woh equal hn, tu un ki input bhi equal honi cheye

also called injective mapping.

**Recall a one to one and onto mapping**: A mapping (function) $f : A \to B$ is said to be one to one if

(injective)one to one function:
all the elements of domain map all the elements of range

$$f(x_1) = f(x_2) \quad \Rightarrow \quad x_1 = x_2.$$

A mapping (function) $f : A \to B$ is said to be onto if for every $y \in B$ there exists at least an $x \in A$ such that

onto kosurjective function bhi khty h

$$f(x) = y.$$

A mapping (function) is said to be bijective if the mapping is one to one and onto.

agr eak mapping f define hy from A to B tu jab hum B mein sy koi(orbitrary) element lein or woh element hum function mein dalein tu answer Y aye.

Difference b/w onto and one to one

One-to-one functions map distinct elements to distinct images, while onto may map multiple elements to the same image.






A set is finite if there's a initial segment from which we can define a bijective mapping from that initial segment to the set of Natural No.

## 1.4 Finite and Infinite Set

A set $A$ is finite if either $A$ is empty or there exists $n \in \mathbb{N}$ and a bijective mapping $c : I_n \to A$. Thus the finite sequence $(c_1, ..., c_n)$ lists the elements of $A$, without repetition.

A set is infinite if there"s no initial segment such that we can define a bijection from that initial segment to set of Natural No.

A set is infinite if it is not finite.

**Proposition**: If $g : I_m \to I_n$ is an one to one (injective) mapping then $m \leq n$.

This proposition says if we've any initial segment then from given initial segment to another initial segment we've an injective mapping, then the No. of elements in the domain will be <= to the No. of elements in the co-domain.

**Proof**:

- The proof is by induction on $m$. The result is trivially true if $m = 1$.
- Suppose that it holds for $m$, and that $f : I_{m+1} \to I_n$ is injective. Then $m+1 > 2$, so that $f(I_{m+1})$ contains at least two points, and so $n = k+1$, for some $k \in \mathbb{N}$.
- Let $\tau : I_n \to I_n$ be the mapping that transposes $f(m+1)$ and $n$ and leaves the other elements of $I_n$ fixed.
- Then $\tau \circ f : I_{m+1} \to I_n$ is injective, and $\tau(f(I_m)) \subseteq I_k$.
- By the inductive hypothesis, $m \leq k$, and so $m + 1 \leq k + 1 = n$.

Fira Neehson

**Corollary**: If $A$ is a non-empty finite set, there exists a unique $n \in \mathbb{N}$ for which there exists a bijection $c : I_n \to A$.

**Proof**: Suppose that $c : I_n \to A$ and $c' : I_{n'} \to A$ are bijections.
Then $c^{-1} \circ c' : I_{n'} \to I_n$ is a bijection, and so $n' \leq n$.

Similarly, $n \leq n'$.

**Remark**: The number $n$ is known as cardinality of $A$ and is written as $| A |$.

**Proposition**: Suppose that $A$ is a finite set, and that $f : A \to B$ is a bijection. Then $B$ is finite, and $| B | = | A |$.

**Proof**: For if $C : I_{|A|} \to A$ is a bijection, then the mapping

$$f \circ C : I_{|A|} \to B$$

is the bijection.

Finite (countable numbers without repetition)
infinite (uncountable without repetition)

Any non-empty set(let say F) under two binary operations(+ .) is called field iff
1.(F,+) abelian
2.(F-{0}, .) abelian
3.Distributive law hods.



## 1.5 Field

A field is a set $F$, together with two laws of composition, addition $(+)$ and multiplication $(.)$ such that for all $a, b, c \in F$ the following properties holds

- $a + b = b + a$ and $ab = ba$ (commutative laws).
- $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ (associative laws).
- $a(b + c) = ab + ac$ (distributive law).
- There are distinct members 0 and 1 such that $a + 0 = a$ and $a1 = a$ for all $a$.
- For each $a \in F$ there is an element $-a \in F$ such that $a + (-a) = 0$, and if $a \neq 0$, there is an element $1/a$ such that $a(1/a) = 1$.

**Remark**: The left distributive law also holds.
**Examples**:

- Let $z_2 = \{0, 1\}$ such that

$$0 + 0 = 1 + 1 = 0; 0 + 1 = 1 + 0 = 1$$

and

$$0.0 = 0.1 = 1.0 = 0, 1.1 = 1.$$

set of integer is not a field becz there's is no any NO. which we multiply and get the answer 1 as multiplicative inverse.

- Is the set of integers a field?

agr eak set A(injective but not surjective) hy tu A sy A py hi mapping define krein or os mapping ko injective prove ke dein tu woh set Dedekind Infinite set kehlata hy.

## 1.6 Dedekind Infinite Set

Dedekind defined a set $A$ to be infinite if there is an injective map $j : A \to A$ which is not onto (surjective); such sets are now called Dedekind infinite.

**Example**: Show that $\mathbb{N}$ is Dedekind infinite.
The mapping $f : \mathbb{N} \to \mathbb{N}$ defined by $f(n) = 2n$ is injective, and is not surjective.

:) Set of natural number is infinite set.

**Corollary**: The set of natural numbers $\mathbb{N}$ is infinite set.

**Countable sets**: A set $A$ is countable if it is finite or if there is a bijection $c : \mathbb{N} \to A$; otherwise it is uncountable.

Countable set: Agr natural No. ku elawa kisi set ko infinite set prove krna hy tu hum eak bijection map define krein gy ju naturqal No. sy os set py define ho gi or woh one-to-one or onto ho gi.

uncountable set: agr bijection mapping define na ho sky tu.

Fira Neehson

**Remark**: A set is countable if it is empty or if there is a bijection from an initial segment of $\mathbb{N}$ onto $A$. The function $c$ is called an enumeration of $A$. A set is countably infinite if it is infinite and countable.

Thus $A$ is countably infinite if and only if the elements of $A$ can be listed, or enumerated, as an infinite sequence $(c_1, c_2, ...)$, without repetition.

If $A$ is countable (countably infinite) and $j : A \to B$ is a bijection, then $B$ is countable (countably infinite). Not every set is countable. It was Cantor who first showed, in 1873, that there are different sizes of infinite set, showing that the set of real numbers is uncountable.

**Proposition**: Let $A$ be a non-empty set. Then the following are equivalent.

(a) $A$ is countable.

(b) There exists a surjection $f : \mathbb{N} \to A$.

(c) There exists an injection $g : A \to \mathbb{N}$.

**Theorem**: The set $\mathbb{N} \times \mathbb{N}$ is countable.

**Proof**: Define the mapping $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$f(k, l) = 2^k 2^l.$$

Use above proposition to prove that $\mathbb{N} \times \mathbb{N}$ is countable.

Fira Neehson

## 1.7 The Set of Rational Numbers

Jin ko p/q ki form ma likh skty h q≠0

**Construction of set of rational numbers**: Let $\mathbb{Z}^* = \mathbb{Z}\backslash\{0\}$ be the set of nonzero integers. Define a relation on $\mathbb{Z} \times \mathbb{Z}^*$ by setting $(p, q) \sim (r, s)$ if $ps = qr$.

yhn (p,q) Z sy belong krty hn or (r,s) Z^* sy ju non-zero hon gy

**Proposition**: The relation $(p, q) \sim (r, s)$ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^*$.

**Proof**:

**Transitive**: Suppose $(p, q) \sim (r, s)$ and $(r, s) \sim (t, u)$, we need to show that $(p, q) \sim (t, u)$. Consider

$$pusr = (ps)(ru) = (qr)(ts) = qtsr$$

$\mathbb{Q}$ is **abelian group under addition**: We define addition on $\mathbb{Q}$ by

$$\frac{p}{q} + \frac{r}{s} = \frac{ps + qr}{qs}.$$

We define the multiplication as $\left(\frac{p}{q}\right)\left(\frac{r}{s}\right) = \frac{pr}{qs}$.

**Proposition**: Let $\mathbb{Q}^* = \mathbb{Q}/\{0/1\}$. Then, $(\mathbb{Q}^*, .)$ and $(\mathbb{Q}^*, +)$ are abelian group.

transitive property: if a related to b and b related to c then a also related to c.

set of rational number is a set in which di-nominator can't be zero.

Fira Neehson

Relatively prime NO. jin ko hum(m,n)=1 likty hn
or agr m,n ky factors krein gy tu common factor sirf 1 ho ga.



The set of rational numbers is defined as

$$\mathbb{Q} = \{\frac{m}{n} \ : \ m, n \in \mathbb{Z}, n \neq 0\}.$$

**Example**: Find solution of the equation $p = \sqrt{2}$ in the set of rational numbers if possible.

**Solution**: See Lecture.

mcqs

**Example**: Let $A = \{p \in \mathbb{Q} : \ p^2 < 2\}$. We will show that for every $p \in A$, we can find a rational number $q$ such that $p < q$.

To do this, we associate with each rational $p > 0$ the number

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \tag{1.2}$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \tag{1.3}$$

:)p^2=2 does not have any rational solution.

- If $p$ is in $A$ then $p^2 - 2 < 0$, (1.2) shows that $q > p$ and (1.3) shows that $q^2 < 2$. Thus $q$ is in $A$.

Let $B = \{p \in \mathbb{Q} : \ p^2 > 2\}$. We will show that for every $p \in B$, we can find a rational number $q$ such that $q < p$.

- If $p$ is in $B$ then $p^2 - 2 > 0$, (1.2) shows that $0 < q < p$ and (1.3) shows that $q^2 > 2$. Thus $q$ is in $B$.

set together with a relation
that arranges the elements in a specific linear order.

## 1.8 Ordered Set

ordered set pehly poset(partially order) hona cheye.

Let $S$ be a set. An order on $S$ is a relation , denoted by $<$, with the following properties

e. g Real numbers, Natural numbers etc

- For each pair $a, b \in S$, exactly one of the following is true:

kam sy kam eak hold kry. $a = b$, $a < b$, or $b < a$.

- If $a, b, c \in S$ such that If $a < b$ and $b < c$, then $a < c$. (The relation $<$ is *transitive*.)

- If $a < b$, then $a + c < b + c$ for any $c$, and if $0 < c$, then $ac < bc$.

**Example**: The set of rational numbers $\mathbb{Q}$ is an order set, if $r < s$ is defined to means that $s - r$ is a positive rational number.

An ordered set is a set $S$ in which an order is defined.

The notation $x \leq y$ indicates that $x < y$ or $x = y$, without specifying which one of these two is hold.

What is the negation of $x > y$?

eak set agr poset hy tu os ky koi sy bhi 2 elements otha lein or woh element comparable ho yani ya choty hon ya bary hon ya barabr hon.

:) every TOS is poset but but a poset maynot be TOS(total order set)

Fira Neehson

a partially ordered set (poset) is a set with a binary relation that allows for partial comparability among elements, following reflexivity, antisymmetry, and transitivity rules.



### 1.8.1 Partial Order

: A partial order is a relation $R$ on a set $S$ such that for all $a, b, c \in S$, we have the following koi bhi set ly lo (1,2,3...) or os py relation hy R, es R ko poset bolein

- $aRa$ (Reflexivity). gy jab yeh 3 properties hold krein
- If $aRb$ and $bRa$ implies $a = b$ (Anti Symmetric).
- If $aRb$, $bRc$ then $aRc$ (Transitve).

small element

## 1.9 Lower Bound given set mein sy sab sy chuty elemets find kr lo

Suppose the set $S$ is an ordered set, and $E \subset S$. If there exists a $b$ such that $x \geq b$ whenever $x \in E$. We say that $E$ is bounded below. In this case, $b$ is a lower bound of $E$. set ( {2, 3, 5} ):The number 2 is the lower bound because it is the smallest number in the set.

If $b$ is a lower bound of $E$, then so is any smaller number.

greatest lower bound Greatest lower bound of a set. yaniju lower bound find kiya hy os ka sab sy bara element .

### 1.9.1 Infimum

If $\alpha$ is a lower bound of $E$, but no number greater than $\alpha$ is a lower bound of $E$, then $\alpha$ is an infimum of $E$, and we write

$$\alpha = \inf E.$$

e.g***A= { 2,3,5} The greatest lower bound of A is 2 because 2 is less than or equal to all elements in A.inf(A) =2

**Example**: Recall the set $A = \{p \in \mathbb{Q} : p^2 > 2\}$.

large element

## 1.10 Upper Bound woh element ju har element ko succeed krty hn .

Suppose the set $S$ is an ordered set, and $E \subset S$. If there exists a $b$ such that $x \leq b$ whenever $x \in E$. We say that $E$ is bounded above In this case, $b$ is an upper bound of $E$.

If $b$ is an upper bound of $E$, then so is any larger number

tu ap given subset ka sub sy bara element(usually last one) or phir given set mein sy os last one sy bhi bary elemets ko find kerlo.

### 1.10.1 Supremum upper bound ka chota element

If $\beta$ is an upper bound of $E$, but no number less than $\beta$ is an upper bound of $E$, then $\beta$ is a supremum of $E$, and we write

$$\beta = \sup E.$$

supremum also called least upper bound of a set yani ju upper bound find kiyen hn kisi bhi set ky on mein sy sab sy chuta element hi supremum ho ga.

**Example**: Recall the set $A = \{p \in \mathbb{Q} : p^2 < 2\}$.

set ( {2, 3, 5} ):The number 5 is the upper bound because it is the largest number in the set.

Fira Neehson

e g***A={2,3,5}The lowest upper bound of A is 5 because 5 is greater than or equal to all elements in A. sup(A) = 5

**Example**: If $S$ is the set of negative numbers, then any nonnegative number is an upper bound of $S$, and $\sup S = 0$.

**Example**: If $S_1$ is the set of negative integers, then any number $a$ such that $a \geq -1$ is an upper bound of $S_1$, and $\sup S_1 = -1$.

**Remark**: The supremum or infimum of a set may or may not belong to that set.

**Example**: Consider the set

$$E = \{p \in \mathbb{Q} \ : p < 0\}.$$

What are the upper bounds? All +ve Q numbers
What is the supremum of this set? 0
Is it belong to the set? not belonging

**Example**: Consider the set

$$E_1 = \{p \in \mathbb{Q} \ : p \leq 0\}.$$

What are the upper bounds? All +ve Q numbers and 0 as well
What is the supremum of this set? 0
Is it belong to the set? yes

**Example**: Consider the set

$$E_2 = \{1/n \ : n \in \mathbb{N}\}.$$

upper bound ka least upper bound set ma ho

## 1.11 Least Upper Bound Property

An ordered set $S$ is said to have the least-upper-bound property if the following is true:

If $E \subset S$, $E$ is not empty, and $E$ is bounded above, then $\sup E$ exists in $S$.

Recall the set $A = \{p \in \mathbb{Q} : \ p^2 > 2\}$.

Recall the set $B = \{p \in \mathbb{Q} : \ p^2 < 2\}$.

Consider the set

$$E_2 = \{1/n \ : n \in \mathbb{N}\}.$$

**Theorem**: Suppose $S$ is an ordered set with the least-upper-bound property, $B \subseteq S$, $B$ is not empty, and $B$ is bounded below. Let $L$ be the set of all lower bounds of $B$. Then

$$a = \sup L$$

eak set X least upper bound property hold kry ga agr X ka koi bhi no-empty subset(other than Phi) ky upper bound ka least upper bound X mein mojud ho.
or
every non-empty subset of X with an upper bound has least upper bound in X.
ex: A={1,2,3} subset a1={1,2}=>subset ka upper bound 2 hy or woh as (LUB) A mein bhi hy.

exists in $S$, and $a = \inf B$. In particular $\inf B$ exists in $S$.

**Proof**: $L$ is not empty set.

$L$ is bounded above.

$S$ satisfy least-upper-bound property therefore $L$ has a supremum in $S$; call it $a$.

If $\gamma < a$ then $\gamma$ is not an upper bound of $L$, hence $\gamma \notin B$. It follows that $a \leq x$ for every $x \in B$.

Thus $a \in L$.

If $a < \beta$ then $\beta \notin L$, since a is an upper bound of $L$.

We have shown that $a \in L$ but $\beta \notin L$ if $\beta > a$.

In other words, $a$ is a lower bound of $B$, but $\beta$ is not if $\beta > a$.

This means that $a = \inf B$.

**Theorem**: There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property

Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

upper bound mein sy sab sy bra element hi supremum ho ga agr yeh exist key tu completeness Axiom hold kry ga.

## 1.12 The Completeness Axiom

If a nonempty set of real numbers is bounded above, then it has a supremum.

The above property is called completeness, and we say that the real number system is a complete ordered field.

**Theorem**: If a nonempty set $S$ of real numbers is bounded above, then $\sup S$ is the unique real number $\beta$ such that

- $x \leq \beta$ for all $x$ in $S$;
- if $\varepsilon > 0$ (no matter how small), there is an $x_0$ in $S$ such that $x_0 > \beta - \varepsilon$.

**Proof**: We first show that $\beta = \sup S$ has first and second properties. Since $\beta$ is an upper bound of $S$, it must satisfy the first property. Since any real number $a$ less than $\beta$ can be written as $\beta - \varepsilon$ with $\varepsilon = \beta - a > 0$, second property is just another way of saying that no number less than $\beta$ is an upper bound of $S$. Hence, $\beta = \sup S$ satisfies first and second properties.

Now we show that there cannot be more than one real number with first and second properties. Suppose that $\beta_1 < \beta_2$ and $\beta_2$ has the second property ; thus, if $\varepsilon > 0$, there is an $x_0$ in $S$ such that $x_0 > \beta_2 - \varepsilon$. Then, by taking $\varepsilon = \beta_2 - \beta_1$, we see that there is an $x_0$ in $S$ such that

$$x_0 > \beta_2 - (\beta_2 - \beta_1) = \beta_1,$$

so $\beta_1$ cannot have the first property. Therefore, there cannot be more than one real number that satisfies both parts.

ex:Interval [0, 1]: Consider the closed interval [0, 1]. This interval contains all real numbers between 0 and 1, including 0 and 1 themselves. Any non-empty subset of [0, 1] that is bounded above (for example, the set {0.1, 0.5, 0.9}) will have a least upper bound (supremum) within [0, 1].

any two positive real numbers a and b, there exists a positive integer n such that na > b.



## 1.13 The Archimedean Property of $\mathbb{R}$

**Theorem**: If $\rho$ and $\varepsilon$ are positive, then $n\varepsilon > \rho$ for some integer $n$.

**Proof**: The proof is by contradiction. If the statement is false, $\rho$ is an upper bound of the set

agr a or b koi sy 2 NO. hn tu, b ki jab bhi koi -ve value lein gy(b<=0) answer na>b ho ga. n=1,2,3,....

$$S = \{x = n\varepsilon,\ n \in \mathbb{Z}\}.$$

Therefore, $S$ has a supremum $\beta$ (Why?), by definition of least upper bound property of real numbers.

Therefore,

$$n\varepsilon \leq \beta \quad \text{for all integers } n. \tag{1.4}$$

Since $n+1$ is an integer whenever $n$ is, (1.4) implies that

$$(n+1)\varepsilon \leq \beta$$

and therefore

$$n\varepsilon \leq \beta - \varepsilon$$

for all integers $n$.

A set A in the real numbers R is said to be dense if, between any two distinct points in A, there exists another point in A.

## 1.14 Dense Set in $\mathbb{R}$

if every real number is either an element of set or a limit point of set.

A set $D$ is said to be dense in the set of real numbers if every open interval $(a, b)$ contains a member of $D$. The set of rational numbers Q is dense in R. beacuse

**Theorem**: The rational numbers are dense in $\mathbb{R}$, that is, if $a$ and $b$ are real numbers with $a < b$, there is a rational number $p/q$ such that $a < p/q < b$.

between √2&√3, there are infinitely many rational numbers like 5/4, 11/8, etc.

**Recall the The Archimedean property**

**Theorem**: If $\rho$ and $\varepsilon$ are positive, then $n\varepsilon > \rho$ for some integer $n$.

**Proof of the theorem**: The Archimedean property with $\rho = 1$ and $\varepsilon = b - a$, there is a positive integer $q$ such that $q(b-a) > 1$.

There is also an integer $j$ such that $j > qa$. This is obvious if $a \leq 0$, and it follows from Archimedean property with $\varepsilon = 1$ and $\rho = qa$ if $a > 0$.

Let $p$ be the smallest integer such that $p > qa$. Then $p - 1 \leq qa$, so

$$qa < p \leq qa + 1.$$

complete set : subset of a set bounded above or non empty agr non empty ho ga tbhi supremum ly sky h or Jo value ay wo set ma exist krni chahiya / koi bhi set jis ka supremum bhi using k set ma exist krta ho.

Since $1 < q(b-a)$, this implies that

$$qa < p < qa + q(b-a) = qb.$$

so $qa < p < qb$. Therefore, $a < p/q < b$.

Rational number: 2
Irrational number: √2
Their product: 2×√2 = 2√2
If 2√2 were rational, it could be written as a fraction, say a/b. Then, we would have:
2√2 = a/b
Squaring both sides gives:
8 = (a/b)²
8 = a²/b²
Since the left side is an integer, the right side must also be an integer. This means b must be 1 (since b cannot be 0).
So, we have:
8 = a²/1
a² = 8
This means a must be ±2√2, which is not an integer!
Therefore, our assumption that 2√2 is irrational



because of least upper bound

## 1.15 The Set of Rational Numbers is not Complete

The rational number system is not complete; that is, a set of rational numbers may be bounded above (by rational numbers), but not have a rational upper bound less than any other rational upper bound, that is, that set does not have a rational supremum.

**Recall**: **Theorem**: If a nonempty set $S$ of real numbers is bounded above, then $\sup S$ is the unique real number $\beta$ such that

- $x \leq \beta$ for all $x$ in $S$;
- if $\varepsilon > 0$ (no matter how small), there is an $x_0$ in $S$ such that $x_0 > \beta - \varepsilon$.

Consider the set

$$A = \{p \in \mathbb{Q} : \ p^2 < 2\}.$$

If $p \in A$, then $p < \sqrt{2}$.

Then using the fact that there is a rational number between every two real numbers implies that if $\varepsilon > 0$ there is a rational number $r_0$ such that $\sqrt{2} - \varepsilon < r_0 < \sqrt{2}$, so using above Theorem implies that $\sqrt{2} = \sup A$. However, $\sqrt{2}$ is *irrational*; that is, it cannot be written as the ratio of integers.

Therefore, if $r_1$ is any rational upper bound of $A$, then $\sqrt{2} < r_1$. Since there is a rational number between every two real numbers, there is a rational number $r_2$ such that $\sqrt{2} < r_2 < r_1$. Since $r_2$ is also a rational upper bound of $A$, this shows that $A$ has no rational supremum.

**Example**Product of a rational and irrational number is irrational.

**Solution**: See lecture.

**Example**: Sum of a rational and an irrational number is irrational.

**Proof**: See lecture.

**Theorem**: The set of irrational numbers is dense in the reals; that is, if $a$ and $b$ are real numbers with $a < b$, there is an irrational number $t$ such that $a < t < b$.

**Proof**: Since between every two real numbers there is a rational number, therefore there are rational numbers $r_1$ and $r_2$ such that

$$a < r_1 < r_2 < b. \tag{1.5}$$

Let

$$t = r_1 + \frac{1}{\sqrt{2}}(r_2 - r_1).$$

Then $t$ is irrational (why?) and $r_1 < t < r_2$, so $a < t < b$, from (1.5).

Rational number: 2
Irrational number: √2
Their sum: 2 + √2
If 2 + √2 were rational, it could be written as a fraction, say a/b. Then, we would have:
2 + √2 = a/b
Subtracting 2 from both sides gives:
√2 = (a/b) - 2
Since (a/b) is rational, (a/b) - 2 is also rational.
But this means √2 is rational, which is a contradiction!
Therefore, our assumption that 2 + √2 is rational leads to a contradiction, and we conclude that 2 + √2 is irrational.



## 1.16 Applications of Properties of Field

McQs

**Proposition**: The axioms of addition imply the following

(a) If $x + y = x + z$ then $y = z$ (**Cancellation law**).

(b) If $x + y = x$ then $y = 0$ (**Uniqueness of the additive identity**).

(c) If $x + y = 0$ then $y = -x$ (**Uniqueness of the additive inverse**).

(d) $-(-x) = x$.

**Proof**: If $x + y = x + z$, then from the axioms of field we have

$$\begin{aligned} y = 0 + y &= (-x + x) + y = -x + (x + y) \\ &= -x + (x + z) = (-x + x) + z = 0 + z = z. \end{aligned}$$

This proves (a).

- Take $z = 0$ in (a) to obtain (b).
- Take $z = -x$ in (a) to obtain (c).
- Since $-x + x = 0$, (c) with $-x$ in place of $x$ gives (d).

mcqs

**Proposition**: The axioms of multiplication imply the following

(a) If $x \neq 0$ and $xy = xz$ then $y = z$ (**Cancellation law**).

(b) If $x \neq 0$ and $xy = x$ then $y = 1$ (**Uniqueness of the multiplicative identity**).

(c) If $x \neq 0$ and $xy = 1$ then $y = 1/x$ (**Uniqueness of the additive inverse**).

(d) If $x \neq 0$ then $1/(1/x) = x$.

**Proposition**: The field axioms imply the following statements for any $x, y, z \in F$.

(a) $0x = 0$.

(b) If $x \neq 0$ and $y \neq 0$ then $xy \neq 0$.

(c) $(-x)y = -(xy) = x(-y)$.

(d) $(-x)(-y) = xy$.

The manipulative properties of the real numbers, such as the relations

mcqs

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2, \\ (3a+2b)(4c+2d) &= 12ac + 6ad + 8bc + 4bd, \\ (-a) &= (-1)a, \quad a(-b) = (-a)b = -ab, \\ \frac{a}{b} + \frac{c}{d} &= \frac{ad+bc}{bd} \quad (b, d \neq 0), \end{aligned}$$

all follow from the axioms of field.

**Theorem**: For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real $y$ such that $y^n = x$.

**Proof**: The uniqueness is clear, since if there are two positive numbers $y_1$ and $y_2$ then

$$0 < y_1 < y_2 \quad \Rightarrow \quad y_1^n < y_2^n.$$

Let $E$ be the set containing of all positive real numbers $t$ such that $t^n < x$.

The set $E$ is not empty, as if $t = x/(x+1)$ then $0 \leq t < 1$ and $t^n < t < x$.

If $t > 1 + x$ then $t^n \geq t > x$, so that $t \notin E$. Thus $t \in E$ is an upper bound of $E$.

Then by least upper bound property there exists $y \in \mathbb{R}$ such that

$$y = \sup E.$$

We need to prove that $y^n = x$ we will show that each of the inequalities $y^n < x$ and $y^n > x$ leads to a contradiction.

Recall the identity $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + ... + a^{n-1})$ yields the identity

$$b^n - a^n = (b-a)nb^{n-1},$$

when $0 < a < b$.

Assume $y^n < x$, choose $h$ so that $0 < h < 1$ and

$$h < \frac{x - y^n}{n(y+1)^{n-1}}.$$

Put $a = y$, $b = y + h$. Then

$$(y+h)^n - y^n < hn(y+h)^{n-1} < hn(y+1)^{n-1} < x - y^n.$$

Thus $(y+h)^n < x$, and $y + h \in E$, which is contradiction to the fact that $y$ is an upper bound of $E$.

Assume $y^n > x$, put

$$k = \frac{y^n - x}{ny^{n-1}}.$$

Then $0 < k < y$. If $t \geq y - k$, we conclude that

$$y^n - t^n \leq y^n - (y-k)^n < kny^{n-1} = y^n - x.$$

Thus $t^n > x$, and $t \notin E$. It follows that $y - k$ is an upper bound of $E$. But $y - k < y$, which contradicts the fact that $y$ is the least upper bound of $E$.

**Theorem**: If $a$ and $b$ are any two real numbers, then

$$|a + b| \leq |a| + |b|. \tag{1.6}$$

**Proof**: There are four possibilities:

McQs

(a) If $a \geq 0$ and $b \geq 0$, then $a + b \geq 0$, so $|a + b| = a + b = |a| + |b|$.

(b) If $a \leq 0$ and $b \leq 0$, then $a + b \leq 0$, so $|a + b| = -a + (-b) = |a| + |b|$.

(c) If $a \geq 0$ and $b \leq 0$, then $a + b = |a| - |b|$.

(d) If $a \leq 0$ and $b \geq 0$, then $a + b = -|a| + |b|$.

Eq. 1.6 holds in cases **(c)** and **(d)**, since

$$|a+b| = \begin{cases} |a| - |b| & \text{if } |a| \geq |b|, \\ |b| - |a| & \text{if } |b| \geq |a|. \end{cases} \tag{1.7}$$

**Corollary**: If $a$ and $b$ are any two real numbers, then

$$|a-b| \geq \big||a| - |b|\big|, \qquad \text{and} \qquad |a+b| \geq \big||a| - |b|\big|. \tag{1.8}$$

**Proof**: Replacing $a$ by $a - b$ in (1.6) yields

$$|a| \leq |a-b| + |b| \quad \Rightarrow \quad |a-b| \geq |a| - |b|.$$

Interchanging $a$ and $b$ here yields

$$|b-a| \geq |b| - |a|,$$

which is equivalent to

$$|a-b| \geq |b| - |a|, \tag{1.9}$$

since $|b - a| = |a - b|$. Since

$$\big||a| - |b|\big| = \begin{cases} |a| - |b| & \text{if} \quad |a| > |b|, \\ |b| - |a| & \text{if} \quad |b| > |a|. \end{cases}$$

(**??**) and (1.9) imply (1.8). Replacing $b$ by $-b$ in (1.8) yields (**??**), since $|-b| = |b|$.

## 1.17 The Extended Real Number System

A nonempty set $S$ of real numbers is *unbounded above* if it has no upper bound, or *unbounded below* if it has no lower bound.

It is convenient to adjoin to the real number system two fictitious points, $+\infty$ (which we usually write more simply as $\infty$) and $-\infty$, and to

Define the order relationships between them and any real number $x$ by

$$-\infty < x < \infty. \tag{1.10}$$

We call $\infty$ and $-\infty$ *points at infinity*.

The extended real number system is an extension of the real number system that includes two additional elements: +∞ (positive infinity) and -∞ (negative infinity). It is often denoted as [-∞, ∞] or $\mathbb{R} \cup \{-\infty, \infty\}$.

asa number jis lower bound , upper bound, supremum and infimum na mtlb infinity ho us k Lia extended real number system use krty h



If $S$ is a nonempty set of reals, we write

$$\sup S = \infty \tag{1.11}$$

to indicate that $S$ is unbounded above, and

$$\inf S = -\infty \tag{1.12}$$

to indicate that $S$ is unbounded below.

$$\text{If} \quad S = \{x \ : \ x < 2\},$$

then $\sup S = 2$ and $\inf S = -\infty$.

$$\text{If} \quad S = \{x \ : \ x \geq -2\},$$

then $\sup S = \infty$ and $\inf S = -2$.

If $S$ is the set of all integers, then $\sup S = \infty$ and $\inf S = -\infty$.

A member of the extended reals differing from $-\infty$ and $\infty$ is *finite*; that is, an ordinary real number is finite. However, the word "finite" in "finite real number" is redundant and used only for emphasis, since we would never refer to $\infty$ or $-\infty$ as real numbers. The real number system with $\infty$ and $-\infty$ adjoined is called the *extended real number system*, or simply the *extended reals*.

We must defined arithmetic operations with $\pm\infty$. A member of the extended reals differing from $-\infty$ and $\infty$ is *finite*; that is, an ordinary real number is finite. However, the word "finite" in "finite real number" is redundant and used only for emphasis, since we would never refer to $\infty$ or $-\infty$ as real numbers.

The arithmetic relationships among $\infty$, $-\infty$, and the real numbers are defined as follows.

- If $a$ is any real number, then McQs

$$\begin{aligned} a + \infty &= \infty + a = \infty, \\ a - \infty &= -\infty + a = -\infty, \\ \frac{a}{\infty} &= \frac{a}{-\infty} = 0. \end{aligned}$$

- If $a > 0$, then McQs

$$\begin{aligned} a\,\infty &= \infty a = \infty, \\ a\,(-\infty) &= (-\infty)\,a = -\infty. \end{aligned}$$

- If $a < 0$, then McQs

$$\begin{aligned} a\infty &= \infty a = -\infty, \\ a(-\infty) &= (-\infty)a = \infty. \end{aligned}$$

We also define

$$\infty + \infty = \infty\infty = (-\infty)(-\infty) = \infty$$

and

$$-\infty - \infty = \infty(-\infty) = (-\infty)\infty = -\infty.$$

Finally, we define

$$|\infty| = |-\infty| = \infty.$$

It is not useful to define $\infty - \infty$, $0 \cdot \infty$, $\infty/\infty$, and $0/0$. They are called *indeterminate forms*, and left undefined.

## 1.18 Principle of Mathematical Induction

The rigorous construction of the real number system starts with a set $\mathbb{N}$ of undefined elements called natural numbers, with the following properties. The set of natural number $\mathbb{N}$ satisfy the following:

- $\mathbb{N}$ is nonempty.
- Associated with each natural number $n$ there is a unique natural number $n'$ called the *successor of* $n$.
- There is a natural number $\overline{n}$ that is not the successor of any natural number.

The set of natural number $\mathbb{N}$ satisfy the following:

- Distinct natural numbers have distinct successors; that is, if $n \neq m$, then $n' \neq m'$.
- The only subset of $\mathbb{N}$ that contains $\overline{n}$ and the successors of all its elements is $\mathbb{N}$ itself.

**Theorem**: Let $P_1$, $P_2$,..., $P_n$, ... be propositions, one for each positive integer, such that

- $P_1$ is true;
- for each positive integer $n$, $P_n$ implies $P_{n+1}$.

Then $P_n$ is true for each positive integer $n$.

**Proof**: Let

$$\mathbb{M} = \{n \in \mathbb{N} \text{ and } P_n \text{ is true}\}.$$

From first axiom of Peano's, $1 \in \mathbb{M}$, and from second axiom, $n + 1 \in \mathbb{M}$ whenever $n \in \mathbb{M}$.

Therefore, $\mathbb{M} = \mathbb{N}$, by fourth axiom.

1. P(1) is true (Put The 1st value of LHS is equal to the RHS) base step
2. For any natural number k, if P(k) is true, then P(k+1) is also true (Inductive Step)

Then, P(n) is true for all natural numbers n.

Example:
* Base Case (n = 1):
1 = 1(1 + 1)/2 = 1
* Inductive Step:
Assume it's true for n = k:
1 + 2 + ... + k = k(k + 1)/2
it's true for n = k + 1:
1 + 2 + ... + k + (k + 1) = (k + 1)((k + 1) + 1)/2
= (k + 1)(k + 2)/2
= k(k + 1)/2 + (k + 1)
= (1 + 2 + ... + k) + (k + 1)
LHS=RHS
so, the formula is true for n = k + 1.



**Example**: Let $P_n$ be the proposition that

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

**Solution**: Then $P_1$ is the proposition that $1 = 1$, which is certainly true.

If $P_n$ is true, then adding $n + 1$ to both sides of the equation yields

$$\begin{aligned}(1 + 2 + \cdots + n) + (n + 1) &= \frac{n(n+1)}{2} + (n + 1)\\ &= (n + 1)\left(\frac{n}{2} + 1\right)\\ &= \frac{(n+1)(n+2)}{2} = \frac{(n+1)(n+2)}{2},\end{aligned}$$

**Example**: For each nonnegative integer $n$, let $x_n$ be a real number and suppose that

$$|x_{n+1} - x_n| \le r|x_n - x_{n-1}|, \quad n \ge 1,$$

where $r$ is a fixed positive number.

Show that

$$|x_n - x_{n-1}| \le r^{n-1}|x_1 - x_0| \quad \text{if} \quad n \ge 1.$$

**Solution**: For $n = 1$, 2, and 3, we find that

$$\begin{aligned}|x_2 - x_1| &\le r|x_1 - x_0|,\\ |x_3 - x_2| &\le r|x_2 - x_1| \le r^2|x_1 - x_0|,\\ |x_4 - x_3| &\le r|x_3 - x_2| \le r^3|x_1 - x_0|.\end{aligned}$$

agr basis step hi satisfy na kry to definitely inductive step bhi satisfy nhi kry ga

- It is important to verify $P_1$, since $P_n$ may imply $P_{n+1}$ even if some or all of the propositions $P_1, P_2, \ldots, P_n, \ldots$ are false.

- Let $P_n$ be the proposition that $2n - 1$ is divisible by 2. If $P_n$ is true then $P_{n+1}$ is also, since

  $$2n + 1 = (2n - 1) + 2.$$

  However, we cannot conclude that $P_n$ is true for $n \ge 1$. In fact, $P_n$ is false for every $n$.

Base Case (n = 1):
2(1) + 1 = 3, not divisible by 2.
so, this proposition is actually always false, since 2n + 1 is an odd number and cannot be divisible by 2.

### 1.18.1 Principle of Mathematical Induction For $\mathbb{Z}$

**Theorem**: Let $n_0$ be any integer (positive, negative, or zero).

Let $P_{n_0}$, $P_{n_0+1}$, ..., $P_n$, ... be propositions, one for each integer $n \geq n_0$, such that

- $P_{n_0}$ is true;
- for each integer $n \geq n_0$, $P_n$ implies $P_{n+1}$.

Then $P_n$ is true for every integer $n \geq n_0$.

**Proof**: For $m \geq 1$, let $Q_m$ be the proposition defined by $Q_m = P_{m+n_0-1}$. Then $Q_1 = P_{n_0}$ is true by first part.

If $m \geq 1$ and $Q_m = P_{m+n_0-1}$ is true, then $Q_{m+1} = P_{m+n_0}$ is true by second part with $n$ replaced by $m + n_0 - 1$.

Therefore, $Q_m$ is true for all $m \geq 1$ by Mathematical induction Theorem with $P$ replaced by $Q$ and $n$ replaced by $m$. This is equivalent to the statement that $P_n$ is true for all $n \geq n_0$.

Base Case (n = -5):
3(-5) + 16 = -15 + 16 = 1 > 0
Inductive Step:
3k + 16 > 0
3(k+1) + 16 = 3k + 3 + 16
= (3k + 16)+3
P(n) is true for all integers
n ≥ -5: 3n + 16 > 0

**Example**: Prove the proposition $P_n$ given by

$$3n + 16 > 0, \qquad n \geq -5.$$

**Example**: Let $P_n$ be the proposition that

$$n! - 3^n > 0, \qquad n \geq 7.$$

Base Case (n = 7):
7! - 3(7) = 5040 - 21 = 5019 > 0 (True)

If $P_n$ is true, then

$$\begin{aligned} (n+1)! - 3^{n+1} &= n!(n+1) - 3^{n+1} \\ &> 3^n(n+1) - 3^{n+1} \quad \text{(by the induction assumption)} \\ &= 3^n(n-2). \end{aligned}$$

Therefore, $P_n$ implies $P_{n+1}$ if $n > 2$. By trial and error, $n_0 = 7$ is the smallest integer such that $P_{n_0}$ is true; hence, $P_n$ is true for $n \geq 7$, by Theorem **??**.

**Theorem**: Let $n_0$ be any integer (positive, negative, or zero). Let $P_{n_0}$, $P_{n_0+1}$,..., $P_n$, ... be propositions, one for each integer $n \geq n_0$, such that

- $P_{n_0}$ is true;
- for $n \geq n_0$, $P_{n+1}$ is true if $P_{n_0}$, $P_{n_0+1}$,..., $P_n$ are all true.

S ⊃ T (S is a superset of T and In other words, we can say that T is a subset of S and represented as (⊃) that means set S has more elements than T.



S ⊃ T

Then $P_n$ is true for $n \geq n_0$.

**Example**: Let $S = \{x \in \mathbb{R} : 0 < x < 1\}$, $T = \{x \in (0,1) : x \text{ is rational}\}$, and $U = \{x \in (0,1) : x \text{ is irrational}\}$. Then $S \supset T$ and $S \supset U$, and the inclusion is strict in both cases. The unions of pairs of these sets are

$$S \cup T = S, \quad S \cup U = S, \quad \text{and} \quad T \cup U = S,$$

and their intersections are

$$S \cap T = T, \quad S \cap U = U, \quad \text{and} \quad T \cap U = \emptyset.$$

## 1.19 Generalization of Union and Intersection

jis ma finite and infinite elements aty h

: If $\mathcal{F}$ is an arbitrary collection of sets, then $\cup\{S : S \in \mathcal{F}\}$ is the set of all elements that are members of at least one of the sets in $\mathcal{F}$, and $\cap\{S : S \in \mathcal{F}\}$ is the set of all elements that are members of every set in $\mathcal{F}$.

The union and intersection of finitely many sets $S_1, \ldots, S_n$ are also written as $\bigcup_{k=1}^{n} S_k$ and $\bigcap_{k=1}^{n} S_k$.

The union and intersection of an infinite sequence $\{S_k\}_{k=1}^{\infty}$ of sets are written as $\bigcup_{k=1}^{\infty} S_k$ and $\bigcap_{k=1}^{\infty} S_k$.

**Example**: If $\mathcal{F}$ is the collection of sets

$$S_\rho = \{x : \rho < x \leq 1 + \rho\},$$

where $0 < \rho \leq 1/2$, then

$$\bigcup\{S_\rho : S_\rho \in \mathcal{F}\} = \{x : 0 < x \leq 3/2\}$$

$$\bigcap\{S_\rho : S_\rho \in \mathcal{F}\} = \{x : 1/2 < x \leq 1\}.$$

**Example**: If, for each positive integer $k$, the set $S_k$ is the set of real numbers that can be written as $x = m/k$ for some integer $m$,

then $\bigcup_{k=1}^{\infty} S_k$ is the set of rational numbers and $\bigcap_{k=1}^{\infty} S_k$ is the set of integers.

If $a$ and $b$ are in the extended reals and $a < b$, then the *open interval* $(a, b)$ is defined by

$$(a, b) = \{x : a < x < b\}.$$

The open intervals $(a, \infty)$ and $(-\infty, b)$ are *semi-infinite* if $a$ and $b$ are finite, and $(-\infty, \infty)$ is the entire real line.

as pass waly elements

**$\varepsilon$-neighborhood**: If $x_0$ is a real number and $\varepsilon > 0$, then the open interval $(x_0 - \varepsilon, x_0 + \varepsilon)$ is an *$\varepsilon$-neighborhood* of $x_0$. If a set $S$ contains an $\varepsilon$-neighborhood of $x_0$, then $S$ is a *neighborhood* of $x_0$, and $x_0$ is an *interior point* of $S$.

The set of interior points of $S$ is the *interior* of $S$, denoted by $S^0$. If every point of $S$ is an interior point (that is, $S^0 = S$), then $S$ is *open*.

Open set
* No boundary elements (0,1)
Closed set
* Includes Boundary points [a,b]
Complement of open set is closed

A set $S$ is *closed* if $S^c$ is open.

**Example**: An open interval $(a, b)$ is an open set, because if $x_0 \in (a, b)$ and $\varepsilon \leq \min\{x_0 - a, b - x_0\}$, then

$$(x_0 - \varepsilon, x_0 + \varepsilon) \subset (a, b).$$

The entire line $\mathbb{R} = (-\infty, \infty)$ is open, and therefore $\emptyset = \mathbb{R}^c$ is closed. However, $\emptyset$ is also open. for to deny this is to say that $\emptyset$ contains a point that is not an interior point, which is absurd because $\emptyset$ contains no points. Since $\emptyset$ is open, $\mathbb{R}(= \emptyset^c)$ is closed. Thus, $\mathbb{R}$ and $\emptyset$ are both open and closed. They are the only subsets of $\mathbb{R}$ with this property

A *deleted neighborhood* of a point $x_0$ is a set that contains every point of some neighborhood of $x_0$ except for $x_0$ itself.

For example,

$$S = \{x : 0 < |x - x_0| < \varepsilon\}$$

is a deleted neighborhood of $x_0$. We also say that it is a *deleted $\varepsilon$-neighborhood* of $x_0$.

x0− ε and x0+ε are the neighbors of x0. x0 is the inter point of x0− ε and x0+ε then i we delete the interior point x0 that is called delete neighborhood

**Theorem**: The following statements are true for arbitrary collections, finite or infinite, of open and closed sets.

- The union of open sets is open.
- The intersection of closed sets is closed.

**Proof**: Let $\mathcal{G}$ be a collection of open sets and

$$S = \cup\{G : G \in \mathcal{G}\}.$$

If $x_0 \in S$, then $x_0 \in G_0$ for some $G_0$ in $\mathcal{G}$, and since $G_0$ is open, it contains some $\varepsilon$-neighborhood of $x_0$. Since $G_0 \subset S$, this $\varepsilon$-neighborhood is in $S$, which is consequently a neighborhood of $x_0$. Thus, $S$ is a neighborhood of each of its points, and therefore open, by definition.

Let $\mathcal{F}$ be a collection of closed sets and $T = \cap\{F : F \in \mathcal{F}\}$.

Then $T^c = \cup F^c : F \in \mathcal{F}$ and, since each $F^c$ is open, $T^c$ is open, from the first part. Therefore, $T$ is closed, by definition.

**Examples**: If $-\infty < a < b < \infty$, the set

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

this set is closed because of using closed intervals

is closed, since its complement is the union of the open sets $(-\infty, a)$ and $(b, \infty)$. We say that $[a, b]$ is a *closed interval*.

In open set no boundary elements (0,1) included in closed set Boundary points [a,b] are included it contains half closed and half open so, it's not fully closed and open set

**Examples**: The set

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

is a *half-closed* or *half-open interval*

If $-\infty < a < b < \infty$, as is

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\};$$

however, neither of these sets is open or closed. (Why not?)

**Examples**: *Semi-infinite closed intervals* are sets of the form

$$[a, \infty) = \{x \in \mathbb{R} : a \leq x\}$$

$$(-\infty, a] = \{x \in \mathbb{R} : x \leq a\},$$

where $a$ is finite. They are closed sets, since their complements are the open intervals $(-\infty, a)$ and $(a, \infty)$, respectively.

Let $S$ be a subset of $\mathbb{R}$. Then

agr x0 k neighbors ko delete kry to wo s ma contain kry

- $x_0$ is a *limit point* of $S$ if every deleted neighborhood of $x_0$ contains a point of $S$.

- $x_0$ is a *boundary point* of $S$ if every neighborhood of $x_0$ contains at least one point in $S$ and one not in $S$. The set of boundary points of $S$ is the *boundary* of $S$, denoted by $\partial S$. The *closure* of $S$, denoted by $\overline{S}$, is $\overline{S} = S \cup \partial S$.

Let $S$ be a subset of $\mathbb{R}$. Then

- $x_0$ is an *isolated point* of $S$ if $x_0 \in S$ and there is a neighborhood of $x_0$ that contains no other point of $S$.

Let $S$ be a subset of $\mathbb{R}$. Then

- $x_0$ is *exterior* to $S$ if $x_0$ is in the interior of $S^c$. The collection of such points is the *exterior* of $S$.

Let $S = (-\infty, -1] \cup (1, 2) \cup \{3\}$. Then

- The set of limit points of $S$ is $(-\infty, -1] \cup [1, 2]$.
- $\partial S = \{-1, 1, 2, 3\}$ and $\overline{S} = (-\infty, -1] \cup [1, 2] \cup \{3\}$.

Let $S = (-\infty, -1] \cup (1, 2) \cup \{3\}$. Then

- 3 is the only isolated point of $S$.

- The exterior of $S$ is $(-1, 1) \cup (2, 3) \cup (3, \infty)$.

Exterior: whose points that don't belong to a set e.g The point 2 is an exterior point of the set [0, 1] because it doesn't belong to the set.

For $n \geq 1$, let

$$I_n = \left[\frac{1}{2n+1}, \frac{1}{2n}\right] \quad \text{and} \quad S = \bigcup_{n=1}^{\infty} I_n.$$

Then

- The set of limit points of $S$ is $S \cup \{0\}$.
- $\partial S = x : x = 0$ or $x = 1/n$ $(n \geq 2)$ and $\overline{S} = S \cup \{0\}$.
- $S$ has no isolated points.
- The exterior of $S$ is

$$(-\infty, 0) \cup \left[\bigcup_{n=1}^{\infty} \left(\frac{1}{2n+2}, \frac{1}{2n+1}\right)\right] \cup \left(\frac{1}{2}, \infty\right).$$

open set is asi collection k agr UN ka union ly to usi set ma contain kry

## 1.20 Open Coverings

A collection $\mathcal{H}$ of open sets is an *open covering* of a set $S$ if every point in $S$ is contained in a set $H$ belonging to $\mathcal{H}$; that is, if

$$S \subset \cup\{H \in \mathcal{H} : H \in \mathcal{H}\}.$$

if n=1 (x-1, x+1)
if n=2 ( x-1/2,x+1/2)
that cover the interval of 0,1
it contains finite real numbers

**Example**: The set

$$S_1 = [0, 1],$$

is covered by the family of open intervals

$$\mathcal{H}_1 = \left\{\left(x - \frac{1}{N}, x + \frac{1}{N}\right) \middle| 0 < x < 1\right\},$$

$(N =$ positive integer$)$, .

**Example**: The set

$$S_2 = \{1, 2, \ldots, n, \ldots\}$$

if n=1 then (1-1/4,1+1/4)
if n=2 then (2-1/4,2+1/4) so on
each member is contained in s2

is covered by the family of open intervals

$$\mathcal{H}_2 = \left\{\left(n - \frac{1}{4}, n + \frac{1}{4}\right) \middle| n = 1, 2, \ldots\right\}.$$

**Example**: The sets

$$S_3 = \left\{1, \frac{1}{2}, \ldots, \frac{1}{n}, \ldots\right\}, \quad \text{and} \quad S_4 = (0, 1)$$

are covered by the families of open intervals

$$\mathcal{H}_3 = \left\{\left(\frac{1}{n+\frac{1}{2}}, \frac{1}{n-\frac{1}{2}}\right) \middle| n = 1, 2, \ldots\right\},$$

and

if n=1 then (1/2+1/2, 1/1-1/2)
if n=2 then (1/1+1/2, 1/2-1/2)
all the members contained in S3

S is closed: S contains all its limit points.
S is bounded: S is contained in a finite interval [a, b].



[0,1] closed or bounded so it's compact set

$$\mathcal{H}_4 = \{(0, \rho) | \, 0 < \rho < 1\},$$

respectively.

**Theorem**: If $\mathcal{H}$ is an open covering of a closed and bounded subset $S$ of the real numbers then $S$ has an open covering $\widetilde{\mathcal{H}}$ consisting of finitely many open sets belonging to $\mathcal{H}$.

**Compact set**: A closed and bounded set of real numbers is called *compact*.

R^1 = real numbers
R^2= plane
R^3= space
so on

The Heine-Borel theorem says that any open covering of a compact set $S$ contains a finite collection that also covers $S$.

This theorem and its converse show that we could just as well define a set $S$ of reals to be compact if it has the Heine-Borel property; that is, if every open covering of $S$ contains a finite subcovering.

The same is true of $\mathbb{R}^n$. This definition generalizes to more abstract spaces (called *topological spaces*) for which the concept of boundedness need not be defined.

0,1 are closed and bounded so it contains finite open collection

**Example**: The set

$$S_1 = [0, 1],$$

is covered by the family of open intervals

$$\mathcal{H}_1 = \left\{ \left( x - \frac{1}{N}, x + \frac{1}{N} \right) \middle| \, 0 < x < 1 \right\},$$
$$(N = \text{positive integer}), \quad .$$

The $2N$ intervals from $\mathcal{H}_1$ centered at the points $x_k = k/2N \, (0 \leq k \leq 2N - 1)$ cover $S_1$.

**Example**: The set

$$S_2 = \{1, 2, \ldots, n, \ldots\},$$

$$S_3 = \left\{ 1, \frac{1}{2}, \ldots, \frac{1}{n}, \ldots \right\}, \quad \text{and} \quad S_4 = (0, 1)$$

are not compact sets. because it is not closed and bounded

**Theorem**: A set $S$ is closed if and only if no point of $S^c$ is a limit point of $S$.

**Corollary**: A set $S$ is closed if and only if $S$ contains all of its limit points.

**Remark**: Hence, a set with no limit points is closed according to the corollary, in agreement with Theorem. For example, any finite set is closed. More generally, $S$ is closed if there is a $\delta > 0$ such $|x - y| \geq \delta$ for every pair $\{x, y\}$ of distinct points in $S$.

S is closed so S' is open
x0∈ S'
sε subset of S'
that mean sε totally contained in S complement.
so, sε ∩S' = φ
thus, this empty relation doesn't satisfied the property of the limit points.

Sequence:
A sequence is an ordered list of numbers, denoted by {an} or (an),
Example: {1, 2, 4, 8, ...} is a sequence which each term is twice the previous
Series:(it may be finite or infinite)
the sum of the terms of a sequence
Σan = a1 + a2 + a3 + ... + an + ...
Example: 1 + 2 + 4 + 8 + ... is a series,

CHAPTER 2

# Sequences and Series

## 2.1 Sequences

An *infinite sequence* (more briefly, a *sequence*) of real numbers is a real-valued function defined on a set of integers $n : n \geq k$. We call the values of the function the *terms* of the sequence.

sequence We denote a sequence by listing its terms in order; thus,

asa real valued function jiski domain set of integers ay

$$\{s_n\}_k^\infty = \{s_k, s_{k+1}, \dots\}.$$

start from 3 to ∞

**Examples**: Consider the following sequences

$$\left\{\frac{1}{n^2+1}\right\}_0^\infty = \left\{1, \frac{1}{2}, \frac{1}{5}, \dots, \frac{1}{n^2+1}, \dots\right\},$$

$$\{(-1)^n\}_0^\infty = \{1, -1, 1, \dots, (-1)^n, \dots\},$$

$$\left\{\frac{1}{n-2}\right\}_3^\infty = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-2}, \dots\right\}.$$

S1={1/3-2,1/4-2,1/5-2---}
= {1,1/2,1/3---}
start from 1
from ∞
S2={1,1/2,1/3---}

The real number $s_n$ is the $n$th *term* of the sequence.

**Examples**: Usually we are interested only in the terms of a sequence and the order in which they appear, but not in the particular value of $k$ in (**??**). Therefore, we regard the sequences The sequences are

both are identical

$$\left\{\frac{1}{n-2}\right\}_3^\infty \quad \text{and} \quad \left\{\frac{1}{n}\right\}_1^\infty$$

identical.

In the absence of any indication to the contrary, we take $k = 0$ unless $s_n$ is given by a rule that is invalid for some nonnegative integer, in which case $k$ is understood to be the smallest positive integer such that $s_n$ is defined for all $n \geq k$. For example, if

$$s_n = \frac{1}{(n-1)(n-5)},$$

then $k = 6$.

a sequence is said to be convergent if it has a finite limit (unique limit L ) that lies between L- ε and L+ ε



### 2.1.1 Convergent Sequence

A sequence $\{s_n\}$ *converges to a limit* $L$ if for every $\varepsilon > 0$ there is an integer $N$ such that

$$|s_n - L| < \varepsilon \quad \text{if} \quad n \geq N.$$

Sn =c

In this case we say that $\{s_n\}$ is *convergent* and write

|Sn -c|= |c-c| =0

$$\lim_{n\to\infty} s_n = L.$$

JB Sn hi constant Lia h to hmari limit Sn bhi constant hi ho gi

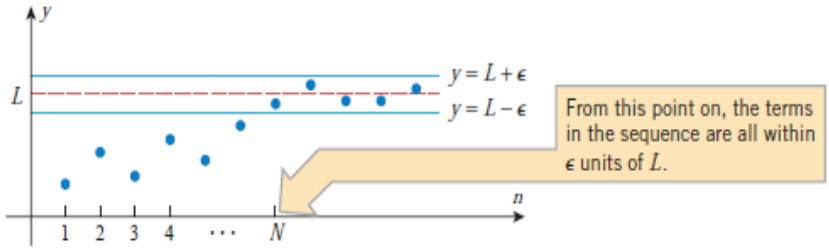



Figure 2.1: Convergence of the sequence

A ~~sequence that does not converge *diverges*, or is *divergent*~~

**Example**: If $s_n = c$ for $n \geq k$, then $|s_n - c| = 0$ for $n \geq k$, and $\lim_{n\to\infty} s_n = c$.

**Example**: If

$$s_n = \left\{\frac{2n+1}{n+1}\right\},$$

then $\lim_{n\to\infty} s_n = 2$, since

$$\begin{aligned} |s_n - 2| &= \left|\frac{2n+1}{n+1} - \frac{2n+2}{n+1}\right| \\ &= \frac{1}{n+1}; \end{aligned}$$

terms between 2-ε and 2 + ε

1 sa zyada limits point nhi hoty so the limit of convergence is unique

**Theorem**: The limit of a convergent sequence is unique.

**Proof**: Suppose that

$$\lim_{n\to\infty} s_n = s \quad \text{and} \quad \lim_{n\to\infty} s_n = s'.$$

imp

We must show that $s = s'$. Let $\varepsilon > 0$. From definition, there are integers $N_1$ and $N_2$ such that

$$|s_n - s| < \varepsilon \quad \text{if} \quad n \geq N_1$$

(because $\lim_{n\to\infty} s_n = s$), and

$$|s_n - s'| < \varepsilon \quad \text{if} \quad n \geq N_2$$

(because $\lim_{n\to\infty} s_n = s'$). These inequalities both hold if $n \geq N = \max(N_1, N_2)$, which implies that

$$\begin{aligned} |s - s'| &= |(s - s_N) + (s_N - s')| \\ &\leq |s - s_N| + |s_N - s'| < \varepsilon + \varepsilon = 2\varepsilon. \end{aligned}$$

Since this inequality holds for every $\varepsilon > 0$ and $|s - s'|$ is independent of $\varepsilon$, we conclude that $|s - s'| = 0$; that is, $s = s'$.

A sequence $\{s_n\}$ is said to be divergent to $\infty$ if for any real number $a$, $s_n > a$ for large $n$ and written as

$$\lim_{n\to\infty} s_n = \infty$$

Similarly,

$$\lim_{n\to\infty} s_n = -\infty$$

if for any real number $a$, $s_n < a$ for large $n$. not regard $\{s_n\}$ as convergent unless $\lim_{n\to\infty} s_n$ is finite, as required by Definition of limit. To emphasize this distinction, we say that $\{s_n\}$ *diverges to* $\infty$ $(-\infty)$ if $\lim_{n\to\infty} s_n = \infty$ $(-\infty)$.

**Example**: The sequence $\{n/2 + 1/n\}$ diverges to $\infty$, since, if $a$ is any real number, then

$$\frac{n}{2} + \frac{1}{n} > a \quad \text{if} \quad n \geq 2a.$$

Therefore, we write

$$\lim_{n\to\infty} \left(\frac{n}{2} + \frac{1}{n}\right) = \infty$$

diverge kr rha h

**Example**: The sequence $\{n - n^2\}$ diverges to $-\infty$, since, if $a$ is any real number, then

$$-n^2 + n = -n(n-1) < a \quad \text{if} \quad n > 1 + \sqrt{|a|}.$$

diverge kr rha h

Therefore, we write

$$\lim_{n\to\infty} (-n^2 + n) = -\infty.$$

**Example**: The sequence $\{(-1)^n n^3\}$ diverges, but not to $-\infty$ or $\infty$.

ye na to converge kr rha h or na hi diverge

## 2.2 Bounded Sequence

A sequence $\{s_n\}$ is *bounded above* if there is a real number $b$ such that

$$s_n \leq b \quad \text{for all } n,$$

*bounded below* if there is a real number $a$ such that

$$s_n \geq a \quad \text{for all } n,$$

or *bounded* if there is a real number $r$ such that

$$|s_n| \leq r \quad \text{for all } n.$$

A sequence is bounded above if it has an upper bound.

A sequence is bounded below if it has a lower bound.

If a sequence is both bounded above and below, then it is said to be bounded.

if the sequence doesn't diverges at -∞ and +∞ then it oscillate b/w -∞ and +∞ so this sequence is oscillating sequence

**Example**: If $s_n = [1+(-1)^n]n$, then $\{s_n\}$ is bounded below ($s_n \geq 0$) but unbounded above.

$\{-s_n\}$ is bounded above ($-s_n \leq 0$) but unbounded below.

If $s_n = (-1)^n$, then $\{s_n\}$ is bounded.

If $s_n = (-1)^n n$, then $\{s_n\}$ is not bounded above or below.

**Theorem**: A convergent sequence is bounded.
**Proof**: By taking $\varepsilon = 1$, we see that if $\lim_{n\to\infty} s_n = s$, then there is an integer $N$ such that

$$|s_n - s| < 1 \quad \text{if} \quad n \geq N.$$

Therefore,

$$|s_n| = |(s_n - s) + s| \leq |s_n - s| + |s| < 1 + |s| \quad \text{if} \quad n \geq N,$$

and

$$|s_n| \leq \max\{|s_0|, |s_1|, \ldots, |s_{N-1}|, 1 + |s|\}$$

for all $n$, so $\{s_n\}$ is bounded.

**Theorem**: A sequence $\{s_n\}$ converges to $s$ if and only if every neighborhood of $s$ contains $s_n$ for all but finitely many $n$.

## 2.3 Monotonic Sequences

A sequence $\{s_n\}$ is *nondecreasing* if $s_n \geq s_{n-1}$ for all $n$, or *nonincreasing* if $s_n \leq s_{n-1}$ for all $n$.

A *monotonic sequence* is a sequence that is either nonincreasing or nondecreasing. If $s_n > s_{n-1}$ for all $n$, then $\{s_n\}$ is *increasing*, while if $s_n < s_{n-1}$ for all $n$, $\{s_n\}$ is *decreasing*.

**Theorem**:

- If $\{s_n\}$ is nondecreasing, then $\lim_{n\to\infty} s_n = \sup\{s_n\}$.
- If $\{s_n\}$ is nonincreasing, then $\lim_{n\to\infty} s_n = \inf\{s_n\}$.

**Recall**: The definition of supremum and infimum of a set.
**Proof**: Let $\beta = \sup\{s_n\}$. If $\beta < \infty$, then by definition if $\varepsilon > 0$ then

$$\beta - \varepsilon < s_N \leq \beta$$

for some integer $N$.

Since $s_N \leq s_n \leq \beta$ if $n \geq N$, it follows that

$$\beta - \varepsilon < s_n \leq \beta \quad \text{if} \quad n \geq N.$$

The sequence {1, 2, 3, ...} is bounded below (by 1) but not bounded above.
- The sequence {-1, -2, -3, ...} is bounded above (by -1) but not bounded below.
- The sequence {0, 1, 0, 1, ...} is bounded (by 0 and 1).

{1,-1}

definition

supremum is the least upper bound
infimum is the greatest lower bound

This implies that $|s_n - \beta| < \varepsilon$ if $n \geq N$, so $\lim_{n\to\infty} s_n = \beta$, by definition.

If $\beta = \infty$ and $b$ is any real number, then $s_N > b$ for some integer $N$. Then $s_n > b$ for $n \geq N$, so $\lim_{n\to\infty} s_n = \infty$.

For the proof of the second part try yourself.

**Example**: If $s_0 = 1$ and $s_n = 1 - e^{-s_{n-1}}$, then $0 < s_n \leq 1$ for all $n$, by induction.

$$\text{Since} \quad s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}}) \quad \text{if} \quad n \geq 1.$$

The mean value theorem implies that

$$s_{n+1} - s_n = e^{-t_n}(s_n - s_{n-1}) \quad \text{if} \quad n \geq 1, \tag{2.1}$$

where $t_n$ is between $s_{n-1}$ and $s_n$. Since $s_1 - s_0 = -1/e < 0$, it follows by induction from (2.1) that $s_{n+1} - s_n < 0$ for all $n$.

Hence, $\{s_n\}$ is bounded and decreasing, and therefore convergent.

**Remark**: Let $\{x_n\}$ and $\{s_n\}$ be two sequences. If $0 \leq x_n \leq s_n$ for $n \geq N$, where $N$ is some fixed number, and if

$$s_n \to 0,$$

then

$$x_n \to 0.$$

## 2.4 Some Special Sequences

**Theorem**: If $p > 0$, then

$$\lim_{n\to\infty} \frac{1}{n^p} = 0.$$

**Proof**: Before presenting the proof let us recall the following:

**Recall the binomial theorem**:

$$(1+x)^n = 1 + nx + x(n-1)\frac{x^2}{2!} + ....$$

We have the following inequality

$$1 + nx \leq (1+x)^n, x > 0.$$

**Theorem**: If $p > 0$, then

$$\lim_{n\to\infty} \sqrt[n]{p} = 1.$$

**Proof**: We will discuss the three cases when $p = 1$, when $p > 1$ and when $0 < p < 1$. If $p > 1$, put $x_n = \sqrt[n]{p} - 1$. Then $x_n > 0$, and, by the binomial theorem

$$1 + nx_n \leq (1 + x_n)^n = p,$$

so that

$$0 < x_n \leq \frac{p-1}{n}.$$

**Theorem**: Show that

$$\lim_{n\to\infty} \sqrt[n]{n} = 1.$$

**Proof**: Take $x_n = \sqrt[n]{n} - 1$.

Then $x_n \geq 0$, and, by the binomial theorem

$$n = (1+x_n)^n \geq \frac{n(n-1)}{2} x_n^2.$$

Hence

$$0 \leq x_n \leq \sqrt{\frac{2}{n-1}}, \qquad n \geq 2.$$

**Theorem**: If $p > 0$ and $\alpha$ is real, then

$$\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0.$$

**Proof**: Let $k$ be an integer such that $k > \alpha$, $k > 0$.

For $n > 2k$,

$$(1+p)^n > \frac{n(n-1)...(n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}.$$

Hence

$$0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k}, \quad n > 2k.$$

Since $\alpha - k < 0, n^{\alpha-k} \to 0$ by

$$\lim_{n\to\infty} \frac{1}{n^p} = 0.$$

**Theorem**: If $|x| < 1$, then

$$\lim_{n\to\infty} x^n = 0.$$

**Theorem**: Let $\lim_{n\to\infty} s_n = s$ and $\lim_{n\to\infty} t_n = t$, where $s$ and $t$ are finite. Then

$$\lim_{n\to\infty} (cs_n) = cs \tag{2.2}$$

if $c$ is a constant;

$$\begin{aligned} \lim_{n\to\infty} (s_n + t_n) &= s + t, \\ \lim_{n\to\infty} (s_n - t_n) &= s - t, \\ \lim_{n\to\infty} (s_n t_n) &= st, \end{aligned}$$

$$\lim_{n\to\infty} \frac{s_n}{t_n} = \frac{s}{t}$$

if $t_n$ is nonzero for all $n$ and $t \neq 0$.

**Proof**: We write

$$s_n t_n - st = s_n t_n - st_n + st_n - st = (s_n - s)t_n + s(t_n - t);$$

hence,

$$|s_n t_n - st| \leq |s_n - s|\,|t_n| + |s|\,|t_n - t|.$$

Since $\{t_n\}$ converges, it is bounded.

Therefore, there is a number $R$ such that $|t_n| \leq R$ for all $n$, and (??) implies that

$$|s_n t_n - st| \leq R|s_n - s| + |s|\,|t_n - t|. \tag{2.3}$$

By definition, if $\varepsilon > 0$ there are integers $N_1$ and $N_2$ such that

$$\begin{aligned} |s_n - s| &< \varepsilon \quad \text{if} \quad n \geq N_1 \\ |t_n - t| &< \varepsilon \quad \text{if} \quad n \geq N_2. \end{aligned}$$

If $N = \max(N_1, N_2)$, then both inequalities hold when $n \geq N$, and the (2.3) implies that

$$|s_n t_n - st| \leq (R + |s|)\varepsilon \quad \text{if} \quad n \geq N.$$

This proves (2.3).

We consider the special case where $s_n = 1$ for all $n$ and $t \neq 0$; thus, we want to show that

$$\lim_{n\to\infty} \frac{1}{t_n} = \frac{1}{t}.$$

First, observe that since $\lim_{n\to\infty} t_n = t \neq 0$, there is an integer $M$ such that $|t_n| \geq |t|/2$ if $n \geq M$.

By definition with $\varepsilon = |t|/2$; thus, there is an integer $M$ such that $|t_n - t| < |t/2|$ if $n \geq M$.

Therefore,

$$|t_n| = |t + (t_n - t)| \geq ||t| - |t_n - t|| \geq \frac{|t|}{2} \quad \text{if} \quad n \geq M.$$

If $\varepsilon > 0$, choose $N_0$ so that $|t_n - t| < \varepsilon$ if $n \geq N_0$, and let $N = \max(N_0, M)$. Then

$$\left|\frac{1}{t_n} - \frac{1}{t}\right| = \frac{|t - t_n|}{|t_n|\,|t|} \leq \frac{2\varepsilon}{|t|^2} \quad \text{if} \quad n \geq N.$$

hence, $\lim_{n\to\infty} 1/t_n = 1/t$. Now we obtain (2.3) in the general case from (2.3) with $\{t_n\}$ replaced by $\{1/t_n\}$.

**Example**: Find the limit of the sequence

$$s_n = \frac{1}{n}\sin\frac{n\pi}{4} + \frac{2(1 + 3/n)}{1 + 1/n}.$$

**Solution**: We apply the applicable parts of Theorem as follows:

$$\begin{aligned}\lim_{n\to\infty} s_n &= \lim_{n\to\infty}\frac{1}{n}\sin\frac{n\pi}{4} + \frac{2\left[\lim_{n\to\infty}1+3\lim_{n\to\infty}(1/n)\right]}{\lim_{n\to\infty}}1+\lim_{n\to\infty}(1/n)\\ &= 0+\frac{2(1+3\cdot 0)}{1+0}=2.\end{aligned}$$

converge

**Example**: Find the limit of the sequence $s_n = \lim_{n\to\infty}\frac{(n/2)+\log n}{3n+4\sqrt{n}}$.
**Solution**:

$$\begin{aligned}\lim_{n\to\infty}\frac{(n/2)+\log n}{3n+4\sqrt{n}} &= \lim_{n\to\infty}\frac{1/2+(\log n)/n}{3+4n^{-1/2}}\\ &= \frac{\lim_{n\to\infty}1/2+\lim_{n\to\infty}(\log n)/n}{\lim_{n\to\infty}3+4\lim_{n\to\infty}n^{-1/2}}\\ &= \frac{1/2+0}{3+0}\\ &= \frac{1}{6}.\end{aligned}$$

converge

## 2.5 Subsequence

jis my set ko orderly arrange krty h

A sequence $\{t_k\}$ is a *subsequence* of a sequence $\{s_n\}$ if

$$t_k = s_{n_k}, \quad k \geq 0,$$

where $\{n_k\}$ is an increasing infinite sequence of integers in the domain of $\{s_n\}$. We denote the subsequence $\{t_k\}$ by $\{s_{n_k}\}$.
**Example**: If

$$\{s_n\} = \left\{\frac{1}{n}\right\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\right\},$$

then letting $n_k = 2k$ yields the subsequence

$$\{s_{2k}\} = \left\{\frac{1}{2k}\right\} = \left\{\frac{1}{2}, \frac{1}{4}, \ldots, \frac{1}{2k}, \ldots\right\},$$

and letting $n_k = 2k+1$ yields the subsequence

$$\{s_{2k+1}\} = \left\{\frac{1}{2k+1}\right\} = \left\{1, \frac{1}{3}, \ldots, \frac{1}{2k+1}, \ldots\right\}.$$

**Example**: The sequence $\{s_n\}$ defined by

$$s_n = (-1)^n\left(1+\frac{1}{n}\right)$$

does not converge, but $\{s_n\}$ has subsequences that do. For example,

$$\{s_{2k}\} = \left\{1+\frac{1}{2k}\right\}$$

A subsequence is a sequence that is derived from another sequence by selecting certain terms in a specific order.

$$\lim_{k\to\infty} s_{2k} = 1.$$

**Example**: The sequence $\{s_n\}$ defined by

$$s_n = (-1)^n \left(1 + \frac{1}{n}\right)$$

does not converge, but $\{s_n\}$ has subsequences that do.

For example,

$$\{s_{2k+1}\} = \left\{-1 - \frac{1}{2k+1}\right\}$$

$$\lim_{k\to\infty} s_{2k+1} = -1.$$

It can be shown that a subsequence $\{s_{n_k}\}$ of $\{s_n\}$ converges to 1 if and only if $n_k$ is even for $k$ sufficiently large, or to $-1$ if and only if $n_k$ is odd for $k$ sufficiently large. Otherwise, $\{s_{n_k}\}$ diverges.

**Theorem**: If

$$\lim_{n\to\infty} s_n = s \quad (-\infty \le s \le \infty),$$

then every subsequence of convergent sequence is convergent $\lim_{k\to\infty} s_{n_k} = s$

for every subsequence $\{s_{n_k}\}$ of $\{s_n\}$.

**Proof**: By definition for every $\varepsilon > 0$, there is an integer $N$ such that

$$|s_n - s| < \varepsilon \quad \text{if} \quad n \ge N.$$

Since $\{n_k\}$ is an increasing sequence, there is an integer $K$ such that $n_k \ge N$ if $k \ge K$. Therefore,

$$|s_{n_k} - L| < \varepsilon \quad \text{if} \quad k \ge K.$$

**Theorem**: If $\{s_n\}$ is monotonic and has a subsequence $\{s_{n_k}\}$ such that

$$\lim_{k\to\infty} s_{n_k} = s,$$

if a sequence is monotonic and ha convergent subsequence then the sequence is convergent

then

$$\lim_{n\to\infty} s_n = s.$$

**Recall the following theorem**:

quiz If $\{s_n\}$ is nondecreasing, then $\lim_{n\to\infty} s_n = \sup\{s_n\}$.

If $\{s_n\}$ is nonincreasing, then $\lim_{n\to\infty} s_n = \inf\{s_n\}$.

**Proof**: Since $\{s_{n_k}\}$ is also nondecreasing in this case, it is sufficient to show that

$$\sup\{s_{n_k}\} = \sup\{s_n\}$$

Since the set of terms of $\{s_{n_k}\}$ is contained in the set of terms of $\{s_n\}$,

$$\sup\{s_n\} \geq \sup\{s_{n_k}\}.$$

Since $\{s_n\}$ is nondecreasing, there is for every $n$ an integer $n_k$ such that $s_n \leq s_{n_k}$. This implies that

$$\sup\{s_n\} \leq \sup\{s_{n_k}\}.$$

**Theorem**: A point $\overline{x}$ is a limit point of a set $S$ **if and only if** there is a sequence $\{x_n\}$ of points in $S$ such that $x_n \neq \overline{x}$ for $n \geq 1$, and

$$\lim_{n\to\infty} x_n = \overline{x}.$$

**Proof**: By definition for each $\varepsilon > 0$, there is an integer $N$ such that $0 < |x_n - \overline{x}| < \varepsilon$ if $n \geq N$.

Therefore, every $\varepsilon$-neighborhood of $\overline{x}$ contains infinitely many points of $S$. This means that $\overline{x}$ is a limit point of $S$.

For necessity, let $\overline{x}$ be a limit point of $S$. Then, for every integer $n \geq 1$, the interval $(\overline{x} - 1/n, \overline{x} + 1/n)$ contains a point $x_n$ $(\neq \overline{x})$ in $S$.

Since $|x_m - \overline{x}| \leq 1/n$ if $m \geq n$, $\lim_{n\to\infty} x_n = \overline{x}$.

**Theorem**:

Mcqs

- If $\{x_n\}$ is bounded, then $\{x_n\}$ has a convergent subsequence.
- If $\{x_n\}$ is unbounded above, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that

$$\lim_{k\to\infty} x_{n_k} = \infty.$$

- If $\{x_n\}$ is unbounded below, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that

$$\lim_{k\to\infty} x_{n_k} = -\infty.$$

**Proof**: Let $S$ be the set of distinct numbers that occur as terms of $\{x_n\}$.

(For example, if $\{x_n\} = \{(-1)^n\}$, $S = \{1, -1\}$; if $\{x_n\} = \{1, \frac{1}{2}, 1, \frac{1}{3}, \ldots, 1, 1/n, \ldots\}$, $S = \{1, \frac{1}{2}, \ldots, 1/n, \ldots\}$.)

If $S$ contains only finitely many points, then some $\overline{x}$ in $S$ occurs infinitely often in $\{x_n\}$; that is, $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that $x_{n_k} = \overline{x}$ for all $k$. Then $\lim_{k\to\infty} x_{n_k} = \overline{x}$, and we are finished in this case.

If $S$ is infinite, then, since $S$ is bounded (by assumption), the Bolzano-Weierstrass theorem implies that $S$ has a limit point $\overline{x}$.

There is a sequence of points $\{y_j\}$ in $S$, distinct from $\overline{x}$, such that

$$\lim_{j\to\infty} y_j = \overline{x}.$$

Although each $y_j$ occurs as a term of $\{x_n\}$, $\{y_j\}$ is not necessarily a subsequence of $\{x_n\}$, because if we write

$$y_j = x_{n_j}.$$

There is no reason to expect that $\{n_j\}$ is an increasing sequence as required in definition of subsequence.

However, it is always possible to pick a subsequence $\{n_{j_k}\}$ of $\{n_j\}$ that is increasing, and then the sequence $\{y_{j_k}\} = \{s_{n_{j_k}}\}$ is a subsequence of both $\{y_j\}$ and $\{x_n\}$.

- The limit superior of a sequence {an} is the largest number that is simultaneously an upper bound for all but finitely many terms of the sequence.

## 2.6 Limit Superior and Limit Inferior

**Theorem**:

- If $\{s_n\}$ is bounded above and does not diverge to $-\infty$, then there is a unique real number $\overline{s}$ such that, if $\varepsilon > 0$,

$$s_n < \overline{s} + \varepsilon \quad \text{for large } n \tag{2.4}$$

and

$$s_n > \overline{s} - \varepsilon \quad \text{for infinitely many } n. \tag{2.5}$$

- If $\{s_n\}$ is bounded below and does not diverge to $\infty$, then there is a unique real number $\underline{s}$ such that, if $\varepsilon > 0$,

$$s_n > \underline{s} - \varepsilon \quad \text{for large } n \tag{2.6}$$

and

$$s_n < \underline{s} + \varepsilon \quad \text{for infinitely many } n. \tag{2.7}$$

**Proof**: Proof of the first part. Since $\{s_n\}$ is bounded above, there is a number $\beta$ such that $s_n < \beta$ for all $n$. Since $\{s_n\}$ does not diverge to $-\infty$, there is a number $\alpha$ such that $s_n > \alpha$ for infinitely many $n$. If we define

$$M_k = \sup\{s_k, s_{k+1}, \ldots, s_{k+r}, \ldots\},$$

then $\alpha \le M_k \le \beta$, so $\{M_k\}$ is bounded. Since $\{M_k\}$ is nonincreasing (why?), it converges. Let

$$\overline{s} = \lim_{k\to\infty} M_k. \tag{2.8}$$

If $\varepsilon > 0$, then $M_k < \overline{s} + \varepsilon$ for large $k$, and since $s_n \le M_k$ for $n \ge k$, $\overline{s}$ satisfies (2.4).

If (2.5) were false for some positive $\varepsilon$, there would be an integer $K$ such that

$$s_n \le \overline{s} - \varepsilon \quad \text{if} \quad n \ge K.$$

However, this implies that

$$M_k \le \overline{s} - \varepsilon \quad \text{if} \quad k \ge K,$$

The limit inferior of a sequence {an} is the smallest number that is simultaneously a lower bound for all but finitely many terms of the sequence.
- If

which contradicts (2.8). Therefore, $\overline{s}$ has the stated properties.

Now we must show that $\overline{s}$ is the only real number with the stated properties. If $t < \overline{s}$, the inequality

$$s_n < t + \frac{\overline{s} - t}{2} = \overline{s} - \frac{\overline{s} - t}{2}$$

cannot hold for all large $n$, because this would contradict (2.5) with $\varepsilon = (\overline{s} - t)/2$. If $\overline{s} < t$, the inequality

$$s_n > t - \frac{t - \overline{s}}{2} = \overline{s} + \frac{t - \overline{s}}{2}$$

cannot hold for infinitely many $n$, because this would contradict (2.4) with $\varepsilon = (t - \overline{s})/2$. Therefore, $\overline{s}$ is the only real number with the stated properties.

The numbers $\overline{s}$ and $\underline{s}$ defined in the previous Theorem are called the *limit superior* and *limit inferior*, respectively, of $\{s_n\}$, and denoted by

$$\overline{s} = \limsup_{n\to\infty} s_n \quad \text{and} \quad \underline{s} = \liminf_{n\to\infty} s_n.$$

We also define

Mcqs

$$\begin{aligned}
\limsup_{n\to\infty} s_n &= \infty \quad \text{if } \{s_n\} \text{ is not bounded above,} \\
\limsup_{n\to\infty} s_n &= -\infty \quad \text{if } \lim_{n\to\infty} s_n = -\infty, \\
\liminf_{n\to\infty} s_n &= -\infty \quad \text{if } \{s_n\} \text{ is not bounded below,} \\
\liminf_{n\to\infty} s_n &= \infty \quad \text{if } \lim_{n\to\infty} s_n = \infty.
\end{aligned}$$

**Theorem**: Every sequence $\{s_n\}$ of real numbers has a unique limit superior, $\overline{s}$, and a unique limit inferior, $\underline{s}$, in the extended reals, and

$$\underline{s} \le \overline{s}.$$

**Examples**:

mcqs

$$\limsup_{n\to\infty} r^n = \begin{cases} \infty, & |r| > 1, \\ 1, & |r| = 1, \\ 0, & |r| < 1; \end{cases}$$

and

$$\liminf_{n\to\infty} r^n = \begin{cases} \infty, & r > 1, \\ 1, & r = 1, \\ 0, & |r| < 1, \\ -1, & r = -1, \\ -\infty, & r < -1. \end{cases}$$

**Example**:

$$\begin{aligned}
\limsup_{n\to\infty} n^2 = \liminf_{n\to\infty} n^2 &= \infty, \\
\limsup_{n\to\infty} (-1)^n \left(1 - \frac{1}{n}\right) &= 1, \\
\liminf_{n\to\infty} (-1)^n \left(n - \frac{1}{n}\right) &= -1,
\end{aligned}$$

and

$$\begin{aligned}
\limsup_{n\to\infty} \left[1 + (-1)^n\right] n^2 &= \infty, \\
\liminf_{n\to\infty} \left[1 + (-1)^n\right] n^2 &= 0.
\end{aligned}$$

**Theorem**: If $\{s_n\}$ is a sequence of real numbers, then

$$\lim_{n\to\infty} s_n = s \tag{2.9}$$

if and only if

$$\limsup_{n\to\infty} s_n = \liminf_{n\to\infty} s_n = s. \tag{2.10}$$

**Proof**: If $s = \pm\infty$, the equivalence of (2.9) and (2.10) follows immediately from their definitions. If $\lim_{n\to\infty} s_n = s$ (finite), then definition of a convergent sequence implies that (2.4)–(2.7) hold with $\overline{s}$ and $\underline{s}$ replaced by $s$. Hence, (2.10) follows from the uniqueness of $\overline{s}$ and $\underline{s}$. For the converse, suppose that $\overline{s} = \underline{s}$ and let $s$ denote their common value. Then (2.4) and (2.6) imply that

$$s - \varepsilon < s_n < s + \varepsilon$$

for large $n$, and (2.9) follows from Definition and the uniqueness of $\lim_{n\to\infty} s_n$.

## 2.7 Cauchy Sequence

A sequence $\{s_n\}$ of real numbers is called a Cauchy sequence if for every $\varepsilon > 0$, there is an integer $N$ such that

$$|s_n - s_m| < \varepsilon \quad \text{if} \quad m, n \geq N.$$

**Theorem**: If $\{s_n\}$ is a Cauchy sequence of real numbers, then $\{s_n\}$ is bounded.

**Proof**: See Lecture

**Theorem** (**Cauchy's convergence criterion**): A sequence $\{s_n\}$ of real numbers converges **if and only if**, for every $\varepsilon > 0$, there is an integer $N$ such that

$$|s_n - s_m| < \varepsilon \quad \text{if} \quad m, n \geq N.$$

**Proof**: Suppose that $\lim_{n\to\infty} s_n = s$ and $\varepsilon > 0$.

There is an integer $N$ such that

$$|s_r - s| < \frac{\varepsilon}{2} \quad \text{if} \quad r \geq N.$$

Therefore,

$$|s_n - s_m| = |(s_n - s) + (s - s_m)| \leq |s_n - s| + |s - s_m| < \varepsilon,$$

whenever $n, m \geq N$. Therefore, the stated condition is necessary for convergence of $\{s_n\}$.

Recall a Cauchy sequence $\{s_n\}$ is bounded. So $\overline{s}$ and $\underline{s}$ are finite.

Now suppose that $\varepsilon > 0$ and $N$ satisfies $|s_n - s_m| < \varepsilon$, $n, m \geq N$.

From definition of limit superior and limit inferior

$$|s_n - \overline{s}| < \varepsilon, \qquad |s_m - \underline{s}| < \varepsilon$$

for some integer $m, n > N$.

Since

$$\begin{aligned} |\overline{s} - \underline{s}| &= |(\overline{s} - s_n) + (s_n - s_m) + (s_m - \underline{s})| \\ &\leq |\overline{s} - s_n| + |s_n - s_m| + |s_m - \underline{s}|. \end{aligned}$$

We have

$$|\overline{s} - \underline{s}| < 3\varepsilon.$$

Since $\varepsilon$ is an arbitrary positive number, this implies that $\overline{s} = \underline{s}$, so $\{s_n\}$ converges.

## 2.8 Series

sum of the terms of sequence

If $\{a_n\}_k^\infty$ is an ~~infinite sequence~~ of real numbers, the symbol

$$\sum_{n=k}^{\infty} a_n$$

is an *infinite series*, and $a_n$ is the *nth term* of the series.

We say that $\sum_{n=k}^{\infty}$ ~~$a_n$ converges~~ *to the sum* $A$, and write

$$\sum_{n=k}^{\infty} a_n = A,$$

if the sequence $\{A_n\}_k^\infty$ defined by

$$A_n = a_k + a_{k+1} + \cdots + a_n, \quad n \geq k,$$

converges to $A$.

### 2.8.1 Sequence of Partial Sums

The finite sum $A_n$ is the *nth partial sum of* $\sum_{n=k}^{\infty} a_n$.

If $\{A_n\}_k^{\infty}$ diverges, we say that $\sum_{n=k}^{\infty} a_n$ *diverges*.

In particular, if $\lim_{n\to\infty} A_n = \infty$ or $-\infty$, we say that $\sum_{n=k}^{\infty} a_n$ *diverges to* $\infty$ *or* $-\infty$, and write

$$\sum_{n=k}^{\infty} a_n = \infty \quad \text{or} \quad \sum_{n=k}^{\infty} a_n = -\infty.$$

### 2.8.2 Oscillatory Series

A divergent infinite series that does not diverge to $\pm\infty$ is said to *oscillate*, or *be oscillatory*.

**Example**: Consider the series

$$\sum_{n=0}^{\infty} r^n, \quad -1 < r < 1.$$

Here $a_n = r^n$ $(n \geq 0)$. The $n$th term of sequence of partial sum is

$$A_n = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r},$$

which converges to $1/(1-r)$ as $n \to \infty$. Thus, we write

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1 - r}, \quad -1 < r < 1.$$

**Example**: The $n$th term of sequence of partial sum is

$$A_n = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r},$$

If $|r| > 1$, then $\sum_{n=0}^{\infty} r^n$ diverges; if $r > 1$, then

$$\sum_{n=0}^{\infty} r^n = \infty,$$

**Example**: The $n$th term of sequence of partial sum is

$$A_n = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r},$$

If $r < -1$, $\sum_{n=0}^{\infty} r^n$ oscillates, since its partial sums alternate in sign and their magnitudes become arbitrarily large for large $n$.

If $r = -1$, then $A_{2m+1} = 0$ and $A_{2m} = 1$ for $m \geq 0$, while if $r = 1$, $A_n = n + 1$; in both cases the series diverges.

**Theorem**: The sum of a convergent series is unique.
**Proof**: See Lecture.

**Theorem**: Let $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where $A$ and $B$ are finite. Then

$$\sum_{n=k}^{\infty}(ca_n) = cA$$

if $c$ is a constant,

$$\sum_{n=k}^{\infty}(a_n + b_n) = A + B,$$

and

$$\sum_{n=k}^{\infty}(a_n - b_n) = A - B.$$

These relations also hold if one or both of $A$ and $B$ is infinite, provided that the right sides are not indeterminate.

finite series di ho or us ma sy Kuch terms nikl dy or nikly sa series ki divergence or convergence py kor effect na ho

## 2.9 Dropping Finitely Many Terms

For example, suppose that we drop the first $k$ terms of a series $\sum_{n=0}^{\infty} a_n$, and consider the new series $\sum_{n=k}^{\infty} a_n$. sums of the two series by

$$\begin{aligned} A_n &= a_0 + a_1 + \cdots + a_n, \quad n \geq 0, \\ A'_n &= a_k + a_{k+1} + \cdots + a_n, \quad n \geq k. \end{aligned}$$

$$A_n = (a_0 + a_1 + \cdots + a_{k-1}) + A'_n, \quad n \geq k,$$

it follows that $A = \lim_{n\to\infty} A_n$ exists (in the extended reals) if and only if $A' = \lim_{n\to\infty} A'_n$ does, and in this case

$$A = (a_0 + a_1 + \cdots + a_{k-1}) + A'.$$

An important principle follows from this.

**Lemma**: Suppose that for $n$ sufficiently large (that is, for $n \geq N$) the terms of $\sum_{n=k}^{\infty} a_n$ satisfy some condition that implies convergence of an infinite series. Then $\sum_{n=k}^{\infty} a_n$ converges.

Similarly, suppose that for $n$ sufficiently large the terms $\sum_{n=k}^{\infty} a_n$ satisfy some condition that implies divergence of an infinite series. Then $\sum_{n=k}^{\infty} a_n$ diverges.

**Example**: *The series* $\sum_{k}^{\infty} a_n$ *converges if* $(-1)^n a_n > 0$, $|a_{n+1}| < |a_n|$, *and* $\lim_{n\to\infty} a_n = 0$.

The terms of

$$\sum_{n=1}^{\infty} \frac{16+(-2)^n}{n2^n}$$

do not satisfy these conditions for all $n \geq 1$, but they do satisfy them for sufficiently large $n$.

**Theorem**: A series $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is an integer $N$ such that

$$|a_n + a_{n+1} + \cdots + a_m| < \varepsilon \quad \text{if} \quad m \geq n \geq N.$$

**Proof**: In terms of the partial sums $\{A_n\}$ of $\sum a_n$,

$$a_n + a_{n+1} + \cdots + a_m = A_m - A_{n-1}.$$

Therefore, (**??**) can be written as

$$|A_m - A_{n-1}| < \varepsilon \quad \text{if} \quad m \geq n \geq N.$$

Since $\sum a_n$ converges if and only if $\{A_n\}$ converges, Theorem implies the conclusion.

**Theorem**: Show that the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

**Proof**: Consider the sequence of partial sums

$$s_1 = 1, s_2 = 1 + \frac{1}{2}, s_3 = 1 + \frac{1}{2} + \frac{1}{3}, s_4 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}, ...$$

form strictly increasing sequence

$$s_1 < s_2 < s_3 < ... < s_n < ...$$

hr pichli term agly sa less than h

$$\begin{aligned}
s_{2^1} &= 1 + \frac{1}{2} > \frac{1}{2} + \frac{1}{2} \\
s_{4=2^2} &= s_2 + \frac{1}{3} + \frac{1}{4} > s_2 + \frac{1}{4} + \frac{1}{4} > \frac{4}{2} \\
s_{8=3^3} &= s_4 + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > s_4 + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} > \frac{4}{2} \\
...s_{2^n} &> \frac{n+1}{2}.
\end{aligned}$$

If $M$ is any constant, we can find a positive integer $n$ such that

$$\frac{n+1}{2} > M.$$

But for this $n$, we have

$$s_{2^n} > \frac{n+1}{2} > M.$$

so that no constant $M$ is greater than or equal to every partial sum of the harmonic series.

**Example**: Consider the geometric series $\sum r^n$.

If $|r| < 1$ and $m \geq n$, then

$$\begin{aligned} |A_m - A_n| &= |r^{n+1} + r^{n+2} + \cdots + r^m| \\ &\leq |r|^{n+1}(1 + |r| + \cdots + |r|^{m-n-1}) \\ &= |r|^{n+1}\frac{1 - |r|^{m-n}}{1 - |r|} < \frac{|r|^{n+1}}{1 - |r|}. \end{aligned}$$

If $\varepsilon > 0$, choose $N$ so that

$$\frac{|r|^{N+1}}{1 - |r|} < \varepsilon.$$

Then Cauchy's convergence criterion implies that

$$|A_m - A_n| < \varepsilon \quad \text{if} \quad m \geq n \geq N.$$

Now Theorem implies that $\sum r^n$ converges if $|r| < 1$, as in Example **??**.

**Corollary**: If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$

**Proof**: If $\sum a_n$ converges, then for each $\varepsilon > 0$ there is an integer $K$ such that

$$\left|\sum_{n=k}^{\infty} a_n\right| < \varepsilon \quad \text{if} \quad k \geq K;$$

that is,

$$\lim_{k\to\infty} \sum_{n=k}^{\infty} a_n = 0.$$

**Be careful**: $\lim_{n\to\infty} a_n = 0$ Necessary condition.

Not sufficient.

**Example**: For the harmonic series $\lim_{n\to\infty} \frac{1}{n} = 0$.

But we have proved that

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is divergent.

**Corollary**: If $\sum a_n$ converges, then for each $\varepsilon > 0$ there is an integer $K$ such that

$$\left|\sum_{n=k}^{\infty} a_n\right| < \varepsilon \quad \text{if} \quad k \geq K;$$

that is,

$$\lim_{k\to\infty} \sum_{n=k}^{\infty} a_n = 0.$$

**Example**: If $|r| < 1$, then

$$\left|\sum_{n=k}^{\infty} r^n\right| = \left|r^k \sum_{n=k}^{\infty} r^{n-k}\right| = \left|r^k \sum_{n=0}^{\infty} r^n\right| = \frac{|r|^k}{1-r}.$$

Therefore, if

$$\frac{|r|^K}{1-r} < \varepsilon,$$

then

$$\left|\sum_{n=k}^{\infty} r^n\right| < \varepsilon \quad \text{if} \quad k \geq K,$$

which implies that $\lim_{k\to\infty} \sum_{n=k}^{\infty} r^n = 0$.

## 2.10 Series of Nonnegative Terms

The series $\sum a_n$ is said to be series of nonnegative terms if $a_n \geq 0$ for $n \geq k$. The theory of series $\sum a_n$ with terms that are nonnegative for sufficiently large $n$ is simpler than the general theory, since such

The series of nonnegative terms either converges to a finite limit or diverges to $\infty$.

**Recall the following theorem**:

- If $\{s_n\}$ is nondecreasing, then $\lim_{n\to\infty} s_n = \sup\{s_n\}$.
- If $\{s_n\}$ is nonincreasing, then $\lim_{n\to\infty} s_n = \inf\{s_n\}$.

**Theorem**: If $a_n \geq 0$ for $n \geq k$, then $\sum a_n$ converges if its partial sums are bounded, or diverges to $\infty$ if they are not. These are the only possibilities and, in either case,

$$\sum_{n=k}^{\infty} a_n = \sup\{A_n : n \geq k\},$$

where

$$A_n = a_k + a_{k+1} + \cdots + a_n, \quad n \geq k.$$

**Proof**: Since $A_n = A_{n-1} + a_n$ and $a_n \geq 0$ $(n \geq k)$, the sequence $\{A_n\}$ is nondecreasing, so the conclusion follows from Theorem and definition of convergence of a series.

**Recall the theorem**: If $a_n \geq 0$ for $n \geq k$, then $\sum a_n$ converges if its partial sums are bounded, or diverges to $\infty$ if they are not. These are the only possibilities and, in either case,

$$\sum_{n=k}^{\infty} a_n = \sup\{A_n : n \geq k\},$$

) ye converge tbhi kry ga JB sum of smaller series a_n less than infinity ho mtlb 0 ho ya constant ho tbhi to Sum of larger series b_n bhi less than infinity tbhi ye converge kry ga.
2) ye diverge tbhi kry ga JB sum of a_n equal to infinity ho to Sum of b_n bhi equal to infinity ho ga



where

$$A_n = a_k + a_{k+1} + \cdots + a_n, \quad n \geq k.$$

## 2.11 The Comparison Test

**Theorem**: Suppose that

$$0 \leq a_n \leq b_n, \quad n \geq k.$$

Then

- $\sum a_n < \infty$ if $\sum b_n < \infty$.
- $\sum b_n = \infty$ if $\sum a_n = \infty$.

**Proof**: If

$$A_n = a_k + a_{k+1} + \cdots + a_n \quad \text{and} \quad B_n = b_k + b_{k+1} + \cdots + b_n, \quad n \geq k,$$

then, we have

$$A_n \leq B_n.$$

converge

If $\sum b_n < \infty$, then $\{B_n\}$ is bounded above implies that $\{A_n\}$ is also; therefore, $\sum a_n < \infty$.

diverge

On the other hand, if $\sum a_n = \infty$, then $\{A_n\}$ is unbounded above implies that $\{B_n\}$ is also; therefore, $\sum b_n = \infty$.

**Example**: Discuss the convergence of the series $\sum \frac{r^n}{n}$.

**Solution**: Since

$$\frac{r^n}{n} < r^n, \quad n \geq 1,$$

and $\sum r^n < \infty$ if $0 < r < 1$, the series $\sum r^n/n$ converges if $0 < r < 1$, by the comparison test.

Comparing these two series is **inconclusive** if $r >$ 1, since it does not help to know that the terms of $\sum r^n/n$ are smaller than those of the divergent series $\sum r^n$.

if r is less than 0 and 1 then it converges

If $r < 0$, the comparison test does not apply, since the series then have infinitely many negative terms. Since

$$r^n < nr^n$$

and $\sum r^n = \infty$ if $r \geq 1$, the comparison test implies that $\sum nr^n = \infty$ if $r \geq 1$.

Comparing these two series is inconclusive if $0 < r < 1$, since it does not help to know that the terms of $\sum nr^n$ are larger than those of the convergent series $\sum r^n$.

**Recall the following**: If $a_n \geq 0$ for $n \geq k$, then $\sum a_n$ converges if its partial sums are bounded, or diverges to $\infty$ if they are not. These are the only possibilities and, in either case,

$$\sum_{n=k}^{\infty} a_n = \sup\{A_n : n \geq k\},$$

where

$$A_n = a_k + a_{k+1} + \cdots + a_n, \quad n \geq k.$$

**Theorem**: Suppose $a_1 \geq a_2 \geq \ldots \geq 0$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

**Proof**: Consider the sequence of partial sum

$$s_n = a_1 + a_2 + \ldots + a_n, \qquad t_k = a_1 + 2a_2 + \ldots + 2^k a_{2^k}.$$

For $n < 2^k$, we have

$$\begin{aligned} s_n &\leq a_1 + (a_2 + a_3) + \ldots + (a_{2^k} + \ldots + a_{2^{k+1}-1}) \\ &\leq a_1 + 2a_2 + \ldots + 2^k a_{2^k} = t_k \end{aligned}$$

so that

$$s_n \leq t_k, \qquad (*).$$

On the other hand, if $n > 2^k$, we have

$$\begin{aligned} s_n &\geq a_1 + a_2 + (a_3 + a_4) + \ldots + (a_{2^{k-1}+1} + \ldots + a_{2^k}) \\ &\geq \frac{1}{2}a_1 + a_2 + 2a_4 + \ldots + 2^{k-1}a_{2^k} = \frac{1}{2}t_k \end{aligned}$$

so that

$$2s_n \geq t_k \qquad (**).$$

By (*) and (**) both sequences $\{s_n\}$ and $\{t_k\}$ are either both bounded or both unbounded.

**Recall the following theorem**:

**Theorem**: If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Mcqs

**Theorem**: Suppose $a_1 \geq a_2 \geq \ldots \geq 0$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

**Theorem**: The series $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

**Proof**: If $p = 0$, then $\sum \frac{1}{n^p}$ is divergent. If $p < 0$, then once again $\sum \frac{1}{n^p}$ is divergent. (Why) If $p > 0$ then the series $\sum \frac{1}{n^p}$ and the series

$$\sum_{k=0}^{\infty} 2^k \frac{1}{2^{kp}} = \sum_{k=0}^{\infty} 2^{(1-p)k}.$$

Notice that $2^{1-p} < 1$ if and only if $1 - p < 0$. Take $x = 2^{1-p}$, then we have $|x| < 1$ and the series $\sum_{k=0}^{\infty} 2^{(1-p)k}$ becomes

$$\sum_{k=0}^{\infty} x^k.$$

The monotonicity of the logarithmic function implies $\{\log n\}$ increases. Hence $\{1/n \log n\}$ decreases.

**Theorem**: If $p > 1$,

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p},$$

converges, if $p \leq 1$, the series diverges.

**Proof**: The monotonicity of the logarithmic function implies $\{\log n\}$ increases. Hence $\{1/n \log n\}$ decreases.

The series

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p},$$

and

$$\sum_{k=1}^{\infty} \frac{1}{2^k (k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}.$$

**Remark**: Consider the series

$$\sum_{n=3}^{\infty} \frac{1}{n \log n \log\log n},$$

greater than 1 => converge
less than or equal to => diverge

diverges whereas

$$\sum_{n=3}^{\infty} \frac{1}{n \log n (\log\log n)^2},$$

converges.

**Example**: Show that the series

$$\sum^{\infty} \frac{1}{(n^2 + n)^q}$$

converges if $q > 1/2$.

**Solution**: By comparison with the convergent series $\sum 1/n^{2q}$, since

$$\frac{1}{(n^2 + n)^q} < \frac{1}{n^{2q}}, \quad n \geq 1.$$

This comparison is inconclusive if $q \leq 1/2$, since then

$$\sum \frac{1}{n^{2q}} = \infty,$$

and it does not help to know that the terms of $\sum^{\infty} \frac{1}{(n^2+n)^q}$ are smaller than those of a divergent series. However, we can use the comparison test here, after a little trick.

Observe that

$$\sum_{n=k-1}^{\infty} \frac{1}{(n+1)^{2q}} = \sum_{n=k}^{\infty} \frac{1}{n^{2q}} = \infty, \quad q \leq 1/2,$$

and

$$\frac{1}{(n+1)^{2q}} < \frac{1}{(n^2+n)^q}.$$

Therefore, the comparison test implies that

$$\sum \frac{1}{(n^2+n)^q} = \infty, \quad q \leq 1/2.$$

## 2.12 The Number $e$

The number $e$ is defined by infinite series and is given

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

where $n! = 1.2.3...n$ if $n \geq 1$ and $0! = 1$.

We have

$$\begin{aligned} s_n &= 1 + 1 + \frac{1}{1.2} + \frac{1}{1.2.3} + ... + \frac{1}{1.2.3...n} \\ &< 1 + 1 + \frac{1}{2} + ... + \frac{1}{2^{n-1}} < 3, \end{aligned}$$

the series converges, and the definition makes sense.

**Approximation of the number $e$:** Let

$$s_n = 1 + 1 + \frac{1}{1.2} + \frac{1}{1.2.3} + ... + \frac{1}{1.2.3...n},$$

then we have

$$\begin{aligned} e - s_n &= \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + ... \\ &< \frac{1}{(n+1)!}\left(1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + ...\right) = \frac{1}{n!n}. \end{aligned}$$

We can conclude that

$$0 < e - s_n < \frac{1}{n!n}.$$

**Approximation of the number** $e$: We can conclude that

$$0 < e - s_n < \frac{1}{n!n}.$$

For example , take $n = 10$, thus $s_{10}$ approximate $e$ with an error less than $10^{-7}$.

**Theorem**: $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e.$

**Proof**: Let

$$s_n = \sum_{k=0}^{n}\frac{1}{k!}, \qquad t_n = \left(1+\frac{1}{n}\right)^n.$$

We have by using binomial theorem

$$\begin{aligned} t_n &= 1+1+\frac{1}{2!}\left(1-\frac{1}{n}\right)+\frac{1}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)+\ldots \\ &\quad +\frac{1}{n!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{n-1}{n}\right). \end{aligned}$$

the series converges, and the definition makes sense.

Hence $t_n \le s_n$, so that

$$\lim_{n\to\infty}\sup t_n \le e \tag{2.11}$$

Next if $n \ge m$,

$$t_n \ge 1+1+\frac{1}{2!}\left(1-\frac{1}{n}\right)+\ldots+\frac{1}{m!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{m-1}{n}\right).$$

Let $n \to \infty$, keeping $m$ fixed. We get

$$\lim_{n\to\infty}\inf t_n \ge 1+1+\frac{1}{2!}+\ldots+\frac{1}{m!},$$

so that

$$s_m \le \lim_{n} \to \infty \inf t_n.$$

Letting $m \to \infty$, we finally get

$$e \le \lim_{n\to\infty}\inf t_n \qquad (**).$$

**Theorem**: The number $e$ is an irrational number.

**Proof**: Before we start the proof recall the following identity

$$0 < e - s_n < \frac{1}{n!n}.$$

Suppose on contrary that $e$ is a rational number. Then $e = p/q$, where $p$ and $q$ are positive integers.

Take $n = q$ then the above identity becomes

$$0 < e - s_q < \frac{1}{q!q} \quad \Rightarrow \quad 0 < q!(e - s_q) < \frac{1}{q}.$$

The number $q!e$ will be an integer (Why?)

Since

$$q!s_q = q!\left(1 + 1 + \frac{1}{2!} + ... + \frac{1}{q!}\right)$$

is an integer, we see that $q!(e - s_q)$ is an integer. Since $q \geq 1$, the identity $0 < q!(e - s_q) < \frac{1}{q}$ implies an integer between 0 and 1.

**Theorem**: Suppose that $a_n \geq 0$ and $b_n > 0$ for $n \geq k$. Then

- $\sum a_n < \infty$ if $\sum b_n < \infty$ and $\limsup_{n\to\infty} a_n/b_n < \infty$.
- $\sum a_n = \infty$ if $\sum b_n = \infty$, and $\liminf_{n\to\infty} a_n/b_n > 0$.

**Proof**: If $\limsup_{n\to\infty} a_n/b_n < \infty$, then $\{a_n/b_n\}$ is bounded, so there is a constant $M$ and an integer $k$ such that

$$a_n \leq Mb_n, \quad n \geq k.$$

Since $\sum b_n < \infty$, implies that $\sum(Mb_n) < \infty$. Now $\sum a_n < \infty$, by the comparison test.

If $\liminf_{n\to\infty} a_n/b_n > 0$, there is a constant $m$ and an integer $k$ such that

$$a_n \geq mb_n, \quad n \geq k.$$

Since $\sum b_n = \infty$, implies that $\sum(mb_n) = \infty$. Now $\sum a_n = \infty$, by the comparison test.

**Theorem**: Suppose that $a_n \geq 0$ and $b_n > 0$ for $n \geq k$. Then

- $\sum a_n < \infty$ if $\sum b_n < \infty$ and $\limsup_{n\to\infty} a_n/b_n < \infty$.
- $\sum a_n = \infty$ if $\sum b_n = \infty$, and $\liminf_{n\to\infty} a_n/b_n > 0$.

**Example**: Let

$$\sum b_n = \sum \frac{1}{n^{p+q}} \quad \text{and} \quad \sum a_n = \sum \frac{2 + \sin n\pi/6}{(n+1)^p(n-1)^q}.$$

Then

$$\frac{a_n}{b_n} = \frac{2 + \sin n\pi/6}{(1 + 1/n)^p(1 - 1/n)^q},$$

so

$$\limsup_{n\to\infty} \frac{a_n}{b_n} = 3 \quad \text{and} \quad \liminf_{n\to\infty} \frac{a_n}{b_n} = 1.$$

Since $\sum b_n < \infty$ if and only if $p+q > 1$, the same is true of $\sum a_n$, by the previous Theorem.

**Corollary**: Suppose that $a_n \geq 0$ and $b_n > 0$ for $n \geq k$, and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L,$$

where $0 < L < \infty$.

lim sup a_n/ b_n = lim inf a_n/ b_n= L
so sum a_n and b_ n are converge and diverge

**Example**: Let

$$\sum b_n = \sum \frac{1}{n^{p+q}}$$

and

$$\sum a_n = \sum \frac{2 + \sin n\pi/6}{(n+1)^p (n-1)^q}.$$

**Example**: Recall the series $\lim_{n\to\infty} \frac{1}{(n^2+n)^q}$, we have proved that the series is convergent for $q > 1/2$ by using a trick and applying comparison test.

$$\lim_{n\to\infty} \frac{1}{(n^2+n)^q} \Big/ \frac{1}{n^{2q}} = \lim_{n\to\infty} \frac{1}{(1+1/n)^q} = 1,$$

so

$$\sum \frac{1}{(n^2+n)^q} \quad \text{and} \quad \sum \frac{1}{n^{2q}}$$

converge or diverge together.

**Recall the following**

**Theorem**: Suppose that $a_n \geq 0$ and $b_n > 0$ for $n \geq k$. Then

- $\sum a_n < \infty$ if $\sum b_n < \infty$ and $\limsup_{n\to\infty} a_n/b_n < \infty$.
- $\sum a_n = \infty$ if $\sum b_n = \infty$, and $\liminf_{n\to\infty} a_n/b_n > 0$.

## 2.13 The Ratio Test

**Theorem**: Suppose that $a_n > 0$, $b_n > 0$, and

$$\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}.$$

Then

- $\sum a_n < \infty$ if $\sum b_n < \infty$.

- $\sum b_n = \infty$ if $\sum a_n = \infty$.

**Proof**: We can rewrite

$$\frac{a_{n+1}}{b_{n+1}} \leq \frac{a_n}{b_n},$$

we see that $\{a_n/b_n\}$ is nonincreasing. Therefore, $\limsup_{n\to\infty} a_n/b_n < \infty$, and by applying the Theorem we have proved the first part.

**Theorem**: Suppose that $a_n > 0$, $b_n > 0$, and

$$\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}.$$

Then

- $\sum b_n = \infty$ if $\sum a_n = \infty$.

**Proof**: Suppose that $\sum a_n = \infty$. Since $\{a_n/b_n\}$ is nonincreasing, there is a number $\rho$ such that $b_n \geq \rho a_n$ for large $n$.

Since $\sum(\rho a_n) = \infty$ if $\sum a_n = \infty$, (with $a_n$ replaced by $\rho a_n$) implies that $\sum b_n = \infty$.

**Example**: If $\sum a_n = \sum \left(2 + \sin \frac{n\pi}{2}\right) r^n$. then

$$\frac{a_{n+1}}{a_n} = r\frac{2 + \sin \frac{(n+1)\pi}{2}}{2 + \sin \frac{n\pi}{2}}$$

which assumes the values $3r/2$, $2r/3$, $r/2$, and $2r$, each infinitely many times.

Hence,

$$\limsup_{n\to\infty} \frac{a_{n+1}}{a_n} = 2r \qquad \text{and} \qquad \liminf_{n\to\infty} \frac{a_{n+1}}{a_n} = \frac{r}{2}.$$

Therefore, $\sum a_n$ converges if $0 < r < 1/2$ and diverges if $r > 2$. The ratio test is inconclusive if $1/2 \leq r \leq 2$.

**Corollary**: Suppose that $a_n > 0$ $(n \geq k)$ and

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = L.$$

Then

- $\sum a_n < \infty$ if $L < 1$.
- $\sum a_n = \infty$ if $L > 1$.

The test is inconclusive if $L = 1$.

**Example**: Decide about the series $\sum a_n = \sum nr^{n-1}$.

**Solution**: Since

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)r^n}{nr^{n-1}} = \left(1 + \frac{1}{n}\right) r,$$

so

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = r.$$

The series converges if $0 < r < 1$ and diverges if $r > 1$.

By using corollary the ratio test is inconclusive if $r = 1$. But recall that for a convergent series the nth term must approach towards zero. Hence the series diverges.

**Remark**: The ratio test does not imply that $\sum a_n < \infty$ if merely

$$\frac{a_{n+1}}{a_n} < 1, \tag{2.12}$$

for large $n$, since this could occur with $\lim_{n\to\infty} a_{n+1}/a_n = 1$, in which case the test is inconclusive.

However, the next theorem shows that $\sum a_n < \infty$ if (2.12) is replaced by the stronger condition that

$$\frac{a_{n+1}}{a_n} \le 1 - \frac{p}{n}$$

for some $p > 1$ and large $n$. It also shows that $\sum a_n = \infty$ if

$$\frac{a_{n+1}}{a_n} \ge 1 - \frac{q}{n}$$

for some $q < 1$ and large $n$.

**Theorem (Raabe's test)**: Suppose that $a_n > 0$ for large $n$. Let

$$M = \limsup_{n\to\infty} n\left(\frac{a_{n+1}}{a_n} - 1\right) \quad \text{and} \quad m = \liminf_{n\to\infty} n\left(\frac{a_{n+1}}{a_n} - 1\right).$$

Then

- $\sum a_n < \infty$ if $M < -1$.
- $\sum a_n = \infty$ if $m > -1$.

The test is inconclusive if $m \le -1 \le M$.

**Example**: If

$$\sum a_n = \sum \frac{n!}{\alpha(\alpha+1)(\alpha+2)\cdots(\alpha+n-1)}, \quad \alpha > 0,$$

then

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{n+1}{\alpha+n} = 1,$$

so the ratio test is inconclusive.

However,

$$\lim_{n\to\infty} n\left(\frac{a_{n+1}}{a_n}-1\right) = \lim_{n\to\infty} n\left(\frac{n+1}{\alpha+n}-1\right)$$
$$= \lim_{n\to\infty} \frac{n(1-\alpha)}{\alpha+n} = 1-\alpha,$$

so Raabe's test implies that $\sum a_n < \infty$ if $\alpha > 2$ and $\sum a_n = \infty$ if $0 < \alpha < 2$.

## 2.14 Cauchy's Root Test

Cauchy's integral test for determining the convergence of infinite series

**Theorem**: If $a_n \geq 0$ for $n \geq k$, then

- $\sum a_n < \infty$ if $\limsup_{n\to\infty} a_n^{1/n} < 1$.
- $\sum a_n = \infty$ if $\limsup_{n\to\infty} a_n^{1/n} > 1$.

The test is inconclusive if $\limsup_{n\to\infty} a_n^{1/n} = 1$.

**Proof**: If $\limsup_{n\to\infty} a_n^{1/n} < 1$, there is an $r$ such that $0 < r < 1$ and $a_n^{1/n} < r$ for large $n$. Therefore, $a_n < r^n$ for large $n$. Since $\sum r^n < \infty$, the comparison test implies that $\sum a_n < \infty$.

If $\limsup_{n\to\infty} a_n^{1/n} > 1$, then $a_n^{1/n} > 1$ for infinitely many values of $n$, so $\sum a_n = \infty$.

**Example**: Cauchy's root test is inconclusive if

$$\sum a_n = \sum \frac{1}{n^p},$$

because then

$$\limsup_{n\to\infty} a_n^{1/n} = \lim_{n\to\infty}\left(\frac{1}{n^p}\right)^{1/n} = \lim_{n\to\infty} \exp\left(-\frac{p}{n}\log n\right) = 1$$

for all $p$.

However, we know that $\sum 1/n^p < \infty$ if $p > 1$ and $\sum 1/n^p = \infty$ if $p \leq 1$.

**Example**: If

$$\sum a_n = \sum \left(2+\sin\frac{n\pi}{4}\right)^n r^n,$$

then

$$\limsup_{n\to\infty} a_n^{1/n} = \limsup_{n\to\infty}\left(2+\sin\frac{n\pi}{4}\right) r = 3r,$$

and so $\sum a_n < \infty$ if $r < 1/3$ and $\sum a_n = \infty$ if $r > 1/3$. The test is inconclusive if $r = 1/3$, but then $|a_{8m+2}| = 1$ for $m \geq 0$, so $\sum a_n = \infty$.

The series ∑an ∑ a n is absolutely convergent if ρ<1 ρ < 1 and is divergent if ρ>1 ρ > 1 . If ρ=1 ρ = 1 , then the test is inconclusive.

Absolute convergence" means a series will converge even when you take the absolute value of each term,



## 2.15 Absolute Convergence

## 2.16 Absolute Convergence

A series $\sum a_n$ *converges absolutely*, or is *absolutely convergent*, if $\sum |a_n| < \infty$.

**Example**: A convergent series $\sum a_n$ of nonnegative terms is absolutely convergent, since $\sum a_n$ and $\sum |a_n|$ are the same.

More generally, any convergent series whose terms are of the same sign for sufficiently large $n$ converges absolutely.

**Example**: Consider the series

$$\sum \frac{\sin n\theta}{n^p},$$

where $\theta$ is arbitrary and $p > 1$.

**Solution**: Since

$$\left|\frac{\sin n\theta}{n^p}\right| \le \frac{1}{n^p}$$

and $\sum 1/n^p < \infty$ if $p > 1$, the comparison test implies that

$$\sum \left|\frac{\sin n\theta}{n^p}\right| < \infty, \quad p > 1.$$

Therefore, the given series converges absolutely if $p > 1$.

**Example**: If $0 < p < 1$, then the series

$$\sum \frac{(-1)^n}{n^p}$$

does not converge absolutely, since

$$\sum \left|\frac{(-1)^n}{n^p}\right| = \sum \frac{1}{n^p} = \infty.$$

However, the series converges, by the alternating series test, which we prove below.

**Remark**: Any test for convergence of a series with nonnegative terms can be used to test an arbitrary series $\sum a_n$ for absolute convergence by applying it to $\sum |a_n|$.

**Example**: Determine the series

$$\sum a_n = \sum (-1)^n \frac{n!}{\alpha(\alpha+1)\cdots(\alpha+n-1)}, \quad \alpha > 0,$$

for absolutely convergent or not?

We apply Raabe's test to

$$\sum a_n = \sum \frac{n!}{\alpha(\alpha+1)\cdots(\alpha+n-1)}.$$

From previous example, $\sum |a_n| < \infty$ if $\alpha > 2$ and $\sum |a_n| = \infty$ if $\alpha < 2$. Therefore, $\sum a_n$ converges absolutely if $\alpha > 2$, but not if $\alpha < 2$.

Notice that this does not imply that $\sum a_n$ diverges if $\alpha < 2$.

**Theorem**: If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

**Proof**: See Lecture.

**Example**: For example, the Theorem implies that

$$\sum \frac{\sin n\theta}{n^p}$$

converges if $p > 1$, since it then converges absolutely.

**What about the converse of the theorem**?

**Conditional convergence** test for determining if an infinite series converges to some finite value

## 2.17 Dirichlet's Test for Series

**Theorem**: The series $\sum_{n=k}^{\infty} a_n b_n$ converges if $\lim_{n\to\infty} a_n = 0$,

$$\sum |a_{n+1} - a_n| < \infty,$$

and

$$|b_k + b_{k+1} + \cdots + b_n| \le M, \quad n \ge k,$$

for some constant $M$.

**Proof**: Define $B_n = b_k + b_{k+1} + \cdots + b_n, \quad n \ge k$ and consider the partial sums of $\sum_{n=k}^{\infty} a_n b_n$:

$$S_n = a_k b_k + a_{k+1} b_{k+1} + \cdots + a_n b_n, \quad n \ge k. \tag{2.13}$$

By substituting

$$b_k = B_k \quad \text{and} \quad b_n = B_n - B_{n-1}, \quad n \ge k+1,$$

into (2.13), we obtain

$$S_n = a_k B_k + a_{k+1}(B_{k+1} - B_k) + \cdots + a_n(B_n - B_{n-1}).$$

Rewriting as

$$\begin{aligned} S_n &= (a_k - a_{k+1})B_k + (a_{k+1} - a_{k+2})B_{k+1} + \cdots \\ &\quad + (a_{n-1} - a_n)B_{n-1} + a_n B_n. \end{aligned} \tag{2.14}$$

(The procedure that led from (2.13) to (2.14) is called *summation by parts*. It is analogous to integration by parts.)

Now (2.14) can be viewed as

$$S_n = T_{n-1} + a_n B_n, \tag{2.15}$$

where

$$T_{n-1} = (a_k - a_{k+1})B_k + (a_{k+1} - a_{k+2})B_{k+1} + \cdots + (a_{n-1} - a_n)B_{n-1};$$

that is, $\{T_n\}$ is the sequence of partial sums of the series

$$\sum_{j=k}^{\infty} (a_j - a_{j+1})B_j. \tag{2.16}$$

Since

$$|(a_j - a_{j+1})B_j| \le M|a_j - a_{j+1}|$$

from given conditions and the comparison test imply that the series (2.16) converges absolutely. Absolutely convergence implies that $\{T_n\}$ converges.

Let $T = \lim_{n\to\infty} T_n$. Since $\{B_n\}$ is bounded and $\lim_{n\to\infty} a_n = 0$, we infer from (2.15) that

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} T_{n-1} + \lim_{n\to\infty} a_n B_n = T + 0 = T.$$

Therefore, $\sum a_n b_n$ converges.

**Example**: Apply Dirichlet's test to the following series

$$\sum_{n=2}^{\infty} \frac{\sin n\theta}{n + (-1)^n}, \quad \theta \neq k\pi \quad (k = \text{integer}),$$

**Solution**: We take

$$a_n = \frac{1}{n + (-1)^n} \quad \text{and} \quad b_n = \sin n\theta.$$

Then $\lim_{n\to\infty} a_n = 0$, and

$$|a_{n+1} - a_n| < \frac{3}{n(n-1)}$$

(verify), so

$$\sum |a_{n+1} - a_n| < \infty.$$

Now

$$B_n = \sin 2\theta + \sin 3\theta + \cdots + \sin n\theta.$$

We can write

$$\begin{aligned} B_n &= \frac{(\cos\frac{3}{2}\theta - \cos\frac{5}{2}\theta) + + \cdots + \left(\cos\left(n-\frac{1}{2}\right)\theta - \cos(n+\frac{1}{2})\theta\right)}{2\sin(\theta/2)} \\ &= \frac{\cos\frac{3}{2}\theta - \cos(n+\frac{1}{2})\theta}{2\sin(\theta/2)}, \end{aligned}$$

which implies that $|B_n| \le \left|\frac{1}{\sin(\theta/2)}\right|, \quad n \ge 2.$

Since $\{a_n\}$ and $\{b_n\}$ satisfy the hypotheses of Dirichlet's theorem, $\sum a_n b_n$ converges.

**Remark**: Dirichlet's test takes a simpler form if $\{a_n\}$ is nonincreasing, as follows.

**Corollary**: The series $\sum a_n b_n$ converges if $a_{n+1} \le a_n$ for $n \ge k$, $\lim_{n\to\infty} a_n = 0$, and

$$|b_k + b_{k+1} + \cdots + b_n| \le M, \quad n \ge k,$$

for some constant $M$.

**Proof**: Recall the Dirichlet's test for series, we need to show that

$$\sum_{n=k}^{\infty} |a_{n+1} - a_n| = a_k < \infty.$$

If $a_{n+1} \le a_n$, then $\sum_{n=k}^{m} |a_{n+1} - a_n| = \sum_{n=k}^{m} (a_n - a_{n+1}) = a_k - a_{m+1}$.
Since $\lim_{m\to\infty} a_{m+1} = 0$, it follows that

$$\sum_{n=k}^{\infty} |a_{n+1} - a_n| = a_k < \infty.$$

Therefore, the hypotheses of Dirichlet's test are satisfied, so $\sum a_n b_n$ converges.

**Example**: Discuss the convergence of the series

$$\sum \frac{\sin n\theta}{n^p}.$$

**Solution**: The given series is convergent if $p > 1$.

If we take $a_n = 1/n^p$ and $b_n = \sin n\theta$, then we have

$$a_{n+1} \le a_n, \qquad \lim_{n\to\infty} = 0.$$

and we have proved in previous example that

$$|b_k + b_{k+1} + \cdots + b_n| \le M, \quad n \ge k,$$

Consequently, the given series also converges if $0 < p \le 1$ and this follows from Abel's test.

series where the terms alternate in sign, meaning that each term has a opposite sign to the preceding term.
General Form:
a_1 - a_2 + a_3 - a_4 + ...a_n



## 2.18 Alternating Series

Series whose terms alternate between positive and negative, called alternating series.

if we take the value of n or k is even then we give the answer is positive if we take the value of n or k is odd then we give the answer is negative

$$\sum_{k=1}^{\infty}(-1)^{k+1}\frac{1}{k}.$$

$$\sum_{k=1}^{\infty}(-1)^{k}\frac{1}{k}.$$

$$\sum_{k=1}^{\infty}(-1)^{k}a_k, \qquad \sum_{k=1}^{\infty}(-1)^{k+1}a_k.$$

**Corollary (Alternating series test)**: The series $\sum(-1)^n a_n$ converges if $0 \leq a_{n+1} \leq a_n$ and $\lim_{n\to\infty} a_n = 0$.

**Proof**: Let $b_n = (-1)^n$; recall that

$$B_n = b_k + b_{k+1} + \cdots + b_n, \quad n \geq k.$$

Then $\{|B_n|\}$ is a sequence of zeros and ones and therefore bounded. The conclusion now follows from Abel's test.

**Example**: Show that the series is convergent

$$\sum_{k=1}^{\infty}(-1)^{k+1}\frac{1}{k}.$$

**Example**: Show that the series is convergent

$$\sum_{k=1}^{\infty}(-1)^{k+1}\frac{k+3}{k(k+1)}.$$

## 2.19 Grouping Terms in a Series

- The terms of a finite sum can be grouped by putting parenthesis arbitrarily.
- For example,

$$(1+7)+(6+5)+4 =$$
$$(1+7+6)+(5+4).$$

agr hmy finite series di ho to hm us ki grouping kr k kia is pa convergence or divergence ki property apply kr skty h
answer:
yes we do agr series convergent ho to uski grouping krna k bd result bhi convergent hi ay ga

- Is the same true for infinite series? yes

**Theorem**: Suppose that $\sum_{n=k}^{\infty} a_n = A$, where $-\infty \leq A \leq \infty$. Let $\{n_j\}_1^{\infty}$ be an increasing sequence of integers, with $n_1 \geq k$.

Define

$$\begin{aligned} b_1 &= a_k + \cdots + a_{n_1}, \\ b_2 &= a_{n_1+1} + \cdots + a_{n_2}, \\ &\vdots \\ b_r &= a_{n_{r-1}+1} + \cdots + a_{n_r}. \end{aligned}$$

Then

$$\sum_{j=1}^{\infty} b_{n_j} = A.$$

**Proof**: If $T_r$ is the $r$th partial sum of $\sum_{j=1}^{\infty} b_{n_j}$ and $\{A_n\}$ is the $n$th partial sum of $\sum_{s=k}^{\infty} a_s$.

Then

$$\begin{aligned} T_r &= b_1 + b_2 + \cdots + b_r \\ &= (a_1 + \cdots + a_{n_1}) + (a_{n_1+1} + \cdots + a_{n_2}) + \cdots + \\ &\quad (a_{n_{r-1}+1} + \cdots + a_{n_r}) \\ &= A_{n_r}. \end{aligned}$$

Thus, $\{T_r\}$ is a subsequence of $\{A_n\}$, so

$$\lim_{r\to\infty} T_r = \lim_{n\to\infty} A_n = A.$$

**Example**: If $\sum_{n=0}^{\infty}(-1)^n a_n$ satisfies the hypotheses of the alternating series test and converges to the sum $S$

Then the theorem of grouping terms in a series enables us to write

$$S = \sum_{n=0}^{k}(-1)^n a_n + (-1)^{k+1}\sum_{j=1}^{\infty}(a_{k+2j-1} - a_{k+2j})$$

and

$$S = \sum_{n=0}^{k}(-1)^n a_n + (-1)^{k+1}\left[a_{k+1} - \sum_{j=1}^{\infty}(a_{k+2j} - a_{k+2j-1})\right].$$

Since $0 \leq a_{n+1} \leq a_n$, these two equations imply that $S - S_k$ is between 0 and $(-1)^{k-1}a_{k+1}$.

- Be careful while introducing parenthesis to a divergent series
- Apply carefully the previous theorem

**Example**: For example, it is tempting to write

$$\sum_{n=1}^{\infty}(-1)^{n+1} = (1-1)+(1-1)+\cdots = 0+0+\cdots$$

and conclude that $\sum_{n=1}^{\infty}(-1)^n = 0$.

But equally tempting to write

$$\begin{aligned}\sum_{n=1}^{\infty}(-1)^{n+1} &= 1-(1-1)-(1-1)-\cdots \\ &= 1-0-0-\cdots\end{aligned}$$

and conclude that $\sum_{n=1}^{\infty}(-1)^{n+1} = 1$.

Is there a contradiction with the theorem? Of course, there is no contradiction here, since Theorem does not apply to this series, and neither of these operations is legitimate.

## 2.20 Rearrangements of Series

**Theorem**: If $\sum_{n=1}^{\infty} b_n$ is a rearrangement of an absolutely convergent series $\sum_{n=1}^{\infty} a_n$, then $\sum_{n=1}^{\infty} b_n$ also converges absolutely, and to the same sum.

**Proof**: Let

$$\overline{A}_n = |a_1|+|a_2|+\cdots+|a_n| \quad \text{and} \quad \overline{B}_n = |b_1|+|b_2|+\cdots+|b_n|.$$

For each $n \geq 1$, there is an integer $k_n$ such that $b_1, b_2, \ldots, b_n$ are included among $a_1, a_2, \ldots, a_{k_n}$, so $\overline{B}_n \leq \overline{A}_{k_n}$. Since $\{\overline{A}_n\}$ is bounded, so is $\{\overline{B}_n\}$, and therefore $\sum |b_n| < \infty$.

Now let

$$\begin{aligned} A_n &= a_1+a_2+\cdots+a_n, \quad B_n = b_1+b_2+\cdots+b_n, \\ A &= \sum_{n=1}^{\infty} a_n, \quad \text{and} \quad B = \sum_{n=1}^{\infty} b_n. \end{aligned}$$

We must show that $A = B$. Suppose that $\varepsilon > 0$. From Cauchy's convergence criterion for series and the absolute convergence of $\sum a_n$, there is an integer $N$ such that

$$|a_{N+1}|+|a_{N+2}|+\cdots+|a_{N+k}| < \varepsilon, \quad k \geq 1.$$

Choose $N_1$ so that $a_1, a_2, \ldots, a_N$ are included among $b_1, b_2, \ldots, b_{N_1}$. If $n \geq N_1$, then $A_n$ and $B_n$ both include the terms $a_1, a_2, \ldots, a_N$, which cancel on subtraction; thus, $|A_n - B_n|$ is dominated by the sum of the absolute values of finitely many terms from $\sum a_n$ with subscripts greater than $N$.

Since every such sum is less than $\varepsilon$,

$$|A_n - B_n| < \varepsilon \quad \text{if} \quad n \geq N_1.$$

Therefore, $\lim_{n\to\infty}(A_n - B_n) = 0$ and $A = B$.

## 2.21 Addition and Multiplication of Series

**Theorem** (Addition of series): If $\sum a_n = A$ and $\sum b_n = B$; then their sum $\sum(a_n + b_n) = A + B$ and $\sum ca_n = cA$, for some fixed $c$.

**Proof**: Consider the sequence of partial sums

**Product of infinite series**: Given two series

$$\sum_{n=0}^{\infty} a_n \quad \text{and} \quad \sum_{n=0}^{\infty} b_n.$$

We can arrange all possible products $a_i b_j$ $(i, j \geq 0)$ in a two-dimensional array:

$$\begin{array}{cccc} a_0b_0 & a_0b_1 & a_0b_2 & a_0b_3 & \cdots \\ a_1b_0 & a_1b_1 & a_1b_2 & a_1b_3 & \cdots \\ a_2b_0 & a_2b_1 & a_2b_2 & a_2b_3 & \cdots \\ a_3b_0 & a_3b_1 & a_3b_2 & a_3b_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \end{array} \tag{2.17}$$

where the subscript on $a$ is constant in each row and the subscript on $b$ is constant in each column.

Any sensible definition of the product

$$\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right)$$

clearly must involve every product in this array exactly once; thus, we might define the product of the two series to be the series $\sum_{n=0}^{\infty} p_n$, where $\{p_n\}$ is a sequence obtained by ordering the products in (2.17) according to some method that chooses

every product exactly once. One way to do this is indicated by

$$\begin{array}{ccccccccc}
a_0b_0 & \to & a_0b_1 & & a_0b_2 & \to & a_0b_3 & \cdots \\
 & & \downarrow & & \uparrow & & \downarrow & \\
a_1b_0 & \leftarrow & a_1b_1 & & a_1b_2 & & a_1b_3 & \cdots \\
\downarrow & & & & \uparrow & & \downarrow & \\
a_2b_0 & \to & a_2b_1 & \to & a_2b_2 & & a_2b_3 & \cdots \\
 & & & & & & \downarrow & \\
a_3b_0 & \leftarrow & a_3b_1 & \leftarrow & a_3b_2 & \leftarrow & a_3b_3 & \cdots \\
\downarrow & & & & & & & \\
\vdots & & \vdots & & \vdots & & \vdots &
\end{array} \tag{2.18}$$

**Product of infinite series**: Another by

$$\begin{array}{ccccccccc}
a_0b_0 & \to & a_0b_1 & & a_0b_2 & \to & a_0b_3 & & a_0b_4 & \cdots \\
 & \swarrow & & \nearrow & & \swarrow & & \nearrow & & \\
a_1b_0 & & a_1b_1 & & a_1b_2 & & a_1b_3 & & \cdots & \\
\downarrow & \nearrow & & \swarrow & & \nearrow & & & & \\
a_2b_0 & & a_2b_1 & & a_2b_2 & & a_2b_3 & & \cdots & \\
 & \swarrow & & \nearrow & & & & & & \\
a_3b_0 & & a_3b_1 & & a_3b_2 & & a_3b_3 & & \cdots & \\
\downarrow & \nearrow & & & & & & & & \\
a_4b_0 & & \vdots & & \vdots & & \vdots & & &
\end{array} \tag{2.19}$$

There are infinitely many others, and to each corresponds a series that we might consider to be the product of the given series. This raises a question: If

$$\sum_{n=0}^{\infty} a_n = A \quad \text{and} \quad \sum_{n=0}^{\infty} b_n = B.$$

where $A$ and $B$ are finite, does every product series $\sum_{n=0}^{\infty} p_n$ constructed by ordering the products in (2.17) converge to $AB$?

The next theorem tells us when the answer is yes.

**Theorem**: Let $\sum_{n=0}^{\infty} a_n = A$ and $\sum_{n=0}^{\infty} b_n = B$,

where $A$ and $B$ are finite, and at least one term of each series is nonzero. Then $\sum_{n=0}^{\infty} p_n = AB$ for every sequence $\{p_n\}$ obtained by ordering the products if and only if $\sum a_n$ and $\sum b_n$ converge absolutely. Moreover, in this case, $\sum p_n$ converges absolutely.

**Proof**: Before we start the proof let us recall one way of writing the product is

$$\begin{array}{ccccccc} a_0b_0 & \to & a_0b_1 & & a_0b_2 & \to & a_0b_3 & \cdots \\ & & \downarrow & & \uparrow & & \downarrow & \\ a_1b_0 & \leftarrow & a_1b_1 & & a_1b_2 & & a_1b_3 & \cdots \\ \downarrow & & & & \uparrow & & \downarrow & \\ a_2b_0 & \to & a_2b_1 & \to & a_2b_2 & & a_2b_3 & \cdots \\ & & & & & & \downarrow & \\ a_3b_0 & \leftarrow & a_3b_1 & \leftarrow & a_3b_2 & \leftarrow & a_3b_3 & \cdots \\ \downarrow & & & & & & & \\ \vdots & & \vdots & & \vdots & & \vdots & \end{array} \tag{2.20}$$

First, let $\{p_n\}$ be the sequence obtained by arranging the products $\{a_ib_j\}$ according to the scheme indicated just above, and define

$$\begin{aligned} A_n &= a_0 + a_1 + \cdots + a_n, & \overline{A}_n &= |a_0| + |a_1| + \cdots + |a_n|, \\ B_n &= b_0 + b_1 + \cdots + b_n, & \overline{B}_n &= |b_0| + |b_1| + \cdots + |b_n|, \\ P_n &= p_0 + p_1 + \cdots + p_n, & \overline{P}_n &= |p_0| + |p_1| + \cdots + |p_n|. \end{aligned}$$

From arrangement, we see that

$$P_0 = A_0B_0, \quad P_3 = A_1B_1, \quad P_8 = A_2B_2,$$

and, in general,

$$P_{(m+1)^2-1} = A_mB_m. \tag{2.21}$$

Similarly,

$$\overline{P}_{(m+1)^2-1} = \overline{A}_m\overline{B}_m. \tag{2.22}$$

If $\sum |a_n| < \infty$ and $\sum |b_n| < \infty$, then $\{\overline{A}_m\overline{B}_m\}$ is bounded.

Since $\overline{P}_m \le \overline{P}_{(m+1)^2-1}$, (2.22) implies that $\{\overline{P}_m\}$ is bounded.

Therefore, $\sum |p_n| < \infty$, so $\sum p_n$ converges. Now

$$\begin{aligned} \textstyle\sum_{n=0}^{\infty} p_n &= \lim_{n\to\infty} P_n && \text{(by definition)} \\ &= \lim_{m\to\infty} P_{(m+1)^2-1} \\ &= \lim_{m\to\infty} A_mB_m && \text{(from (2.21))} \\ &= (\lim_{m\to\infty} A_m)\,(\lim_{m\to\infty} B_m) \\ &= AB. \end{aligned}$$

Since any other ordering of the products produces a rearrangement of the absolutely convergent series $\sum_{n=0}^{\infty} p_n$.

Theorem about rearrangements of absolutely convergent series implies that $\sum |q_n| < \infty$ for every such ordering and that $\sum_{n=0}^{\infty} q_n = AB$.

This shows that the stated condition is sufficient.

For necessity, again let $\sum_{n=0}^{\infty} p_n$ be obtained from the ordering indicated in (2.20). Suppose that $\sum_{n=0}^{\infty} p_n$ and all its rearrangements converge to $AB$. Then $\sum p_n$ must converge absolutely (Why).

Therefore, $\{\overline{P}_{m^2-1}\}$ is bounded, and (2.22) implies that $\{\overline{A}_m\}$ and $\{\overline{B}_m\}$ are bounded.

(Here we need the assumption that neither $\sum a_n$ nor $\sum b_n$ consists entirely of zeros. Why?) Therefore, $\sum |a_n| < \infty$ and $\sum |b_n| < \infty$.

## 2.22 Power Series

infinite series that can be thought of as a polynomial with an infinite number of terms,

**Maclaurin and Taylor series**: If $f$ has derivatives of all order at $x_0$, then we call the series

e. g binomial series, geometric series etc

$$\sum_{n=0}^{\infty}(x-x_0)^n = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \ldots + \frac{f^n(x_0)}{n!}(x-x_0)^n + \ldots$$

is known as Taylor series for $f$ at the point $x = x_0$.

The special case of Taylor series when $x_0 = 0$, the series is known as Maclaurin series

$$\sum_{n=0}^{\infty} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^n(0)}{k!}x^n + \ldots.$$

**Examples**:

$$e^x = \sum_{n=0}^{\infty}\frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^n}{n!} + \ldots.$$

$$\sin x = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + (-1)^n\frac{x^{2n+1}}{(2n+1)!} + \ldots.$$

$$\cos x = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots + (-1)^n\frac{x^{2n}}{(2n)!} + \ldots.$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \ldots + x^n + \ldots.$$

**Power Series**: An infinite series of the form

$$\sum_{n=0}^{\infty} a_n(x-x_0)^n,$$

where $x_0$ and $a_0, a_1, \ldots,$ are constants, is called a *power series in* $x - x_0$.

If $x_0 = 0$ then power series becomes

$$\sum_{n=0}^{\infty} a_n x^n.$$

**Theorem**: For the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$, define $R$ in the extended real numbers by

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

In particular, $R = 0$, $R = \infty$ or $0 < R < \infty$. Then the power series converges

- If $\limsup_{n\to\infty} |a_n|^{1/n} = \infty$ then $R = 0$ and the power series converges only for $x = x_0$.
- If $\limsup_{n\to\infty} |a_n|^{1/n} = 0$, then $R = \infty$ and the power series converges for all $x$.
- If $\limsup_{n\to\infty} |a_n|^{1/n} \neq 0$ such that $0 < R < \infty$, then power series converges for $x$ in $(x_0 - R, x_0 + R)$.
- The series diverges if $|x - x_0| > R$.
- No general statement can be made concerning convergence at the endpoints $x = x_0+R$ and $x = x_0-R$ : the series may converge absolutely or conditionally at both, converge conditionally at one and diverge at the other, or diverge at both.

**Theorem**: For the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$, define $R$ in the extended real numbers by

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

**Proof**: Put $y = x - x_0$, $c_n = a_n y^n$ and apply the root test

$$\limsup_{n\to\infty} |c_n|^{1/n} = |y| \limsup_{n\to\infty} |a_n|^{1/n} = \frac{|y|}{R}.$$

**Remark**: $R$ is called the radius of convergence of the power series and $(x_0 - R, x_0 + R)$ is known as interval of convergence.

**Example**: For the power series

$$\sum \frac{\sin n\pi/6}{2^n}(x - 1)^n,$$

what is $a_n$, $x_0$ and $R$?

**Solution**: For $R$, we have

$$\begin{aligned}\limsup_{n\to\infty} |a_n|^{1/n} &= \limsup_{n\to\infty} \left(\frac{|\sin n\pi/6}{2^n}\right)^{1/n} \\ &= \frac{1}{2}\limsup_{n\to\infty}(|\sin n\pi/6|)^{1/n} \\ &= \frac{1}{2}(1) = \frac{1}{2}.\end{aligned}$$

Therefore, $R = 2$ and Theorem about the convergence of power series implies that the series converges absolutely uniformly in closed subintervals of $(-1, 3)$ and diverges if $x < -1$ or $x > 3$.

**Remark**: The Theorem about the convergence of the power series does not tell us what happens when $x = -1$ or $x = 3$. we can see that the series diverges in both these cases since its general term does not approach zero.

**Example**: For the power series

$$\sum \frac{x^n}{n},$$

what is $a_n$, $x_0$ and $R$?

**Solution**: For $R$, we have

$$\limsup_{n\to\infty} |a_n|^{1/n} = \limsup_{n\to\infty} \left(\frac{1}{n}\right)^{1/n} = \limsup_{n\to\infty} \exp\left(\frac{1}{n}\log\frac{1}{n}\right) = e^0 = 1.$$

Therefore, $R = 1$ and the series converges absolutely uniformly in closed subintervals of $(-1, 1)$ and diverges if $|x| > 1$.

For $x = -1$ the series becomes $\sum(-1)^n/n$, which converges conditionally, and at $x = 1$ the series becomes $\sum 1/n$, which diverges.

**Theorem**: The radius of convergence of $\sum a_n(x - x_0)^n$ is given by

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$$

if the limit exists in the extended real number system.

**Proof**: It is sufficient to show that if

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$$

exists in the extended real number system, then

$$L = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Let $0 < L < \infty$, if

$$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

holds with $0 < L < \infty$ and $0 < \varepsilon < L$, there is an integer $N$ such that

$$L - \varepsilon < \left| \frac{a_{m+1}}{a_m} \right| < L + \varepsilon \quad \text{if} \quad m \geq N,$$

so

$$|a_m|(L - \varepsilon) < |a_{m+1}| < |a_m|(L + \varepsilon) \quad \text{if} \quad m \geq N.$$

By induction,

$$|a_N|(L - \varepsilon)^{n-N} < |a_n| < |a_N|(L + \varepsilon)^{n-N} \quad \text{if} \quad n > N.$$

Therefore, if

$$K_1 = |a_N|(L - \varepsilon)^{-N} \quad \text{and} \quad K_2 = |a_N|(L + \varepsilon)^{-N},$$

then

$$K_1^{1/n}(L - \varepsilon) < |a_n|^{1/n} < K_2^{1/n}(L + \varepsilon). \tag{2.23}$$

Since $\lim_{n\to\infty} K^{1/n} = 1$ if $K$ is any positive number, (2.23) implies that

$$L - \varepsilon \leq \liminf_{n\to\infty} |a_n|^{1/n} \leq \limsup_{n\to\infty} |a_n|^{1/n} \leq L + \varepsilon.$$

Since $\varepsilon$ is an arbitrary positive number, it follows that

$$\lim_{n\to\infty} |a_n|^{1/n} = L,$$

hence the proof.

**Example**: Determine the radius of convergence for the power series

$$\sum \frac{x^n}{n!}.$$

**Solution**: We have

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{n!}{(n+1)!} = \lim_{n\to\infty} \frac{1}{n+1} = 0.$$

Therefore, $R = \infty$; that is, the series converges for all $x$.

**Example**: Determine the radius and interval of convergence for the power series

$$\sum n! x^n.$$

**Solution**:We have

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(n+1)!}{n!} = \lim_{n\to\infty}(n+1) = \infty.$$

Therefore, $R = 0$, and the series converges only if $x = 0$.

**Example**: Find the interval of convergence for the power series

$$\sum \frac{(-1)^n}{4^n n^p} x^{2n} \quad (p = \text{constant}), \tag{2.24}$$

**Solution**: The given power series has infinitely many zero coefficients (of odd powers of $x$).

However, by setting $y = x^2$, we obtain the series

$$\sum \frac{(-1)^n}{4^n n^p} y^n, \tag{2.25}$$

which has nonzero coefficients for which

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{4^n n^p}{4^{n+1}(n+1)^p} = \frac{1}{4}\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{-p} = \frac{1}{4}.$$

Therefore, (2.25) converges if $|y| < 4$ and diverges if $|y| > 4$.

Setting $y = x^2$, we conclude that (2.24) converges if $|x| < 2$ and diverges if $|x| > 2$.

CHAPTER 3

# Continuity

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**Function**: A rule $f$ that assigns to each member of a nonempty set $D$ a unique member of a set $Y$ is a *function from $D$ to $Y$*. The relationship between a member $x$ of $D$ and the member $y$ of $Y$ that $f$ assigns to $x$ as

$$y = f(x).$$

The set $D$ is the *domain* of $f$, denoted by $D_f$. The members of $Y$ are the possible *values* of $f$.

If $y_0 \in Y$ and there is an $x_0$ in $D$ such that $f(x_0) = y_0$ then we say that $f$ *attains* or *assumes* the value $y_0$. The set of values attained by $f$ is the *range* of $f$.

A *real-valued function of a real variable* is a function whose domain and range are both subsets of the real numbers.

**Examples**: Consider the functions

$$f(x) = x^2, \quad g(x) = \sin x, \quad \text{and} \quad h(x) = e^x.$$

The functions $f$, $g$, and $h$ defined on the extended real number system $(-\infty, \infty)$.

- The range of f is $[0, \infty)$.
- The range of $g$ is $[-1, 1]$.
- The range of $h$ is $(0, \infty)$.

The equation

$$[f(x)]^2 = x$$

does not define a function except on the singleton set $\{0\}$. If $x < 0$, no real number satisfies the $[f(x)]^2 = x$, while if $x > 0$, two real numbers satisfy $[f(x)]^2 = x$. However, the conditions

$$[f(x)]^2 = x \quad \text{and} \quad f(x) \geq 0$$

define a function $f$ on $D_f = [0, \infty)$ with values $f(x) = \sqrt{x}$.

Similarly, the conditions

$$[g(x)]^2 = x \quad \text{and} \quad g(x) \leq 0$$

define a function $g$ on $D_g = [0, \infty)$ with values $g(x) = -\sqrt{x}$. The ranges of $f$ and $g$ are $[0, \infty)$ and $(-\infty, 0]$, respectively.

We first define the *Cartesian product* $X \times Y$ of two nonempty sets $X$ and $Y$ to be the set of all ordered pairs $(x, y)$ such that $x \in X$ and $y \in Y$; thus,

$$X \times Y = \{(x, y) : x \in X, y \in Y\}.$$

If $(x, y)$ and $(x, y_1)$ are in $f$, then $y = y_1$. The set of $x$'s that occur as first members of $f$ is the of $f$.

If $x$ is in the domain of $f$, then the unique $y$ in $Y$ such that $(x, y) \in f$ is the *value of* $f$ *at* $x$, and we write $y = f(x)$.

The set of all such values, a subset of $Y$, is the *range of* $f$.

If $D_f \cap D_g \neq \emptyset$, then $f + g$, $f - g$, and $fg$ are defined on $D_f \cap D_g$ by

$$\begin{aligned} (f+g)(x) &= f(x) + g(x), \\ (f-g)(x) &= f(x) - g(x), \end{aligned}$$

and

$$(fg)(x) = f(x)g(x).$$

The quotient $f/g$ is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

for $x$ in $D_f \cap D_g$ such that $g(x) \neq 0$.

**Example**: Define $(f+g)(x), (f-g)(x), (fg)(x)$ and $(f/g)$, for the functions $f(x) = \sqrt{4 - x^2}$ and $g(x) = \sqrt{x - 1}$.

**Solution**: First of all observe that $D_f = [-2, 2]$ and $D_g = [1, \infty)$.

Then $f + g$, $f - g$, and $fg$ are defined on $D_f \cap D_g = [1, 2]$ by

$$\begin{aligned} (f+g)(x) &= \sqrt{4 - x^2} + \sqrt{x - 1}, \\ (f-g)(x) &= \sqrt{4 - x^2} - \sqrt{x - 1}, \end{aligned}$$

and

$$(fg)(x) = (\sqrt{4 - x^2})(\sqrt{x - 1}) = \sqrt{(4 - x^2)(x - 1)}.$$

The quotient $f/g$ is defined on $(1, 2]$ by

$$\left(\frac{f}{g}\right)(x) = \sqrt{\frac{4 - x^2}{x - 1}}.$$

Although the last expression in (**??**) is also defined for $-\infty < x < -2$, it does not represent $fg$ for such $x$, since $f$ and $g$ are not defined on $(-\infty, -2]$.

## 3.1 Limits

**The tangent line problem**: Given a function $f$ and a point $P(x_0, y_0)$ on its graph, find an equation of the line that is tangent to the graph at $P$.

**The area problem**: Given a function $f$, find the area between the graph of $f$ and an interval $[a, b]$ on the x-axis.

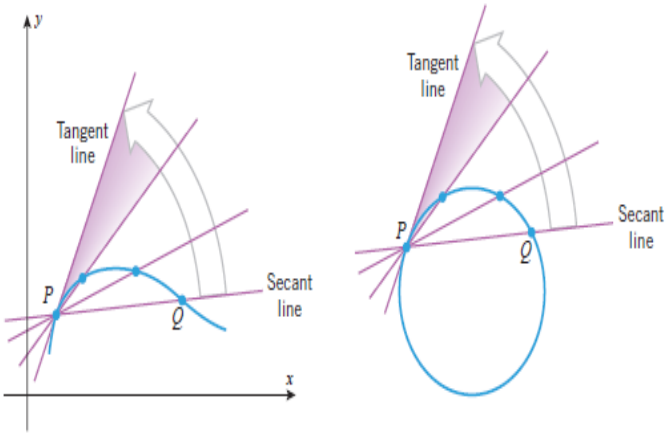



Figure 3.1: Tangent line as a limit of secant line

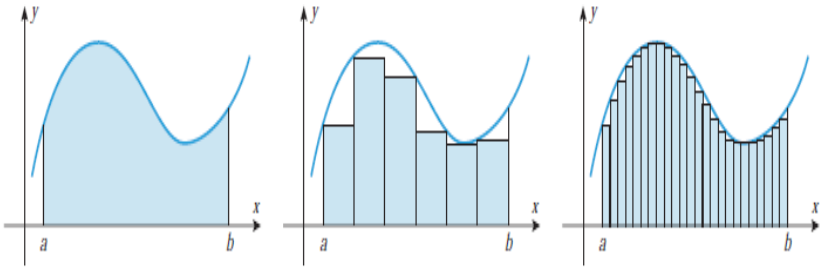



Figure 3.2: Area under the curve as limit

**Example**: Examine the behavior of the function $f(x) = x^2 - x + 1$ for $x$ values closer to 2.

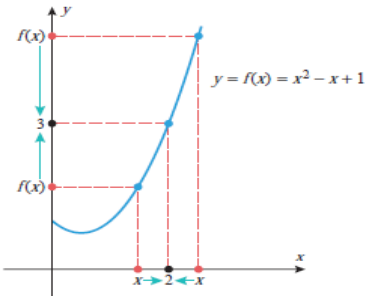



Figure 3.3: Limit $x \to 2$

### 3.1.1 Limits (An informal view)

If the values of $f(x)$ can be made as close as we like to $L$ by taking values of $x$ sufficiently close to a (but not equal to $a$), then we write

$$\lim_{x \to a} f(x) = L.$$

Sometimes it is also written as

$$f(x) \to L, \qquad \text{as} \qquad x \to a.$$

**Example**: Investigate the limit

$$\lim_{x \to 1} \frac{x - 1}{\sqrt{x} - 1}.$$

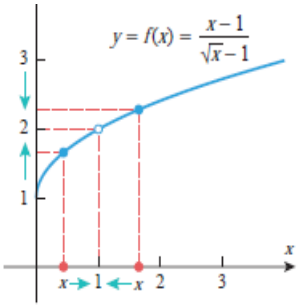



Figure 3.4: Limit $x \to 1$ when $f(x)$ is not defined at $x = 1$

**Example**: Investigate the limit

$$\lim_{x \to 0} \frac{\sin x}{x}.$$

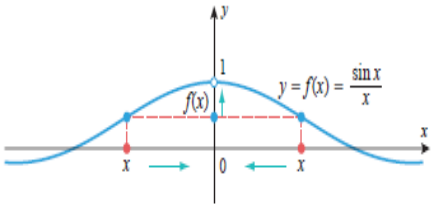



Figure 3.5: Limit $x \to 0$ when $f(x)$ is not defined at $x = 0$

**Example**: Investigate the limit

$$\lim_{x \to 0} \frac{|x|}{x}.$$

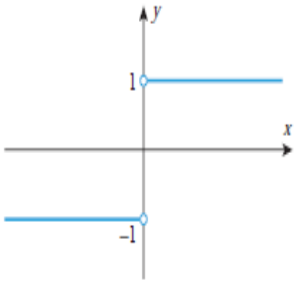



Figure 3.6: Limit $x \to 0$

### 3.1.2 Formal Definition of Limit

We say that $f(x)$ *approaches the limit* $L$ *as* $x$ *approaches* $x_0$, and write

$$\lim_{x \to x_0} f(x) = L,$$

if $f$ is defined on some deleted neighborhood of $x_0$ and, for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon, \qquad \text{whenever} \qquad 0 < |x - x_0| < \delta. \tag{3.1}$$

**Example**: For the function defined by $f(x) = cx$, where $c \in \mathbb{R}$. Show that

$$\lim_{x \to x_0} f(x) = cx_0.$$

**Solution**: We write

$$|f(x) - cx_0| = |cx - cx_0| = |c||x - x_0|.$$

If $c \neq 0$, this yields

$$|f(x) - cx_0| < \varepsilon$$

if

$$|x - x_0| < \delta,$$

where $\delta$ is any number such that $0 < \delta \leq \varepsilon/|c|$.

If $c = 0$, then $f(x) - cx_0 = 0$ for all $x$, so $|f(x) - cx_0| < \varepsilon$ holds for all $x$.

**Example**: Prove that $\lim_{x\to 2}(3x - 5) = 1$.

**Solution**: See Lecture.

**Example**: For the function

$$f(x) = x \sin \frac{1}{x}, \quad x \neq 0,$$

show that

$$\lim_{x\to 0} f(x) = 0.$$

**Solution**: Even though $f$ is not defined at $x_0 = 0$, because if

$$0 < |x| < \delta = \varepsilon,$$

then

$$|f(x) - 0| = \left| x \sin \frac{1}{x} \right| \leq |x| < \varepsilon.$$

On the other hand, the function

$$g(x) = \sin \frac{1}{x}, \quad x \neq 0,$$

has no limit as $x$ approaches 0, since it assumes all values between $-1$ and 1 in every neighborhood of the origin.

**Theorem**: If $\lim_{x\to x_0} f(x)$ exists, then it is unique.

**Proof**: Suppose that limit of the function exists and it is $L_1$ and $L_2$. Let $\varepsilon > 0$. From definition of the limit, there are positive numbers $\delta_1$ and $\delta_2$ such that

$$|f(x) - L_i| < \varepsilon \quad \text{if} \quad 0 < |x - x_0| < \delta_i, \quad i = 1, 2.$$

If $\delta = \min(\delta_1, \delta_2)$, then

$$\begin{aligned} |L_1 - L_2| &= |L_1 - f(x) + f(x) - L_2| \\ &\leq |L_1 - f(x)| + |f(x) - L_2| < 2\varepsilon \quad \text{if} \quad 0 < |x - x_0| < \delta. \end{aligned}$$

We have now established an inequality that does not depend on $x$; that is,

$$|L_1 - L_2| < 2\varepsilon.$$

Since this holds for any positive $\varepsilon$, $L_1 = L_2$.

**Theorem**: If

$$\lim_{x \to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x \to x_0} g(x) = L_2. \tag{3.2}$$

then

$$\begin{aligned} \lim_{x \to x_0} (f+g)(x) &= L_1 + L_2, \\ \lim_{x \to x_0} (f-g)(x) &= L_1 - L_2, \\ \lim_{x \to x_0} (fg)(x) &= L_1 L_2, \\ \lim_{x \to x_0} \left(\frac{f}{g}\right)(x) &= \frac{L_1}{L_2}, \qquad L_2 \neq 0. \end{aligned}$$

**Proof**: From (3.2) and definition of the limit, if $\varepsilon > 0$, there is a $\delta_1 > 0$ such that

$$|f(x) - L_1| < \varepsilon \tag{3.3}$$

if $0 < |x - x_0| < \delta_1$, and a $\delta_2 > 0$ such that

$$|g(x) - L_2| < \varepsilon \tag{3.4}$$

if $0 < |x - x_0| < \delta_2$. Suppose that

$$0 < |x - x_0| < \delta = \min(\delta_1, \delta_2), \tag{3.5}$$

so that (3.3) and (3.4) both hold. Then

$$\begin{aligned} |(f \pm g)(x) - (L_1 \pm L_2)| &= |(f(x) - L_1) \pm (g(x) - L_2)| \\ &\leq |f(x) - L_1| + |g(x) - L_2| < 2\varepsilon, \end{aligned}$$

which proves (3.3) and (3.3).

We write

$$\begin{aligned} |(fg)(x) - L_1 L_2| &= |f(x)g(x) - L_1 L_2| \\ &= |f(x)(g(x) - L_2) + L_2(f(x) - L_1)| \\ &\leq |f(x)||g(x) - L_2| + |L_2||f(x) - L_1| \\ &\leq (|f(x)| + |L_2|)\varepsilon \\ &\leq (|f(x) - L_1| + |L_1| + |L_2|)\varepsilon \\ &\leq (\varepsilon + |L_1| + |L_2|) \\ &\leq (1 + |L_1| + |L_2|)\varepsilon \end{aligned}$$

First observe that if $L_2 \neq 0$, there is a $\delta_3 > 0$ such that

$$|g(x) - L_2| < \frac{|L_2|}{2}, \qquad |g(x)| > \frac{|L_2|}{2} \quad \text{if} \quad 0 < |x - x_0| < \delta_3.$$

Now suppose that $0 < |x - x_0| < \min(\delta_1, \delta_2, \delta_3)$. Then

$$\begin{aligned}
\left|\left(\frac{f}{g}\right)(x) - \frac{L_1}{L_2}\right| &= \left|\frac{f(x)}{g(x)} - \frac{L_1}{L_2}\right| = \frac{|L_2 f(x) - L_1 g(x)|}{|g(x) L_2|} \\
&\leq \frac{2}{|L_2|^2}|L_2 f(x) - L_1 g(x)| \\
&= \frac{2}{|L_2|^2}\,|L_2[f(x) - L_1] + L_1[L_2 - g(x)]| \\
&\leq \frac{2}{|L_2|^2}\,[|L_2||f(x) - L_1| + |L_1||L_2 - g(x)|] \\
&\leq \frac{2}{|L_2|^2}(|L_2| + |L_1|)\varepsilon.
\end{aligned}$$

**Example**: Find

$$\lim_{x\to 2} \frac{9 - x^2}{x + 1} \quad \text{and} \quad \lim_{x\to 2}(9 - x^2)(x + 1).$$

**Solution**: If $c$ is a constant, then $\lim_{x\to x_0} c = c$, and $\lim_{x\to x_0} x = x_0$. Therefore

$$\begin{aligned}
\lim_{x\to 2}(9 - x^2) &= \lim_{x\to 2} 9 - \lim_{x\to 2} x^2 \\
&= \lim_{x\to 2} 9 - (\lim_{x\to 2} x)^2 \\
&= 9 - 2^2 = 5,
\end{aligned}$$

and

$$\lim_{x\to 2}(x + 1) = \lim_{x\to 2} x + \lim_{x\to 2} 1 = 2 + 1 = 3.$$

Therefore,

$$\lim_{x\to 2} \frac{9 - x^2}{x + 1} = \frac{\lim_{x\to 2}(9 - x^2)}{\lim_{x\to 2}(x + 1)} = \frac{5}{3}$$

and

$$\lim_{x\to 2}(9 - x^2)(x + 1) = \lim_{x\to 2}(9 - x^2) \lim_{x\to 2}(x + 1) = 5 \cdot 3 = 15.$$

**Example**: The function

$$f(x) = 2x \sin\sqrt{x}$$

satisfies the inequality

$$|f(x)| < \varepsilon, \qquad \text{whenever} \quad 0 < x < \delta = \varepsilon/2.$$

However, this does not mean that $\lim_{x\to 0} f(x) = 0$, since $f$ is not defined for negative $x$.

**Example**: The function

$$g(x) = x + \frac{|x|}{x}, \quad x \neq 0,$$

can be rewritten as

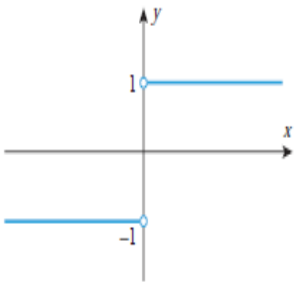
$$g(x) = \begin{cases} x+1, & x > 0, \\ x-1, & x < 0; \end{cases}$$

hence, every open interval containing $x_0 = 0$ also contains points $x_1$ and $x_2$ such that $|g(x_1) - g(x_2)|$ is as close to 2 as we please. Therefore, $\lim_{x\to x_0} g(x)$ does not exist.

Although $f(x)$ and $g(x)$ do not approach limits as $x$ approaches zero, they each exhibit a definite sort of limiting behavior for small positive values of $x$, as does $g(x)$ for small negative values of $x$. The kind of behavior we have in mind is defined precisely as follows.

**Example**: Investigate the limit

$$\lim_{x\to 0} \frac{|x|}{x}.$$




## 3.2 One Sided Limits

We say that $f(x)$ *approaches the left-hand limit* $L$ *as* $x$ *approaches* $x_0$ *from the left*, and write

$$\lim_{x\to x_0-} f(x) = L,$$

if $f$ is defined on some open interval $(a, x_0)$ and, for each $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{if} \quad x_0 - \delta < x < x_0.$$

We say that $f(x)$ *approaches the right-hand limit* $L$ *as* $x$ *approaches* $x_0$ *from the right*, and write

$$\lim_{x\to x_0+} f(x) = L,$$

if $f$ is defined on some open interval $(x_0, b)$ and, for each $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{if} \quad x_0 < x < x_0 + \delta.$$

**Example**: Let

$$g(x) = \frac{x + |x|(1+x)}{x} \sin\frac{1}{x}, \quad x \neq 0.$$

**Solution**: If $x < 0$, then

$$g(x) = -x \sin\frac{1}{x}, \qquad \lim_{x \to 0-} g(x) = 0,$$

since

$$|g(x) - 0| = \left| x \sin\frac{1}{x} \right| \leq |x| < \varepsilon$$

if $-\varepsilon < x < 0$; that is, we have $\delta = \varepsilon$.

If $x > 0$, then

$$g(x) = (2 + x) \sin\frac{1}{x},$$

which takes on every value between $-2$ and 2 in every interval $(0, \delta)$.

Hence, $g(x)$ does not approach a right-hand limit at $x$ approaches 0 from the right. This shows that a function may have a limit from one side at a point but fail to have a limit from the other side.

**Example**: Prove that

$$\lim_{x \to 0^+} \sqrt{x} = 0.$$

**Solution**: See Lecture.

**Example**: Show that

$$\lim_{x \to 0+} \left( \frac{|x|}{x} + x \right) = \quad 1, \qquad \lim_{x \to 0-} \left( \frac{|x|}{x} + x \right) = -1,$$

**Solution**: See Lecture

**Theorem**: A function $f$ has a limit at $x_0$ if and only if it has left- and right-hand limits at $x_0$, and they are equal. More specifically,

$$\lim_{x \to x_0} f(x) = L$$

if and only if

$$\lim_{x \to x_0+} f(x) = \lim_{x \to x_0-} f(x) = L.$$

## 3.3 Limits at $\pm\infty$

If $f$ is defined on an interval $(a, \infty)$, then $f(x)$ *approaches the limit* $L$ *as* $x$ *approaches* $\infty$, and write

$$\lim_{x\to\infty} f(x) = L.$$

For each $\varepsilon > 0$, there is a number $\beta$ such that

$$|f(x) - L| < \varepsilon \qquad \text{if} \quad x > \beta.$$

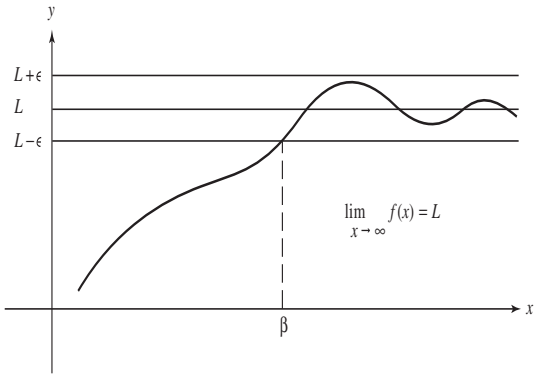



Figure 3.7: Limits at $\pm\infty$

**Examples**: Let

$$f(x) = 1 - \frac{1}{x^2}, \quad g(x) = \frac{2|x|}{1+x}, \quad \text{and} \quad h(x) = \sin x.$$

Find limits when $x \to \pm\infty$.

**Solution**: We have

$$\lim_{x\to\infty} f(x) = 1,$$

since

$$|f(x) - 1| = \frac{1}{x^2} < \varepsilon \quad \text{if} \quad x > \frac{1}{\sqrt{\varepsilon}}.$$

We have

$$\lim_{x\to\infty} g(x) = 2,$$

since

$$|g(x) - 2| = \left|\frac{2x}{1+x} - 2\right| = \frac{2}{1+x} < \frac{2}{x} < \varepsilon \quad \text{if} \quad x > \frac{2}{\varepsilon}.$$

However, $\lim_{x\to\infty} h(x)$ does not exist, since $h$ assumes all values between $-1$ and 1 in any semi-infinite interval $(\tau, \infty)$. Discuss the limits at $x \to -\infty$.

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{x^2},$$

and

$$p(x) = \sin\frac{1}{x},$$
$$q(x) = \frac{1}{x^2}\sin\frac{1}{x}$$

## 3.4 Infinite Limits

**Infinite limits**: We say that $f(x)$ *approaches* $\infty$ *as* $x$ *approaches* $x_0$ *from the left*, and write

$$\lim_{x\to x_0-} f(x) = \infty.$$

If $f$ is defined on an interval $(a, x_0)$ and, for each real number $M$, there is a $\delta > 0$ such that

$$f(x) > M \quad \text{if} \quad x_0 - \delta < x < x_0.$$

Similarly, we can define the following limits

$$\lim_{x\to x_0-} f(x) = -\infty, \quad \lim_{x\to x_0+} f(x) = \infty, \quad \lim_{x\to x_0+} f(x) = -\infty.$$

$$\begin{aligned}
\lim_{x\to 0-}\frac{1}{x} &= -\infty,\\
\lim_{x\to 0+}\frac{1}{x} &= \infty;\\
\lim_{x\to 0-}\frac{1}{x^2} &= \lim_{x\to 0+}\frac{1}{x^2} = \infty\\
\lim_{x\to 0}\frac{1}{x^2} &= \infty;\\
\lim_{x\to\infty} x^2 &= \lim_{x\to-\infty} x^2 = \infty;
\end{aligned}$$

**Example**: If

$$f(x) = e^{2x} - e^x.$$

we cannot obtain $\lim_{x\to\infty} f(x)$ by writing

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} e^{2x} - \lim_{x\to\infty} e^x,$$

because this produces the indeterminate form $\infty - \infty$.

However, by writing

$$f(x) = e^{2x}(1 - e^{-x}),$$

we find that

$$\lim_{x\to\infty} f(x) = \left(\lim_{x\to\infty} e^{2x}\right)\left(\lim_{x\to\infty} 1 - \lim_{x\to\infty} e^{-x}\right) = \infty(1-0) = \infty.$$

**Example**: Find

$$\lim_{x\to\infty} \frac{2x^2 - x + 1}{3x^2 + 2x - 1}.$$

**Example**: Find

$$\lim_{x\to 1} \frac{x^8 - 1}{x^4 - 1}.$$

$$f(x) = \begin{cases} \frac{1}{x+2}, & x < -2, \\ x^2 - 5, & -2 < x \leq 3, \\ \sqrt{x+13}, & x > 3. \end{cases}$$

## 3.5 Continuity

**Continuity (Informal)**:

- We say that $f$ is *continuous at* $x_0$ if $f$ is defined on an open interval $(a, b)$ containing $x_0$ and
$$\lim_{x\to x_0} f(x) = f(x_0).$$

- We say that $f$ is *continuous from the left at* $x_0$ if $f$ is defined on an open interval $(a, x_0)$ and
$$\lim_{x\to x_0-} f(x) = f(x_0).$$

- We say that $f$ is *continuous from the right at* $x_0$ if $f$ is defined on an open interval $(x_0, b)$ and
$$\lim_{x\to x_0+} f(x) = f(x_0).$$

**Continuity (Formal)**:

- A function $f$ is continuous at $x_0$ if and only if $f$ is defined on an open interval $(a, b)$ containing $x_0$ and for each $\varepsilon > 0$ there is a $\delta > 0$ such that
$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad |x - x_0| < \delta.$$

- A function $f$ is continuous from the right at $x_0$ if and only if $f$ is defined on an interval $[x_0, b)$ and for each $\varepsilon > 0$ there is a $\delta > 0$ such that
$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad x_0 \leq x < x_0 + \delta.$$

- A function $f$ is continuous from the left at $x_0$ if and only if $f$ is defined on an interval $(a, x_0]$ and for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad x_0 - \delta < x \leq x_0.$$

**Example**: Discuss the continuity of $f$ defined on $[0, 2]$ by

$$f(x) = \begin{cases} x^2, & 0 \leq x < 1, \\ x + 1, & 1 \leq x \leq 2 \end{cases}$$

**Solution**:

$$\begin{aligned} \lim_{x \to 0+} f(x) &= 0 = f(0), \\ \lim_{x \to 1-} f(x) &= 1 \neq f(1) = 2, \\ \lim_{x \to 1+} f(x) &= 2 = f(1), \\ \lim_{x \to 2-} f(x) &= 3 = f(2). \end{aligned}$$

Therefore, $f$ is continuous from the right at 0 and 1 and continuous from the left at 2, but not at 1.

**Example**: Discuss the continuity of $f$ defined on $[0, 2]$ by

$$f(x) = \begin{cases} x^2, & 0 \leq x < 1, \\ x + 1, & 1 \leq x \leq 2 \end{cases}$$

**Solution**: If $0 < x$, $x_0 < 1$, then

$$\begin{aligned} |f(x) - f(x_0)| &= |x^2 - x_0^2| = |x - x_0|\,|x + x_0| \\ &\leq 2|x - x_0| < \varepsilon \quad \text{if} \quad |x - x_0| < \varepsilon/2. \end{aligned}$$

Hence, $f$ is continuous at each $x_0$ in $(0, 1)$. If $1 < x$, $x_0 < 2$, then

$$\begin{aligned} |f(x) - f(x_0)| &= |(x + 1) - (x_0 + 1) = |x - x_0| \\ &< \varepsilon \quad \text{if} \quad |x - x_0| < \varepsilon. \end{aligned}$$

Hence, $f$ is continuous at each $x_0$ in $(1, 2)$.

A function $f$ is *continuous on an open interval* $(a, b)$ if it is continuous at every point in $(a, b)$.

If, in addition,

$$\lim_{x \to b-} f(x) = f(b)$$

and

$$\lim_{x \to a_+} = f(a)$$

**Example**: Discuss the continuity of $f(x) = \sqrt{x}, \quad 0 \leq x < \infty$.

**Solution**: Consider

$$|f(x) - f(0)| = \sqrt{x} < \varepsilon \quad \text{if} \quad 0 \leq x < \varepsilon^2,$$

so $\lim_{x \to 0_+} f(x) = f(0)$.

If $x_0 > 0$ and $x \geq 0$, then

$$\begin{aligned} |f(x) - f(x_0)| &= |\sqrt{x} - \sqrt{x_0}| = \frac{|x - x_0|}{\sqrt{x} + \sqrt{x_0}} \\ &\leq \frac{|x - x_0|}{\sqrt{x_0}} < \varepsilon \quad \text{if} \quad |x - x_0| < \varepsilon\sqrt{x_0}, \end{aligned}$$

so $\lim_{x \to x_0} f(x) = f(x_0)$. Hence, $f$ is continuous on $[0, \infty)$.

**Example**: Discuss the continuity of $g(x) = \frac{1}{\sin \pi x}$

**Solution**: The function is continuous on

$$S = \bigcup_{n=-\infty}^{\infty} (n, n+1).$$

The function $g$ is discontinuous on at any $x_0 = n$ (integer), since it is not defined at such points.

## 3.6 Piecewise Continuous Functions

**Piecewise continuous function**: A function $f$ is *piecewise continuous* on $[a, b]$ if

- $\lim_{x \to x_{0_+}} f(x)$ exists for all $x_0$ in $[a, b)$;
- $\lim_{x \to x_{0_-}} f(x)$ exists for all $x_0$ in $(a, b]$;
- $\lim_{x \to x_{0_+}} f(x) = \lim_{x \to x_{0_-}} f(x) = f(x_0)$ for all but finitely many points $x_0$ in $(a, b)$.

**Example**: Discuss the following function

$$f(x) = \begin{cases} 1, & x = 0, \\ x, & 0 < x < 1, \\ 2, & x = 1, \\ x, & 1 < x \leq 2, \\ -1, & 2 < x < 3, \\ 0, & x = 3. \end{cases}$$

**Solution** See lecture for explanation.

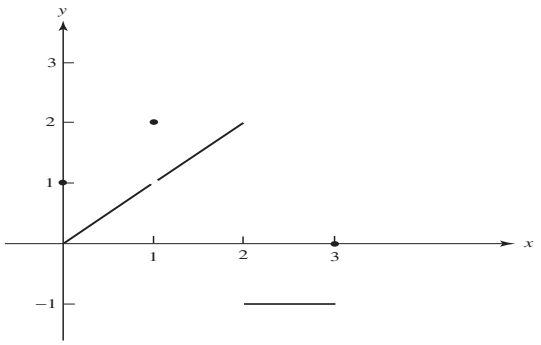



Figure 3.8: Graph of the function

**Example**: Discuss the following function

$$f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0, \\ \\ 0, & x = 0, \end{cases}$$

**Solution**: The function is continuous at all $x_0$ except $x_0 = 0$.

As $x$ approaches 0 from either side, $f(x)$ oscillates between $-1$ and 1 with ever-increasing frequency, so neither $\lim_{x \to 0_+} f(x)$ nor $\lim_{x \to 0_-} f(x)$ exists.

Therefore, the discontinuity of $f$ at 0 is not a jump discontinuity, and if $\rho > 0$, then $f$ is not piecewise continuous on any interval of the form $[-\rho, 0]$, $[-\rho, \rho]$, or $[0, \rho]$.

**Theorem**: If $f$ and $g$ are continuous on a set $S$, then so are $f + g$, $f - g$, and $fg$. In addition, $f/g$ is continuous at each $x_0$ in $S$ such that $g(x_0) \neq 0$.

**Proof**: See Lecture.

**Example**: Discuss the continuity of the following function

$$r(x) = \frac{9 - x^2}{x + 1}.$$

**Solution**: See Lecture.

## 3.7 Removable Discontinuity

It can be shown that if $f_1$, $f_2$, ..., $f_n$ are continuous on a set $S$, then so are $f_1 + f_2 + \cdots + f_n$ and $f_1 f_2 \cdots f_n$.

Therefore, any rational function

$$r(x) = \frac{a_0 + a_1 x + \cdots + a_n x^n}{b_0 + b_1 x + \cdots + b_m x^m}$$

is continuous for all values of $x$ except those for which its denominator vanishes.

**Removable discontinuity**: Let $f$ be defined on a deleted neighborhood of $x_0$ and discontinuous (perhaps even undefined) at $x_0$. We say that $f$ has a removable discontinuity at $x_0$ if $\lim_{x \to x_0} f(x)$ exists.

In this case, the function

$$g(x) = \begin{cases} f(x) & \text{if } x \in D_f \text{ and } x \neq x_0, \\ \lim_{x \to x_0} f(x) & \text{if } x = x_0, \end{cases}$$

is continuous at $x_0$.

**Example**: Consider the function

$$f(x) = x \sin \frac{1}{x}.$$

**Solution**: The function is not defined at $x_0 = 0$, and therefore certainly not continuous there, but $\lim_{x \to 0} f(x) = 0$.

$$g(x) = \begin{cases} f(x) & \text{if } x \in D_f \text{ and } x \neq x_0, \\ \lim_{x \to x_0} f(x) & \text{if } x = x_0, \end{cases}$$

The function

$$f_1(x) = \sin \frac{1}{x}$$

is undefined at 0 and its discontinuity there is not removable, since $\lim_{x \to 0} f_1(x)$ does not exist.

**Composition of functions**: Suppose that $f$ and $g$ are functions with domains $D_f$ and $D_g$.

If $D_g$ has a nonempty subset $T$ such that $g(x) \in D_f$ whenever $x \in T$, then the *composite function* $f \circ g$ is defined on $T$ by

$$(f \circ g)(x) = f(g(x)).$$

**Example**: If

$$f(x) = \log x \quad \text{and} \quad g(x) = \frac{1}{1-x^2},$$

Can we define $f \circ g$?

**Solution**: The domain of the given functions are

$$D_f = (0, \infty) \quad \text{and} \quad D_g = x \in \mathbb{R} / \ x \neq \pm 1.$$

Since $g(x) > 0$ if $x \in T = (-1, 1)$, the composite function $f \circ g$ is defined on $(-1, 1)$ by

$$(f \circ g)(x) = \log \frac{1}{1-x^2}.$$

Can we define $g \circ f$?

The function $g \circ f$ is defined on $(0, 1/e) \cup (1/e, e) \cup (e, \infty)$ by

$$(g \circ f)(x) = \frac{1}{1-(\log x)^2}.$$

**Theorem**: Suppose that $g$ is continuous at $x_0$, $g(x_0)$ is an interior point of $D_f$, and $f$ is continuous at $g(x_0)$. Then $f \circ g$ is continuous at $x_0$.

**Proof**: Since $g(x_0)$ is an interior point of $D_f$ and $f$ is continuous at $g(x_0)$, for every $\varepsilon > 0$ there is a $\delta_1 > 0$ such that $f(t)$ is defined and

$$|f(t) - f(g(x_0))| < \varepsilon \quad \text{if} \quad |t - g(x_0)| < \delta_1.$$

Since $g$ is continuous at $x_0$, there is a $\delta > 0$ such that $g(x)$ is defined and

$$|g(x) - g(x_0)| < \delta_1 \quad \text{if} \quad |x - x_0| < \delta.$$

We can imply that

$$|f(g(x)) - f(g(x_0))| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta.$$

Therefore, $f \circ g$ is continuous at $x_0$.

**Example**: For the functions

$$f(x) = \sqrt{x}, \qquad g(x) = \frac{9-x^2}{x+1}.$$

Check the continuity of $f \circ g$.

**Solution**: The function $f$ is continuous for $x > 0$, and the function $g$ is continuous for $x \neq -1$.

Since $g(x) > 0$ if $x < -3$ or $-1 < x < 3$, the composite function

$$(f \circ g)(x) = \sqrt{\frac{9 - x^2}{x + 1}}$$

is continuous on $(-\infty, -3) \cup (-1, 3)$.

It is also continuous from the left at $-3$ and 3.

## 3.8 Bounded Functions

**Bounded below function**: A function $f$ is *bounded below* on a set $S$ if there is a real number $m$ such that

$$f(x) \geq m \text{ for all } x \in S.$$

In this case, the set

$$V = \{f(x) : x \in S\}$$

has an infimum $\alpha$, and we write

$$\alpha = \inf_{x \in S} f(x).$$

If there is a point $x_1$ in $S$ such that $f(x_1) = \alpha$, we say that $\alpha$ is the *minimum of $f$ on $S$*, and write

$$\alpha = \min_{x \in S} f(x).$$

**Bounded above function**: A function $f$ is *bounded above on $S$* if there is a real number $M$ such that $f(x) \leq M$ for all $x$ in $S$.

In this case, $V$ has a supremum $\beta$, and we write

$$\beta = \sup_{x \in S} f(x).$$

If there is a point $x_2$ in $S$ such that $f(x_2) = \beta$, we say that $\beta$ is the *maximum of $f$ on $S$*, and write

$$\beta = \max_{x \in S} f(x).$$

If $f$ is bounded above and below on a set $S$, we say that $f$ is *bounded* on $S$.

**Example**: The function

$$g(x) = \begin{cases} \frac{1}{2}, & x = 0 \quad \text{or} \quad x = 1, \\ 1 - x, & 0 < x < 1, \end{cases} +$$

is bounded on $[0,1]$, and

$$\sup_{0\le x\le 1} g(x) = 1, \quad \inf_{0\le x\le 1} g(x) = 0.$$

Therefore, $g$ has no maximum or minimum on $[0,1]$, since it does not assume either of the values 0 and 1.

**Example**: The function

$$h(x) = 1 - x, \quad 0 \le x \le 1,$$

which differs from $g$ only at 0 and 1, has the same supremum and infimum as $g$, but it attains these values at $x = 0$ and $x = 1$, respectively.

Therefore,

$$\max_{0\le x\le 1} h(x) = 1 \quad \text{and} \quad \min_{0\le x\le 1} h(x) = 0.$$

**Example**: The function

$$f(x) = e^{x(x-1)} \sin \frac{1}{x(x-1)}, \quad 0 < x < 1,$$

oscillates between $\pm e^{x(x-1)}$ infinitely often in every interval of the form $(0,\rho)$ or $(1-\rho,1)$, where $0 < \rho < 1$.

Furthermore, we have

$$\sup_{0<x<1} f(x) = 1, \quad \inf_{0<x<1} f(x) = -1.$$

However, $f$ does not assume these values, so $f$ has no maximum or minimum on $(0,1)$.

**Theorem**: If $f$ is continuous on a finite closed interval $[a,b]$, then $f$ is bounded on $[a,b]$.

**Proof**: Suppose that $t \in [a,b]$. Since $f$ is continuous at $t$, there is an open interval $I_t$ containing $t$ such that

$$|f(x) - f(t)| < 1 \quad \text{if} \quad x \in I_t \cap [a,b].$$

The collection $\mathcal{H} = \{I_t : a \le t \le b\}$ is an open covering of $[a,b]$.

Since $[a,b]$ is compact, the Heine-Borel theorem implies that there are finitely many points $t_1, t_2, \ldots, t_n$ such that the intervals $I_{t_1}, I_{t_2}, \ldots, I_{t_n}$ cover $[a,b]$.

According to continuity of the function and with $t = t_i$,

$$|f(x) - f(t_i)| < 1 \quad \text{if} \quad x \in I_{t_i} \cap [a,b].$$

Therefore,

$$\begin{aligned} |f(x)| &= |(f(x) - f(t_i)) + f(t_i)| \le |f(x) - f(t_i)| + |f(t_i)| \\ &\le 1 + |f(t_i)| \quad \text{if} \quad x \in I_{t_i} \cap [a, b]. \end{aligned}$$

Let

$$M = 1 + \max_{1 \le i \le n} |f(t_i)|.$$

Since $[a, b] \subset \bigcup_{i=1}^{n} (I_{t_i} \cap [a, b])$, and we have $|f(x)| \le M$ if $x \in [a, b]$.

**Theorem**: Suppose that $f$ is continuous on a finite closed interval $[a, b]$. Let

$$\alpha = \inf_{a \le x \le b} f(x) \quad \text{and} \quad \beta = \sup_{a \le x \le b} f(x).$$

Then $\alpha$ and $\beta$ are respectively the minimum and maximum of $f$ on $[a, b]$; that is, there are points $x_1$ and $x_2$ in $[a, b]$ such that

$$f(x_1) = \alpha \quad \text{and} \quad f(x_2) = \beta.$$

**Proof**: Suppose that there is no $x_1$ in $[a, b]$ such that $f(x_1) = \alpha$. Then $f(x) > \alpha$ for all $x \in [a, b]$. We will show that this leads to a contradiction.

Suppose that $t \in [a, b]$. Then $f(t) > \alpha$, so

$$f(t) > \frac{f(t) + \alpha}{2} > \alpha.$$

Since $f$ is continuous at $t$, there is an open interval $I_t$ about $t$ such that

$$f(x) > \frac{f(t) + \alpha}{2} \quad \text{if} \quad x \in I_t \cap [a, b]. \tag{3.6}$$

The collection $\mathcal{H} = \{I_t : a \le t \le b\}$ is an open covering of $[a, b]$.

Since $[a, b]$ is compact, the Heine–Borel theorem implies that there are finitely many points $t_1, t_2, \ldots, t_n$ such that the intervals $I_{t_1}, I_{t_2}, \ldots, I_{t_n}$ cover $[a, b]$.

Define

$$\alpha_1 = \min_{1 \le i \le n} \frac{f(t_i) + \alpha}{2}.$$

Then, since $[a, b] \subset \bigcup_{i=1}^{n} (I_{t_i} \cap [a, b])$, (3.6) implies that

$$f(t) > \alpha_1, \quad a \le t \le b.$$

But $\alpha_1 > \alpha$, so this contradicts the definition of $\alpha$. Therefore, $f(x_1) = \alpha$ for some $x_1$ in $[a, b]$.

Consider the function

$$g(x) = 1 - (1 - x) \sin \frac{1}{x}, \qquad (0, 1].$$

The function $g(x)$ is continuous and has supremum 2 on the noncompact interval $(0, 1]$.

Since

$$\begin{aligned} g(x) &\leq 1 + (1-x)\left|\sin\frac{1}{x}\right| \\ &\leq 1 + (1-x) < 2 \quad \text{if} \quad 0 < x \leq 1. \end{aligned}$$

But does not assume its supremum on $(0, 1]$.

Consider the function

$$f(x) = e^{-x}.$$

The function $f(x)$ is continuous and has infimum 0, which it does not attain, on the noncompact interval $(0, \infty)$.

## 3.9 The Intermediate Value Theorem

**Theorem**: Suppose that $f$ is continuous on $[a, b]$, $f(a) \neq f(b)$, and $\mu$ is between $f(a)$ and $f(b)$. Then $f(c) = \mu$ for some $c$ in $(a, b)$.

**Proof**: Suppose that $f(a) < \mu < f(b)$. The set

$$S = x : a \leq x \leq b \quad \text{and} \quad f(x) \leq \mu$$

is bounded and nonempty.

Let $c = \sup S$. We will show that $f(c) = \mu$.

If $f(c) > \mu$, then $c > a$ and, since $f$ is continuous at $c$, there is an $\varepsilon > 0$ such that $f(x) > \mu$ if $c - \varepsilon < x \leq c$.

Therefore, $c - \varepsilon$ is an upper bound for $S$, which contradicts the definition of $c$ as the supremum of $S$.

If $f(c) < \mu$, then $c < b$ and there is an $\varepsilon > 0$ such that $f(x) < \mu$ for $c \leq x < c+\varepsilon$, so $c$ is not an upper bound for $S$.

This is also a contradiction. Therefore, $f(c) = \mu$.

## 3.10 Uniform Continuity

A function $f$ is continuous on a subset $S$ of its domain if for each $\varepsilon > 0$ and each $x_0$ in $S$, there is a $\delta > 0$, *which may depend upon* $x_0$ *as well as* $\varepsilon$, such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta \quad \text{and} \quad x \in D_f.$$

A function $f$ is *uniformly continuous* on a subset $S$ of its domain if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - f(x')| < \varepsilon \text{ whenever } |x - x'| < \delta \text{ and } x, x' \in S.$$

**Example**: Check the uniform continuity of the function

$$f(x) = 2x.$$

**Solution**: For the function $f(x)$, we have

$$|f(x) - f(x')| = 2|x - x'| < \varepsilon \quad \text{if} \quad |x - x'| < \varepsilon/2.$$

The function $f(x)$ is uniformly continuous on $(-\infty, \infty)$,

**Remark**: A function $f$ is *not* uniformly continuous on $S$ if there is an $\varepsilon_0 > 0$ such that if $\delta$ is any positive number, there are points $x$ and $x'$ in $S$ such that

$$|x - x'| < \delta$$

but

$$|f(x) - f(x')| \geq \varepsilon_0.$$

**Example**: Show that the function $g(x) = x^2$ is uniformly continuous on $[-r, r]$ for any finite $r$.

**Solution**: To see this, note that

$$|g(x) - g(x') = |x^2 - (x')^2| = |x - x'|\,|x + x'| \leq 2r|x - x'|,$$

so

$$|g(x) - g(x')| < \varepsilon \quad \text{if} \quad |x - x'| < \delta = \frac{\varepsilon}{2r} \quad \text{and} \quad -r \leq x, x' \leq r.$$

**Example**: The function $g(x) = x^2$ is not uniformly continuous on $(-\infty, \infty)$.

**Solution**: To see this, we will show that if $\delta > 0$ there are real numbers $x$ and $x'$ such that

$$|x - x'| = \delta/2 \quad \text{and} \quad |g(x) - g(x')| \geq 1.$$

To this end, we write

$$|g(x) - g(x')| = |x^2 - (x')^2| = |x - x'|\,|x + x'|.$$

If $|x - x'| = \delta/2$ and $x, x' > 1/\delta$, then

$$|x - x'|\,|x + x'| > \frac{\delta}{2}\left(\frac{1}{\delta} + \frac{1}{\delta}\right) = 1.$$

**Example**: The function

$$f(x) = \cos \frac{1}{x}$$

is continuous on $(0, 1]$

However, $f$ is not uniformly continuous on $(0, 1]$, since

$$\left| f\left(\frac{1}{n\pi}\right) - f\left(\frac{1}{(n+1)\pi}\right) \right| = 2, \quad n = 1, 2, \ldots.$$

**Theorem**: If $f$ is continuous on a closed and bounded interval $[a, b]$, then $f$ is uniformly continuous on $[a, b]$.

**Proof**: Suppose that $\varepsilon > 0$. Since $f$ is continuous on $[a, b]$, for each $t$ in $[a, b]$ there is a positive number $\delta_t$ such that

$$|f(x) - f(t)| < \frac{\varepsilon}{2} \quad \text{if} \quad |x - t| < 2\delta_t \quad \text{and} \quad x \in [a, b].$$

If $I_t = (t - \delta_t, t + \delta_t)$, the collection

$$\mathcal{H} = \{I_t : t \in [a, b]\}$$

is an open covering of $[a, b]$.

Since $[a, b]$ is compact, the Heine–Borel theorem implies that there are finitely many points $t_1$, $t_2$, $\ldots$, $t_n$ in $[a, b]$ such that $I_{t_1}$, $I_{t_2}$, $\ldots$, $I_{t_n}$ cover $[a, b]$.

Now define

$$\delta = \min\{\delta_{t_1}, \delta_{t_2}, \ldots, \delta_{t_n}\}. \tag{3.7}$$

We will show that if

$$|x - x'| < \delta \quad \text{and} \quad x, x' \in [a, b], \tag{3.8}$$

then $|f(x) - f(x')| < \varepsilon$.

From the triangle inequality,

$$\begin{aligned} |f(x) - f(x')| &= |\,(f(x) - f(t_r)) + (f(t_r) - f(x'))\,| \\ &\leq |f(x) - f(t_r)| + |f(t_r) - f(x')|. \end{aligned} \tag{3.9}$$

Since $I_{t_1}$, $I_{t_2}$, $\ldots$, $I_{t_n}$ cover $[a, b]$, $x$ must be in one of these intervals. Suppose that $x \in I_{t_r}$; that is,

$$|x - t_r| < \delta_{t_r}. \tag{3.10}$$

From continuity condition with $t = t_r$,

$$|f(x) - f(t_r)| < \frac{\varepsilon}{2}. \tag{3.11}$$

From (3.8), (3.10), and the triangle inequality,

$$|x' - t_r| = |(x' - x) + (x - t_r)| \le |x' - x| + |x - t_r| < \delta + \delta_{t_r} \le 2\delta_{t_r}.$$

Therefore, from continuity condition with $t = t_r$ and $x$ replaced by $x'$ implies that

$$|f(x') - f(t_r)| < \frac{\varepsilon}{2}.$$

This, (3.9), and (3.11) imply that $|f(x) - f(x')| < \varepsilon$.

## 3.11 Monotonic Functions

**Monotonic functions**: A function $f$ is *nondecreasing* on an interval $I$ if

$$f(x_1) \le f(x_2) \quad \text{whenever} \quad x_1, x_2 \in I, \quad x_1 < x_2. \tag{3.12}$$

or *nonincreasing* on $I$ if

$$f(x_1) \ge f(x_2) \quad \text{whenever } x_1 \text{ and } x_2 \text{ are in } I \text{ and } x_1 < x_2. \tag{3.13}$$

In either case, $f$ is on $I$.

If $\le$ can be replaced by $<$ in (3.12), $f$ is *increasing* on $I$. If $\ge$ can be replaced by $>$ in (3.13), $f$ is *decreasing* on $I$. In either of these two cases, $f$ is *strictly monotonic* on $I$.

**Example**: The function

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ 2, & 1 \le x \le 2, \end{cases}$$

is nondecreasing on $I = [0, 2]$.

**Example**: The function $g(x) = x^2$ is increasing on $[0, \infty)$. The function $h(x) = -x^3$ is decreasing on $(-\infty, \infty)$.

**Theorem**: Suppose that $f$ is monotonic on $(a, b)$ and define

$$\alpha = \inf_{a<x<b} f(x) \quad \text{and} \quad \beta = \sup_{a<x<b} f(x).$$

1. If $f$ is nondecreasing, then $\lim_{x\to a_+} f(x) = \alpha$ and $\lim_{x\to b_-} f(x) = \beta$.
2. If $f$ is nonincreasing, then $\lim_{x\to a_+} f(x) = \beta$ and $\lim_{x\to b_-} f(x) = \alpha$. (Here $a+ = -\infty$ if $a = -\infty$ and $b- = \infty$ if $b = \infty$.)

3. If $a < x_0 < b$, then $\lim_{x\to x_{0_+}} f(x)$ and $\lim_{x\to x_{0_-}} f(x)$ exist and are finite; moreover,
$$\lim_{x\to x_{0_-}} f(x) \le f(x_0) \le \lim_{x\to x_{0_+}} f(x)$$
if $f$ is nondecreasing, and
$$\lim_{x\to x_{0_-}} f(x) \ge f(x_0) \ge \lim_{x\to x_{0_+}} f(x)$$
if $f$ is nonincreasing.

**Proof**: We first show that $\lim_{x\to a_+} f(x) = \alpha$.

If $M > \alpha$, there is an $x_0$ in $(a, b)$ such that $f(x_0) < M$. Since $f$ is nondecreasing, $f(x) < M$ if $a < x < x_0$. Therefore, if $\alpha = -\infty$, then $\lim_{x\to a_+} f(x) = -\infty$.

If $\alpha > -\infty$, let $M = \alpha + \varepsilon$, where $\varepsilon > 0$.

Then $\alpha \le f(x) < \alpha + \varepsilon$, so
$$|f(x) - \alpha| < \varepsilon \quad \text{if} \quad a < x < x_0. \tag{3.14}$$

If $a = -\infty$, this implies that $f(-\infty) = \alpha$. If $a > -\infty$, let $\delta = x_0 - a$. Then (3.14) is equivalent to
$$|f(x) - \alpha| < \varepsilon \quad \text{if} \quad a < x < a + \delta,$$
which implies that $f(a+) = \alpha$.

We now show that $\lim_{x\to b_-} f(x) = \beta$.

If $M < \beta$, there is an $x_0$ in $(a, b)$ such that $f(x_0) > M$.

Since $f$ is nondecreasing, $f(x) > M$ if $x_0 < x < b$. Therefore, if $\beta = \infty$, then $\lim_{x\to b_-} f(x) = \infty$.

If $\beta < \infty$, let $M = \beta - \varepsilon$, where $\varepsilon > 0$. Then $\beta - \varepsilon < f(x) \le \beta$, so
$$|f(x) - \beta| < \varepsilon \quad \text{if} \quad x_0 < x < b. \tag{3.15}$$

If $b = \infty$, this implies that $f(\infty) = \beta$. If $b < \infty$, let $\delta = b - x_0$.

Then (3.15) is equivalent to
$$|f(x) - \beta| < \varepsilon \quad \text{if} \quad b - \delta < x < b,$$
which implies that $f(b-) = \beta$.

## 3.12 Limits Inferior and Superior

Suppose that $f$ is bounded on $[a, x_0)$, where $x_0$ may be finite or $\infty$.

For $a \le x < x_0$, define
$$\begin{aligned} S_f(x; x_0) &= \sup_{x\le t<x_0} f(t) \\ I_f(x; x_0) &= \inf_{x\le t<x_0} f(t). \end{aligned}$$

Then the *left limit superior of* $f$ *at* $x_0$ is defined to be

$$\limsup_{x\to x_{0-}} f(x) = \lim_{x\to x_{0-}} S_f(x;x_0).$$

The *left limit inferior of* $f$ *at* $x_0$ is defined to be

$$\liminf_{x\to x_{0-}} f(x) = \lim_{x\to x_{0-}} I_f(x;x_0).$$

(If $x_0 = \infty$, we define $x_0- = \infty$.)

**Theorem**: If $f$ is bounded on $[a,x_0)$, then $\beta = \limsup_{x\to x_{0-}} f(x)$ exists and is the unique real number with the following properties:

1. If $\varepsilon > 0$, there is an $a_1$ in $[a,x_0)$ such that
$$f(x) < \beta + \varepsilon \quad \text{if} \quad a_1 \le x < x_0.$$
2. If $\varepsilon > 0$ and $a_1$ is in $[a,x_0)$, then
$$f(\overline{x}) > \beta - \varepsilon \quad \text{for some } \overline{x} \in [a_1,x_0).$$

**Theorem**: If $f$ is bounded on $[a,x_0)$, then $\alpha = \liminf_{x\to x_{0-}} f(x)$ exists and is the unique real number with the following properties:

1. If $\varepsilon > 0$, there is an $a_1$ in $[a,x_0)$ such that
$$f(x) > \alpha - \varepsilon \quad \text{if} \quad a_1 \le x < x_0.$$
2. If $\varepsilon > 0$ and $a_1$ is in $[a,x_0)$, then
$$f(\overline{x}) < \alpha + \varepsilon \quad \text{for some } \overline{x} \in [a_1,x_0).$$

**Theorem**: Suppose that $f$ is monotonic on $(a,b)$ and define

$$\alpha = \inf_{a<x<b} f(x), \quad \beta = \sup_{a<x<b} f(x).$$

1. If $f$ is nondecreasing, then $\lim_{x\to a_+} f(x) = \alpha$ and $\lim_{x\to b_-} f(x) = \beta$.
2. If $a < x_0 < b$, then $\lim_{x\to x_{0+}} f(x)$ and $\lim_{x\to x_{0-}} f(x)$ exist and are finite; moreover,
$$\lim_{x\to x_{0-}} f(x) \le f(x_0) \le \lim_{x\to x_{0+}} f(x).$$

**Theorem**: If $f$ is monotonic and nonconstant on $[a,b]$, then $f$ is continuous on $[a,b]$ if and only if its range $R_f = \{f(x) : x \in [a,b]\}$ is the closed interval with endpoints $f(a)$ and $f(b)$.

**Proof**: We assume that $f$ is nondecreasing. The theorem implies that the set $\widetilde{R}_f = \{f(x) : x \in (a,b)\}$ is a subset of the open interval $(\lim_{x\to a_+} f(x), \lim_{x\to b_-} f(x))$.

Therefore,

$$R_f = \{f(a)\} \cup \widetilde{R}_f \cup \{f(b)\} \subset \{f(a)\} \cup (\lim_{x\to a_+} f(x), \lim_{x\to b_-} f(x)) \cup \{f(b)\}. \quad (3.16)$$

Now suppose that $f$ is continuous on $[a,b]$. Then $f(a) = \lim_{x\to a_+} f(x)$, $\lim_{x\to b_-} f(x) = f(b)$. So (3.16) implies that $R_f \subset [f(a), f(b)]$.

If $f(a) < \mu < f(b)$, then by Intermediate Value Theorem implies that $\mu = f(x)$ for some $x$ in $(a,b)$. Hence, $R_f = [f(a), f(b)]$.

For the converse, suppose that $R_f = [f(a), f(b)]$. Since $f(a) \le \lim_{x\to a_+} f(x)$ and $\lim_{x\to b_-} f(x) \le f(b)$, (3.16) implies that $f(a) = \lim_{x\to a_+} f(x)$ and $\lim_{x\to b_-} f(x) = f(b)$.

We know that if $f$ is nondecreasing and $a < x_0 < b$, then

$$\lim_{x\to x_0-} f(x) \le f(x_0) \le \lim_{x\to 0_+} f(x).$$

If either of these inequalities is strict, $R_f$ cannot be an interval. Since this contradicts our assumption, $\lim_{x\to x_{0_}} f(x) = f(x_0) = \lim_{x\to x_{0_+}} f(x)$.

Therefore, $f$ is continuous at $x_0$. We can now conclude that $f$ is continuous on $[a,b]$.

**Theorem**: Suppose that $f$ is increasing and continuous on $[a,b]$, and let $f(a) = c$ and $f(b) = d$.

Then there is a unique function $g$ defined on $[c,d]$ such that

$$g(f(x)) = x, \quad a \le x \le b,$$

and

$$f(g(y)) = y, \quad c \le y \le d.$$

Moreover, $g$ is continuous and increasing on $[c,d]$.

**Proof**: **Step I** There is a function $g$ satisfying the above two equations.

**Step II**: Uniqueness of the function $g$.

**Step II**: $g$ is increasing and continuous.

Since $f$ is continuous, then for each $y_0$ in $[c,d]$ there is an $x_0$ in $[a,b]$ such that

$$f(x_0) = y_0,$$

and, since $f$ is increasing, there is only one such $x_0$.

Define

$$g(y_0) = x_0, \tag{3.17}$$

we have

$$f(g(y_0)) = y_0, \qquad g(f(x_0)) = x_0.$$

Since this is true for all $x_0$ and $y_0$, so

$$g(f(x)) = x, \quad a \le x \le b, \quad f(g(y)) = y, \quad c \le y \le d.$$

The uniqueness of $g$ follows from our assumption that $f$ is increasing, and therefore only one value of $x_0$ can satisfy $f(x_0) = y_0$, for each $y_0$.

To see that $g$ is increasing, suppose that $y_1 < y_2$ and let $x_1$ and $x_2$ be the points in $[a, b]$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$.

Since $f$ is increasing, $x_1 < x_2$. Therefore,

$$g(y_1) = x_1 < x_2 = g(y_2),$$

so $g$ is increasing.

Since $R_g = \{g(y) : y \in [c, d]\}$ is the interval $[g(c), g(d)] = [a, b]$, therefore from theorem, we have proved with $f$ and $[a, b]$ replaced by $g$ and $[c, d]$ implies that $g$ is continuous on $[c, d]$.

**Example**: If

$$f(x) = x^2, \quad 0 \le x \le R.$$

**Solution**: The inverse of the given function is

$$f^{-1}(y) = g(y) = \sqrt{y}, \quad 0 \le y \le R^2.$$

**Example**: If

$$f(x) = 2x + 4, \quad 0 \le x \le 2,$$

**Solution**: The inverse of the given function is

$$f^{-1}(y) = g(y) = \frac{y - 4}{2}, \quad 4 \le y \le 8.$$

CHAPTER 4

# Differentiability

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## 4.1 Derivative

**Derivative**: A function $f$ is *differentiable* at an interior point $x_0$ of its domain if the difference quotient

$$\frac{f(x) - f(x_0)}{x - x_0}, \quad x \neq x_0,$$

approaches a limit as $x$ approaches $x_0$, in which case the limit is called the *derivative of $f$ at $x_0$*, and is denoted by $f'(x_0)$.

Thus,

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

If we take $x = x_0 + h$ then

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

- If $f$ is defined on an open set $S$, we say that $f$ is *differentiable on $S$* if $f$ is differentiable at every point of $S$.
- If $f$ is differentiable on $S$, then $f'$ is a function on $S$. We say that $f$ is *continuously differentiable* on $S$ if $f'$ is continuous on $S$.
- If $f$ is differentiable on a neighborhood of $x_0$, it is reasonable to ask if $f'$ is differentiable at $x_0$. If so, we denote the derivative of $f'$ at $x_0$ by $f''(x_0)$.
- This is the *second derivative of $f$ at $x_0$*, and it is also denoted by $f^{(2)}(x_0)$. Continuing inductively, if $f^{(n-1)}$ is defined on a neighborhood of $x_0$, then the $n$th *derivative of $f$ at $x_0$*, denoted by $f^{(n)}(x_0)$, is the derivative of $f^{(n-1)}$ at $x_0$.
- This is the *second derivative of $f$ at $x_0$*, and it is also denoted by $f^{(2)}(x_0)$.
- Continuing inductively, if $f^{(n-1)}$ is defined on a neighborhood of $x_0$, then the $n$th *derivative of $f$ at $x_0$*, denoted by $f^{(n)}(x_0)$, is the derivative of $f^{(n-1)}$ at $x_0$.
- For convenience we define the *zeroth derivative* of $f$ to be $f$ itself; thus

$$f^{(0)} = f.$$

We assume that you are familiar with the other standard notations for derivatives; for example,

$$f^{(2)} = f'', \quad f^{(3)} = f''',$$
$$\frac{d^n f}{dx^n} = f^{(n)}.$$

**Example**: If $n$ is a positive integer, find the derivative of the function

$$f(x) = x^n.$$

**Solution**: We have

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x^n - x_0^n}{x - x_0} = \frac{x - x_0}{x - x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k,$$

$$f'(x_0) = \lim_{x \to x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k = n x_0^{n-1}.$$

Since this holds for every $x_0$, we drop the subscript and write

$$f'(x) = nx^{n-1} \quad \text{or} \quad \frac{d}{dx}(x^n) = nx^{n-1}.$$

**Example**: Find the derivative of the line $y = mx + c$.

**Solution**:

$$\begin{aligned} &\frac{f(x) - f(x_0)}{x - x_0} \\ &= \frac{mx + c - (mx_0 + c)}{x - x_0} \\ &= \frac{m(x - x_0)}{x - x_0} = m \end{aligned}$$

Consequently

$$f'(x_0) = \lim_{x \to x_0} m = m.$$

**Geometrical interpretation of derivative**: The equation of the line through two points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ on the curve $y = f(x)$ is

$$y = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0).$$

Varying $x_1$ generates lines through $(x_0, f(x_0))$ that rotate into the line

$$y = f(x_0) + f'(x_0)(x - x_0)$$

as $x_1$ approaches $x_0$. This is the *tangent* to the curve $y = f(x)$ at the point $(x_0, f(x_0))$.

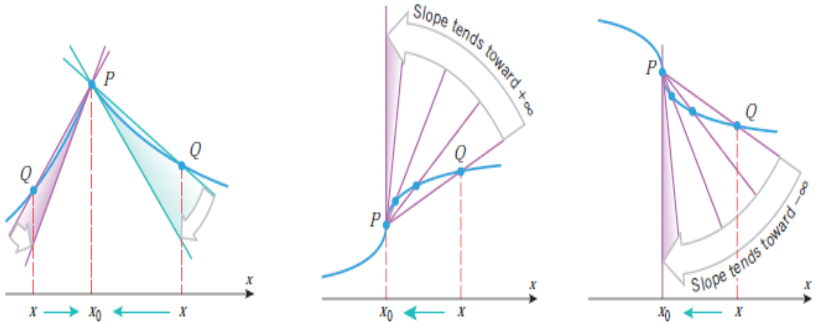



Figure 4.1: The tangent lines

**Lemma**: If $f$ is differentiable at $x_0$, then

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where $E$ is defined on a neighborhood of $x_0$ and

$$\lim_{x \to x_0} E(x) = E(x_0) = 0.$$

**Proof**: Define

$$E(x) = \begin{cases} \frac{f(x)-f(x_0)}{x-x_0} - f'(x_0), & x \in D_f \text{ and } x \neq x_0, \\ 0, & x = x_0. \end{cases}$$

Apply that lemma to $f(x) = x^2$ at $x_0 = 3$.

### 4.1.1 Differentiability Implies Continuity

**Theorem**: If $f$ is differentiable at $x_0$, then $f$ is continuous at $x_0$.

**Proof**:

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where $E$ is defined on a neighborhood of $x_0$ and

$$\lim_{x \to x_0} E(x) = E(x_0) = 0.$$

where

$$E(x) = \begin{cases} \frac{f(x)-f(x_0)}{x-x_0} - f'(x_0), & x \in D_f \text{ and } x \neq x_0, \\ 0, & x = x_0. \end{cases}$$

**Examples**:

- $f(x) = x^2$ has derivative $f'(x) = 4x$.
- $f(x) = \frac{1}{x}$ has derivative $f'(x) = \frac{-1}{x^2}$.
- $f(x) = \sin x$ has derivative $f'(x) = \cos x$.

Is continuity implies differentiability?

**Counter example**: Consider the function

$$f(x) = |x|.$$

The functions can be written as

$$f(x) = \begin{cases} x, & x > 0, \\ -x, & x \leq 0, \end{cases} \quad \Rightarrow \quad f'(x) = \begin{cases} 1, & x > 0, \\ -1, & x \leq 0, \end{cases}$$

$$\lim_{x\to 0+} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0+} \frac{x}{x} \tag{4.1}$$

$$\lim_{x\to 0-} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0-} \frac{-x}{x} = -1 \tag{4.2}$$

are different,

$$\lim_{x\to 0} \frac{f(x)-f(0)}{x-0}$$

does not exist (Theorem **??**); thus, $f$ is not differentiable at 0, even though it is continuous at 0.

**Theorem**: If $f$ and $g$ are differentiable at $x_0$, then so are $f+g$, $f-g$, and $fg$, with

1. $(f+g)'(x_0) = f'(x_0) + g'(x_0)$;
2. $(f-g)'(x_0) = f'(x_0) - g(x_0)$;
3. $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$.

The quotient $f/g$ is differentiable at $x_0$ if $g(x_0) \neq 0$, with

- $\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0)-f(x_0)g'(x_0)}{[g(x_0)]^2}$.

**Proof**: The trick is to add and subtract the right quantity in the numerator of the difference quotient for $(fg)'(x_0)$; thus,

$$\begin{aligned} & \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} \\ = & \frac{f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0} \\ = & \frac{f(x) - f(x_0)}{x - x_0} g(x) + f(x_0) \frac{g(x) - g(x_0)}{x - x_0}. \end{aligned}$$

The difference quotients on the right approach $f'(x_0)$ and $g'(x_0)$ as $x$ approaches $x_0$, and $\lim_{x \to x_0} g(x) = g(x_0)$.

**Example**: Find $\frac{ds}{dt}$ where
$s(t) = (5 - t^2)t^{3/2}$.

**Example**: Find $\frac{dy}{dx}$ where
$y(x) = \frac{x^4 - 5x^2}{x - 5}$.

**Lemma**: If $f$ is differentiable at $x_0$, then

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where $E$ is defined on a neighborhood of $x_0$ and

$$\lim_{x \to x_0} E(x) = E(x_0) = 0.$$

**Theorem** (The chain rule): Suppose that $g$ is differentiable at $x_0$ and $f$ is differentiable at $g(x_0)$.

Then the composite function $h = f \circ g$, defined by

$$h(x) = f(g(x)),$$

is differentiable at $x_0$, with

$$h'(x_0) = f'(g(x_0))g'(x_0).$$

**Proof**: Since $f$ is differentiable at $g(x_0)$, we can write

$$f(t) - f(g(x_0)) = [f'(g(x_0)) + E(t)][t - g(x_0)],$$

where

$$\lim_{t \to g(x_0)} E(t) = E(g(x_0)) = 0. \tag{4.3}$$

Letting $t = g(x)$ yields

$$f(g(x)) - f(g(x_0)) = [f'(g(x_0)) + E(g(x))][g(x) - g(x_0)].$$

Since $h(x) = f(g(x))$, this implies that

$$\frac{h(x) - h(x_0)}{x - x_0} = [f'(g(x_0)) + E(g(x))]\frac{g(x) - g(x_0)}{x - x_0}. \tag{4.4}$$

Since $g$ is continuous at $x_0$ we have

$$\lim_{x \to x_0} E(g(x)) = E(g(x_0)) = 0.$$

Therefore, (4.4) implies that

$$h'(x_0) = \lim_{x \to x_0} \frac{h(x) - h(x_0)}{x - x_0} = f'(g(x_0))g'(x_0),$$

as stated.

**Example**: Calculate the derivative of

$$h(x) = \sin \frac{1}{x}, \quad x \neq 0.$$

If

$$f(x) = \sin x \quad \text{and} \quad g(x) = \frac{1}{x}, \quad x \neq 0,$$

then

$$h(x) = f(g(x)) = \sin \frac{1}{x}, \quad x \neq 0,$$

and

$$h'(x) = f'(g(x))g(x) = \left(\cos \frac{1}{x}\right)\left(-\frac{1}{x^2}\right), \quad x \neq 0.$$

**Example**: What is wrong in the following justification?

$$\begin{aligned} \frac{h(x) - h(x_0)}{x - x_0} &= \frac{f(g(x)) - f(g(x_0))}{x - x_0} \\ &= \frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)} \frac{g(x) - g(x_0)}{x - x_0} \end{aligned}$$

and arguing that

$$\lim_{x \to x_0} \frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)} = f'(g(x_0))$$

(because $\lim_{x \to x_0} g(x) = g(x_0)$) and

$$\lim_{x \to x_0} \frac{g(x) - g(x_0)}{x - x_0} = g'(x_0).$$

Is it a valid proof?

## 4.2 One Sided Derivative

**One sided derivatives**: If $f$ is defined on $[x_0, b)$, the *right-hand derivative of $f$ at* $x_0$ is defined to be

$$f'_+(x_0) = \lim_{x \to x_0+} \frac{f(x) - f(x_0)}{x - x_0}.$$

if the limit exists.

If $f$ is defined on $(a, x_0]$, the *left-hand derivative of $f$ at* $x_0$ is defined to be

$$f'_-(x_0) = \lim_{x \to x_0-} \frac{f(x) - f(x_0)}{x - x_0}$$

if the limit exists.

The definition of limit implies that $f$ is differentiable at $x_0$ if and only if $f'_+(x_0)$ and $f'_-(x_0)$ exist and are equal, in which case

$$f'(x_0) = f'_+(x_0) = f'_-(x_0).$$

The functions defined by $f(x) = |x|$ and $f(x) = \frac{|x|}{x}$ have one sided derivatives.

**Example**: For the piecewise defined function

$$f(x) = \begin{cases} x^3, & x \le 0, \\ x^2 \sin \frac{1}{x}, & x > 0, \end{cases}$$

Investigate the one sided derivatives.

**Solution**: For the given function, we have

$$f'(x) = \begin{cases} 3x^2, & x < 0, \\ 2x \sin \frac{1}{x} - \cos \frac{1}{x}, & x > 0. \end{cases}$$

Since neither formula in $f(x)$ holds for all $x$ in any neighborhood of 0, we cannot simply differentiate either to obtain $f'(0)$.

We will calculate the one sided derivatives

$$\begin{aligned} \lim_{x \to 0+} f(x) &= \lim_{x \to 0+} \frac{x^2 \sin \frac{1}{x} - 0}{x - 0} = \lim_{x \to 0+} x \sin \frac{1}{x} = 0, \\ f'_-(0) &= \lim_{x \to 0-} \frac{x^3 - 0}{x - 0} = \lim_{x \to 0-} x^2 = 0; \end{aligned}$$

hence, $f'(0) = f'_+(0) = f'_-(0) = 0$.

**Remark**: There is a difference between one sided derivatives and one sided limit of derivative

## 4.3 Differentiable Function

**Differentiable function**: A function $f$ is *differentiable on the closed interval* $[a, b]$ if $f$ is differentiable on the open interval $(a, b)$ and $f'_+(a)$ and $f'_-(b)$ both exist.

**Continuously differentiable** We say that $f$ is *continuously differentiable on* $[a, b]$ if $f$ is differentiable on $[a, b]$, $f'$ is continuous on $(a, b)$,

$$f'_+(a) = \lim_{x \to a+} f'(x)$$

, and

$$f'_-(b) = \lim_{x \to b-} f'(x).$$

## 4.4 Extreme Values of a Function

We say that $f(x_0)$ is a *local extreme value* of $f$ if there is a $\delta > 0$ such that $f(x) - f(x_0)$ does not change sign on

$$(x_0 - \delta, x_0 + \delta) \cap D_f.$$

More specifically, $f(x_0)$ is a *local maximum value* of $f$ if

$$f(x) \leq f(x_0)$$

or a *local minimum value* of $f$ if

$$f(x) \geq f(x_0)$$

for all $x$ in the set $(x_0 - \delta, x_0 + \delta) \cap D_f$.

The point $x_0$ is called a *local extreme point* of $f$, or, more specifically, a *local maximum* or *local minimum point* of $f$.

**Example**: For the function

$$f(x) = \begin{cases} 1, & -1 < x \leq -\frac{1}{2} \\ |x|, & -\frac{1}{2} < x \leq \frac{1}{2}, \\ \frac{1}{\sqrt{2}} \sin \frac{\pi x}{2}, & \frac{1}{2} < x \leq 4 \end{cases}$$

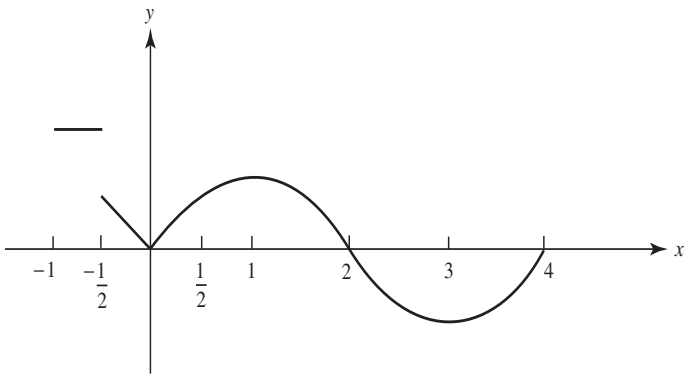



Figure 4.2: Extreme values of a function

**Recall the Lemma**: If $f$ is differentiable at $x_0$, then

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where $E$ is defined on a neighborhood of $x_0$ and

$$\lim_{x \to x_0} E(x) = E(x_0) = 0.$$

**Theorem**: If $f$ is differentiable at a local extreme point $x_0 \in D_f^0$, then $f'(x_0) = 0$.

**Proof**: We will show that $x_0$ is not a local extreme point of $f$ if $f'(x_0) \neq 0$.

From Lemma, we have

$$\frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) + E(x),$$

where $\lim_{x \to x_0} E(x) = 0$.

Therefore, if $f'(x_0) \neq 0$, there is a $\delta > 0$ such that

$$|E(x)| < |f'(x_0)| \quad \text{if} \quad |x - x_0| < \delta, \qquad (*)$$

and the right side of (*) must have the same sign as $f'(x_0)$ for $|x - x_0| < \delta$. Since the same is true of the left side, $f(x) - f(x_0)$ must change sign in every neighborhood of $x_0$ (since $x - x_0$ does).

- If $f'(x_0) = 0$ then $x_0$ is said to be a critical point.

**Recall the following**: If a function $f$ is continuous on the closed interval then $f$ attains its extreme values in the closed interval.

**Theorem**: If $f$ is differentiable at a local extreme point $x_0$, then $f'(x_0) = 0$.

## 4.5 Rolle's Theorem

**Theorem**: Suppose that $f$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval $(a, b)$, and $f(a) = f(b)$. Then $f'(c) = 0$ for some $c$ in the open interval $(a, b)$.

**Proof**: Since $f$ is continuous on $[a, b]$, $f$ attains a maximum and a minimum value on $[a, b]$. If these two extreme values are the same, then $f$ is constant on $(a, b)$, so $f'(x) = 0$ for all $x$ in $(a, b)$.

If the extreme values differ, then at least one must be attained at some point $c$ in the open interval $(a, b)$, and $f'(c) = 0$.

## 4.6 The Mean Value Theorem

**Theorem**: Suppose that $f$ is differentiable on $[a, b]$, $f'(a) \neq f'(b)$, and $\mu$ is between $f'(a)$ and $f'(b)$. Then $f'(c) = \mu$ for some $c$ in $(a, b)$.

**Proof**: Suppose first that

$$f'(a) < \mu < f'(b)$$

and define

$$g(x) = f(x) - \mu x.$$

Then

$$g'(x) = f'(x) - \mu, \quad a \leq x \leq b,$$

and by our supposition $f'(a) < \mu < f'(b)$, we have

$$g'(a) < 0 \quad \text{and} \quad g'(b) > 0.$$

Since $g$ is continuous on $[a, b]$, $g$ attains a minimum at some point $c$ in $[a, b]$.

Due to Lemma and $g'(a) < 0 \quad \text{and} \quad g'(b) > 0$, imply that there is a $\delta > 0$ such that

$$g(x) < g(a), \quad a < x < a + \delta, \quad \text{and} \quad g(x) < g(b), \quad b - \delta < x < b.$$

Therefore $c \neq a$ and $c \neq b$. Hence, $a < c < b$, and therefore $g'(c) = 0$, by the fact that if $f$ is differentiable at a local extreme point $x_0$ then $f'(x_0) = 0$. From , $f'(c) = \mu$. The proof for the case where $f'(b) < \mu < f'(a)$ can be obtained by applying this result to $-f$.

**Theorem**: If $f$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval $(a, b)$, then

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some $c$ in $(a, b)$.

**Proof**: The function

$$h(x) = [b-a]f(x) - [f(b) - f(a)]x$$

is continuous on $[a, b]$ and differentiable on $(a, b)$.

Furthermore

$$h(a) = h(b) = bf(a) - f(b)a.$$

Therefore, Rolle's theorem implies that $h'(c) = 0$ for some $c$ in $(a, b)$.

$$\text{Since} \quad h'(c) = [b-a]f'(c) - [f(b) - f(a)].$$

## 4.7 Generalized Mean Value Theorem

**Theorem**: If $f$ and $g$ are continuous on the closed interval $[a, b]$ and differentiable on the open interval $(a, b)$, then

$$[g(b) - g(a)]f'(c) = [f(b) - f(a)]g'(c)$$

for some $c$ in $(a, b)$.

**Proof**: The function

$$h(x) = [g(b) - g(a)]f(x) - [f(b) - f(a)]g(x)$$

is continuous on $[a, b]$ and differentiable on $(a, b)$.

Furthermore

$$h(a) = h(b) = g(b)f(a) - f(b)g(a).$$

Therefore, Rolle's theorem implies that $h'(c) = 0$ for some $c$ in $(a, b)$.

$$\text{Since} \quad h'(c) = [g(b) - g(a)]f'(c) - [f(b) - f(a)]g'(c).$$

**Theorem**: If $f'(x) = 0$ for all $x$ in $(a, b)$, then $f$ is constant on $(a, b)$.

**Proof**: See Lecture.

**Theorem**: If $f'$ exists and does not change sign on $(a, b)$, then $f$ is monotonic on $(a, b)$ : increasing, nondecreasing, decreasing, or nonincreasing as

$$f'(x) > 0, \quad f'(x) \geq 0, \quad f'(x) < 0, \quad \text{or} \quad f'(x) \leq 0,$$

respectively, for all $x$ in $(a, b)$.

**Proof**: See Lecture.

**Theorem**: If $f$ and $g$ are differentiable on an interval, and if $f'(x) = g'(x)$ for all $x$ in that interval, then $f - g$ is a constant function on the interval.

**Proof**: See Lecture.

## 4.8 Lipschitz Continuity

A function that satisfies inequality

$$|f(x) - f(x')| \leq M|x - x'|, \quad x, x' \in (a, b),$$

for all $x$ and $x'$ in an interval, where $M > 0$ is some real number is said to satisfy a *Lipschitz condition* on the interval.

**Remark**: A differentiable real valued function is Lipschitz continuous, indeed we have

$$\text{differentiability at } x \Rightarrow \text{Lipschits continuity at } x \Rightarrow \text{Continuity at } x$$

**Lipschitz continuity**: Continuity at $x \not\Rightarrow$ Lipschits continuity at $x \not\Rightarrow$ differentiability at $x$

**Example**: The function $f(x) = \sqrt{|x|}$ is continuous at $x = 0$, but not Lipschits continuous at $x = 0$.

The function $f(x) = |x|$ is Lipschits continuous at $x = 0$ but not differentiable at $x = 0$.

**Theorem**: If $f$ and $g$ are continuous on the closed interval $[a, b]$ and differentiable on the open interval $(a, b)$, then

$$[g(b) - g(a)]f'(c)$$

$$= [f(b) - f(a)]g'(c)$$

for some $c$ in $(a, b)$.

**Theorem**: Suppose that $f$ and $g$ are differentiable and $g'$ has no zeros on $(a, b)$.

Let

$$\lim_{x \to b-} f(x) = \lim_{x \to b-} g(x) = 0$$

or

$$\lim_{x \to b-} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \to b-} g(x) = \pm\infty,$$

and suppose that

$$\lim_{x \to b-} \frac{f'(x)}{g'(x)} = L \quad (\text{finite or} \pm\infty).$$

Then

$$\lim_{x \to b-} \frac{f(x)}{g(x)} = L.$$

**Proof**: For $\varepsilon > 0$, due to definition of limit, there is an $x_0$ in $(a, b)$ such that

$$\left| \frac{f'(c)}{g'(c)} - L \right| < \varepsilon \quad \text{if} \quad x_0 < c < b.$$

Generalized mean value theorem implies that if $x$ and $t$ are in $[x_0, b)$, then there is a $c$ between them, and therefore in $(x_0, b)$, such that

$$[g(x) - g(t)]f'(c) = [f(x) - f(t)]g'(c).$$

Since $g'$ has no zeros in $(a, b)$, mean value theorem implies that

$$g(x) - g(t) \neq 0 \quad \text{if} \quad x, t \in (a, b).$$

This means that $g$ cannot have more than one zero in $(a, b)$.

Therefore, we can choose $x_0$ so that, in addition to satisfy

$$\left| \frac{f'(c)}{g'(c)} - L \right| < \varepsilon \quad \text{if} \quad x_0 < c < b.$$

$g$ has no zeros in $[x_0, b)$.

Then we can write

$$\frac{f(x) - f(t)}{g(x) - g(t)} = \frac{f'(c)}{g'(c)},$$

so

$$\left| \frac{f'(c)}{g'(c)} - L \right| < \varepsilon \quad \text{if} \quad x_0 < c < b.$$

implies that

$$\left| \frac{f(x) - f(t)}{g(x) - g(t)} - L \right| < \varepsilon \quad \text{if} \quad x, t \in [x_0, b). \tag{4.5}$$

If $\lim_{x \to b-} f(x) = \lim_{x \to b-} g(x) = 0$ holds, let $x$ be fixed in $[x_0, b)$, and consider the function

$$G(t) = \frac{f(x) - f(t)}{g(x) - g(t)} - L.$$

so

$$\lim_{t \to b-} G(t) = \frac{f(x)}{g(x)} - L.$$

Since

$$|G(t)| < \varepsilon \quad \text{if} \quad x_0 < t < b,$$

we have

$$\left| \frac{f(x)}{g(x)} - L \right| \leq \varepsilon.$$

This holds for all $x$ in $(x_0, b)$, which is the required result.

- For the proof $\lim_{x \to b-} f(x) = \pm\infty \lim_{x \to b-} g(x) = \pm\infty$
- and when $L = \pm\infty$, try yourself.

## 4.9 Indeterminate Forms

The quotient function $f/g$ *is of the form* $0/0$ *as* $x \to b-$ if

$$\lim_{x\to b-} f(x) = \lim_{x\to b-} g(x) = 0,$$

or *of the form* $\infty/\infty$ *as* $x \to b-$ if

$$\lim_{x\to b-} f(x) = \pm\infty$$

and

$$\lim_{x\to b-} g(x) = \pm\infty.$$

The corresponding definitions for $x \to b+$ and $x \to \pm\infty$ are similar.

If $f/g$ is of one of these forms as $x \to b-$ and as $x \to b+$, then we say that it is of that form as $x \to b$.

**Example**: Evaluate the following limit

$$\lim_{x\to 0} \sin x/x.$$

**Solution**: The given limit limit $\lim_{x\to 0} \sin x/x.$ is of the form $0/0$ as $x \to 0$, and L'Hospital's rule yields

$$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\cos x}{1} = 1.$$

**Example**: Evaluate the limit

$$\lim_{x\to -\infty} \frac{e^{-x}}{x}.$$

**Solution**: See Lecture.
**Example**: Evaluate the limit

$$\lim_{x\to +\infty} \frac{x^{-4/3}}{\sin(1/x)}.$$

**Solution**: See Lecture.

**Example**: Using L'Hospital's rule may lead to another indeterminate form; thus,

$$\lim_{x\to\infty} \frac{e^x}{x^2} = \lim_{x\to\infty} \frac{e^x}{2x}$$

if the limit on the right exists in the extended reals. Applying L'Hospital's rule again yields

$$\lim_{x\to\infty}\frac{e^x}{2x}=\lim_{x\to\infty}\frac{e^x}{2}=\infty.$$

Therefore,

$$\lim_{x\to\infty}\frac{e^x}{x^2}=\infty.$$

More generally,

$$\lim_{x\to\infty}\frac{e^x}{x^\alpha}=\infty$$

for any real number $\alpha$

**Example**: Evaluate $\lim_{x\to 0}\frac{4-4\cos x-2\sin^2 x}{x^4}$.

**Solution**:

$$\begin{aligned}\lim_{x\to0}\frac{4-4\cos x-2\sin^2 x}{x^4} &= \lim_{x\to0}\frac{4\sin x-4\sin x\cos x}{4x^3}\\ &= \left(\lim_{x\to0}\frac{\sin x}{x}\right)\left(\lim_{x\to0}\frac{1-\cos x}{x^2}\right)\\ &= \left(\lim_{x\to0}\frac{\sin x}{x}\right)\left(\lim_{x\to0}\frac{\sin x}{2x}\right)\\ &= \frac{1}{2}\left(\lim_{x\to0}\frac{\sin x}{x}\right)^2\\ &= \frac{1}{2}(1)^2=\frac{1}{2}.\end{aligned}$$

**Example**: If

$$f(x)=x-x^2\sin\frac{1}{x}\quad\text{and}\quad g(x)=\sin x,$$

then

$$f'(x)=1-2x\sin\frac{1}{x}+\cos\frac{1}{x}\quad\text{and}\quad g'(x)=\cos x.$$

Therefore, $\lim_{x\to0} f'(x)/g'(x)$ does not exist. However,

$$\lim_{x\to0}\frac{f(x)}{g(x)}=\lim_{x\to0}\frac{1-x\sin(1/x)}{(\sin x)/x}=\frac{1}{1}=1.$$

## 4.10 Indeterminate forms $(0)(\infty)$

A product of functions $fg$ *is said to have the indeterminate form* $0 \cdot \infty$ *as* $x \to b-$ if one of the factors approaches 0 and the other approaches $\pm\infty$ as $x \to b-$.

In this case, it may be useful to apply L'Hospital's rule after writing

$$f(x)g(x) = \frac{f(x)}{1/g(x)} \quad \text{or} \quad f(x)g(x) = \frac{g(x)}{1/f(x)},$$

since one of these ratios is of the form $0/0$ and the other is of the form $\infty/\infty$ as $x \to b-$. Similar statements apply to limits as $x \to b+$, $x \to b$, and $x \to \pm\infty$.

**Example**: Evaluate the limit

$$\lim_{x\to 0+} x \log x.$$

**Solution**: The product $x \log x$ is of the form $0 \cdot \infty$ as $x \to 0+$. Converting it to an $\infty/\infty$ form yields

$$\begin{aligned} \lim_{x\to 0+} x \log x &= \lim_{x\to 0+} \frac{\log x}{1/x} \\ &= \lim_{x\to 0+} \frac{1/x}{-1/x^2} \\ &= -\lim_{x\to 0+} x = 0. \end{aligned}$$

**Example**: Evaluate the limit

$$\lim_{x\to\infty} x \log 1 + 1/x.$$

**Solution**: Converting it to a $0/0$ form yields

$$\begin{aligned} \lim_{x\to\infty} x \log(1 + 1/x) &= \lim_{x\to\infty} \frac{\log(1 + 1/x)}{1/x} \\ &= \lim_{x\to\infty} \frac{[1/(1 + 1/x)]\,(-1/x^2)}{-1/x^2} \\ &= \lim_{x\to\infty} \frac{1}{1 + 1/x} = 1. \end{aligned}$$

In this case, converting to an $\infty/\infty$ form complicates the problem:

$$\begin{aligned} \lim_{x\to\infty} x \log(1 + 1/x) &= \lim_{x\to\infty} \frac{x}{1/\log(1 + 1/x)} \\ &= \lim_{x\to\infty} \frac{1}{\left(\frac{-1}{[\log(1+1/x)]^2}\right)\left(\frac{-1/x^2}{1+1/x}\right)} \\ &= \lim_{x\to\infty} x(x + 1)[\log(1 + 1/x)]^2 = ? \end{aligned}$$

### 4.10.1 Indeterminate form $\infty - \infty$

A difference $f - g$ *is of the form* $\infty - \infty$ *as* $x \to b-$ if

$$\lim_{x \to b-} f(x) = \lim_{x \to b-} g(x) = \pm\infty.$$

In this case, it may be possible to manipulate $f - g$ into an expression that is no longer indeterminate, or is of the form $0/0$ or $\infty/\infty$ as $x \to b-$.

Similar remarks apply to limits as $x \to b+$, $x \to b$, or $x \to \pm\infty$.

**Example**: Evaluate the limit

$$\lim_{x \to 0} \frac{\sin x}{x^2} - \frac{1}{x}.$$

**Solution**: See Lecture.

**Example**: Evaluate the limit

$$\lim_{x \to \infty} x^2 - x.$$

**Solution**: See Lecture.

### 4.10.2 Indeterminate forms $0^0, 1^\infty, \infty^0$

The function $f^g$ is defined by

$$f(x)^{g(x)} = e^{g(x) \log f(x)} = \exp(g(x) \log f(x))$$

for all $x$ such that $f(x) > 0$.

Therefore, if $f$ and $g$ are defined and $f(x) > 0$ on an interval $(a, b)$, implies that

$$\lim_{x \to b-} [f(x)]^{g(x)} = \exp\left( \lim_{x \to b-} g(x) \log f(x) \right) \tag{4.6}$$

if $\lim_{x \to b-} g(x) \log f(x)$ exists in the extended real number system.

If this limit is $\pm\infty$ then (4.6) is valid if we define $e^{-\infty} = 0$ and $e^{\infty} = \infty$.

**Indeterminate forms** $0^0, 1^\infty, \infty^0$: The product $g \log f$ can be of the form $0 \cdot \infty$ in three ways as $x \to b-$:

1. If $\lim_{x \to b-} g(x) = 0$ and $\lim_{x \to b-} f(x) = 0$.
2. If $\lim_{x \to b-} g(x) = \pm\infty$ and $\lim_{x \to b-} f(x) = 1$.
3. If $\lim_{x \to b-} g(x) = 0$ and $\lim_{x \to b-} f(x) = \infty$.

In these three cases, we say that $f^g$ *is of the form* $0^0$, $1^\infty$, *and* $\infty^0$, *respectively, as* $x \to b-$.

Similar definitions apply to limits as $x \to b+$, $x \to b$, and $x \to \pm\infty$.

**Example**: Evaluate the following limit

$$\lim_{x \to 0+} x^x.$$

**Solution**: See Lecture.

**Example**: Evaluate the following limit

$$\lim_{x \to 1} x^{1/(x-1)}.$$

**Solution**:

Since

$$x^{1/(x-1)} = \exp\left(\frac{\log x}{x-1}\right)$$

and

$$\lim_{x \to 1} \frac{\log x}{x-1} = \lim_{x \to 1} \frac{1/x}{1} = 1,$$

it follows that

$$\lim_{x \to 1} x^{1/(x-1)} = e^1 = e.$$

**Example**: Evaluate the following limit

$$\lim_{x \to \infty} x^{1/x}.$$

**Solution**:

Since

$$x^{1/x} = \exp\left(\frac{\log x}{x}\right)$$

and

$$\lim_{x \to \infty} \frac{\log x}{x} = \lim_{x \to \infty} \frac{1/x}{1} = 0,$$

## 4.11 Taylor's Theorem

**Recall the following lemma**: If $f$ is differentiable at $x_0$, then

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where $E$ is defined on a neighborhood of $x_0$ and

$$\lim_{x \to x_0} E(x) = E(x_0) = 0.$$

To generalize this result, we first restate it: the polynomial

$$T_1(x) = f(x_0) + f'(x_0)(x - x_0),$$

which is of degree $\leq 1$ and satisfies

$$T_1(x_0) = f(x_0), \quad T_1'(x_0) = f'(x_0).$$

$T_1(x)$ approximates $f$ so well near $x_0$ that

$$\lim_{x \to x_0} \frac{f(x) - T_1(x)}{x - x_0} = 0.$$

**Taylor polynomial**: Suppose that $f$ has $n$ derivatives at $x_0$ and $T_n$ is the polynomial of degree $\leq n$ such that

$$T_n^{(r)}(x_0) = f^{(r)}(x_0), \quad 0 \leq r \leq n.$$

How well does $T_n$ approximate $f$ near $x_0$?

Since $T_n$ is a polynomial of degree $\leq n$, it can be written as

$$T_n(x) = a_0 + a_1(x - x_0) + \cdots + a_n(x - x_0)^n,$$

where $a_0, \ldots, a_n$ are constants.

Differentiating yields

$$T_n^{(r)}(x_0) = r! a_r, \quad 0 \leq r \leq n,$$

We obtained $a_r$ uniquely as

$$a_r = \frac{f^{(r)}(x_0)}{r!}, \quad 0 \leq r \leq n.$$

Therefore,

$$\begin{aligned} T_n(x) &= f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \\ &= \sum_{r=0}^{n} \frac{f^{(r)}(x_0)}{r!}(x - x_0)^r. \end{aligned}$$

We call $T_n$ *the nthTaylor polynomial of* $f$ *about* $x_0$.

**Theorem**: If $f^{(n)}(x_0)$ exists for some integer $n \geq 1$ and $T_n$ is the $n$th Taylor polynomial of $f$ about $x_0$, then

$$\lim_{x \to x_0} \frac{f(x) - T_n(x)}{(x - x_0)^n} = 0.$$

**Proof**: The proof is by induction. Let $P_n$ be the assertion of the theorem. If $n = 1$; that is, $P_1$ is true. Suppose that $P_n$ is true for some integer $n \geq 1$, and $f^{(n+1)}$ exists.

Since the ratio

$$\frac{f(x) - T_{n+1}(x)}{(x - x_0)^{n+1}}$$

is indeterminate of the form $0/0$ as $x \to x_0$, L'Hospital's rule implies that

$$\lim_{x \to x_0} \frac{f(x) - T_{n+1}(x)}{(x - x_0)^{n+1}} = \frac{1}{n+1} \lim_{x \to x_0} \frac{f'(x) - T'_{n+1}(x)}{(x - x_0)^n} \tag{4.7}$$

if the limit on the right exists.

But $f'$ has an $n$th derivative at $x_0$, and

$$T'_{n+1}(x) = \sum_{r=0}^{n} \frac{f^{(r+1)}(x_0)}{r!}(x - x_0)^r$$

is the $n$th Taylor polynomial of $f'$ about $x_0$.

Therefore, the induction assumption, applied to $f'$, implies that

$$\lim_{x \to x_0} \frac{f'(x) - T'_{n+1}(x)}{(x - x_0)^n} = 0.$$

This and (4.7) imply that

$$\lim_{x \to x_0} \frac{f(x) - T_{n+1}(x)}{(x - x_0)^{n+1}} = 0,$$

which completes the induction.

**Lemma**: If $f^{(n)}(x_0)$ exists, then

$$f(x) = \sum_{r=0}^{n} \frac{f^{(r)}(x_0)}{r!}(x - x_0)^r + E_n(x)(x - x_0)^n, \tag{4.8}$$

where

$$\lim_{x \to x_0} E_n(x) = E_n(x_0) = 0.$$

**Proof**: Define

$$E_n(x) = \begin{cases} \frac{f(x)-T_n(x)}{(x-x_0)^n}, & x \in D_f - \{x_0\}, \\ 0, & x = x_0. \end{cases}$$

Then we know that $\lim_{x\to x_0} E_n(x) = E_n(x_0) = 0$, and it is straightforward to verify (4.8).

**Example**: If $f(x) = e^x$, then $f^{(n)}(x) = e^x$.

Therefore, $f^{(n)}(0) = 1$ for $n \geq 0$, so the $n$th Taylor polynomial of $f$ about $x_0 = 0$ is

$$T_n(x) = \sum_{r=0}^{n} \frac{x^r}{r!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}.$$

According to approximation Theorem we have

$$\lim_{x\to 0} \frac{e^x - \sum_{r=0}^{n} \frac{x^r}{r!}}{x^n} = 0.$$

**Example**: If $f(x) = \log x$, then $f(1) = 0$ and

$$f^{(r)}(x) = (-1)^{(r-1)} \frac{(r-1)!}{x^r}, \quad r \geq 1.$$

so the $n$th Taylor polynomial of $f$ about $x_0 = 1$ is

$$T_n(x) = \sum_{r=1}^{n} \frac{(-1)^{r-1}}{r} (x-1)^r$$

if $n \geq 1$. ($T_0 = 0$.)

According to approximation Theorem we have

$$\lim_{x\to 1} \frac{\log x - \sum_{r=1}^{n} (-1)^{r-1} r(x-1)^r}{(x-1)^n} = 0, \quad n \geq 1.$$

**Example**: If $f(x) = (1+x)^q$, then

$$\begin{aligned} f'(x) &= q(1+x)^{q-1}, \quad f''(x) = q(q-1)(1+x)^{q-2} \\ &\vdots \\ f^{(n)}(x) &= q(q-1)\cdots(q-n+1)(1+x)^{q-n}. \end{aligned}$$

If we define

$$\binom{q}{0} = 1 \quad \text{and} \quad \binom{q}{n} = \frac{q(q-1)\cdots(q-n+1)}{n!}, \quad n \geq 1,$$

then

$$\frac{f^{(n)}(0)}{n!} = \binom{q}{n},$$

and the $n$th Taylor polynomial of $f$ about 0 can be written as

$$T_n(x) = \sum_{r=0}^{n} \binom{q}{r} x^r.$$

According to approximation Theorem we have

$$\lim_{x\to 0} \frac{(1+x)^q - \sum_{r=0}^{n} \binom{q}{r} x^r}{x^n} = 0,$$

for $n \geq 0$.

**Theorem**: Suppose that $f$ has $n$ derivatives at $x_0$ and $n$ is the smallest positive integer such that $f^{(n)}(x_0) \neq 0$.

1. If $n$ is odd, $x_0$ is not a local extreme point of $f$.
2. If $n$ is even, $x_0$ is a local maximum of $f$ if $f^{(n)}(x_0) < 0$, or a local mininum of $f$ if $f^{(n)}(x_0) > 0$.

**Proof**: Since $f^{(r)}(x_0) = 0$ for $1 \leq r \leq n-1$, we have

$$f(x) - f(x_0) = \left[\frac{f^{(n)}(x_0)}{n!} + E_n(x)\right](x - x_0)^n,$$

in some interval containing $x_0$.

Since $\lim_{x\to x_0} E_n(x) = 0$ and $f^{(n)}(x_0) \neq 0$, there is a $\delta > 0$ such that

$$|E_n(x)| < \left|\frac{f^{(n)}(x_0)}{n!}\right| \quad \text{if} \quad |x - x_0| < \delta.$$

We can conclude that

$$\frac{f(x) - f(x_0)}{(x - x_0)^n}$$

has the same sign as $f^{(n)}(x_0)$ if $0 < |x - x_0| < \delta$.

If $n$ is odd the denominator changes sign in every neighborhood of $x_0$, and therefore so must the numerator (since the ratio has constant sign for $0 < |x - x_0| < \delta$).

Consequently, $f(x_0)$ cannot be a local extreme value of $f$. This proves (1).

If $n$ is even, the denominator is positive for $x \neq x_0$, so $f(x) - f(x_0)$ must have the same sign as $f^{(n)}(x_0)$ for $0 < |x - x_0| < \delta$. This proves part (2).

**Example**: Investigate the local extrema for the function $f(x) = e^{x^3}$.

**Solution**: For the given function we have $f'(x) = 3x^2e^{x^3}$, and 0 is the only critical point of $f$.

Since

$$\begin{aligned} f''(x) &= (6x + 9x^4)e^{x^3} \\ \text{and} & \\ f'''(x) &= (6 + 54x^3 + 27x^6)e^{x^3}, \end{aligned}$$

$f''(0) = 0$ and $f'''(0) \neq 0$.

Therefore, 0 is not a local extreme point of $f$. Since $f$ is differentiable everywhere, it has no local maxima or minima.

**Example**: Investigate the local extrema for the function $f(x) = \sin x^2$.

**Solution**: For the given function we have $f'(x) = 2x \cos x^2$, so the critical points of $f$ are 0 and $\pm\sqrt{(k + 1/2)\pi}$, $k = 0, 1, 2, \ldots$.

Since

$$f''(x) = 2\cos x^2 - 4x^2 \sin x^2,$$

$$f''(0) = 2 \quad \text{and} \quad f''\left(\pm\sqrt{(k + 1/2)\pi)}\right) = (-1)^{k+1}(4k + 2)\pi.$$

Therefore, $f$ attains local minima at 0 and $\pm\sqrt{(k + 1/2)\pi}$ for odd integers $k$, and local maxima at $\pm\sqrt{(k + 1/2)\pi}$ for even integers $k$.

**Taylor's Theorem**: Suppose that $f^{(n+1)}$ exists on an open interval $I$ about $x_0$, and let $x$ be in $I$.

Then the remainder

$$R_n(x) = f(x) - T_n(x)$$

can be written as

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n + 1)!}(x - x_0)^{n+1},$$

where $c$ depends upon $x$ and is between $x$ and $x_0$.

**Example**: If $f(x) = e^x$, then $f'''(x) = e^x$, and Taylor's theorem with $n = 2$ implies that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{e^c x^3}{3!},$$

where $c$ is between 0 and $x$. We have

$$e^{0.1} \approx T_2(0.1) = 1 + \frac{0.1}{1!} + \frac{(0.1)^2}{2!} = 1.105$$

hence

$$e^{0.1} = 1.105 + \frac{e^c(0.1)^3}{6},$$

where $0 < c < 0.1$.

Since $0 < e^c < e^{0.1}$, we know from this that

$$1.105 < e^{0.1} < 1.105 + \frac{e^{0.1}(0.1)^3}{6}.$$

**Example**: Since $0 < e^c < e^{0.1}$, we know from this that

$$1.105 < e^{0.1} < 1.105 + \frac{e^{0.1}(0.1)^3}{6}.$$

The second inequality implies that

$$e^{0.1}\left[1 - \frac{(0.1)^3}{6}\right] < 1.105,$$

so

$$e^{0.1} < 1.1052.$$

Therefore,

$$1.105 < e^{0.1} < 1.1052,$$

and the error in approximation is less than 0.0002.

CHAPTER 5

# Riemann Integration

## 5.1 Riemann Sums

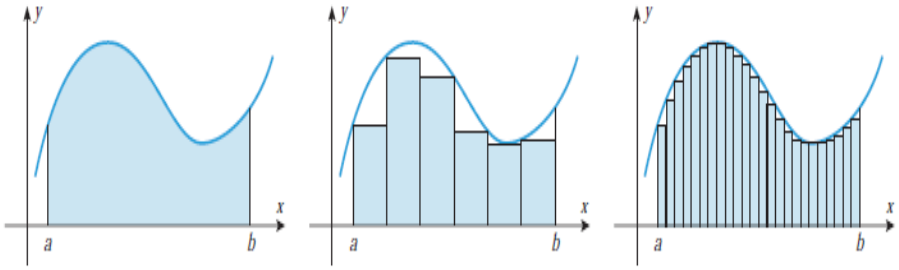



Figure 5.1: Area under the curve

**Partition of an interval**: Let $f$ be a real valued function defined on a finite interval $[a, b]$.

A *partition of* $[a, b]$ is a set of subintervals

$$[x_0, x_1],\ [x_1, x_2], \ldots, [x_{n-1}, x_n],$$

where

$$a = x_0 < x_1 \cdots < x_n = b.$$

Thus, any set of $n + 1$ points satisfying $a = x_0 < x_1 \cdots < x_n = b.$ defines a partition $P$ of $[a, b]$, which we denote by

$$P = \{x_0, x_1, \ldots, x_n\}.$$

The points $x_0$, $x_1$, ..., $x_n$ are the *partition points* of $P$.

The largest of the lengths of the subintervals $[x_0, x_1],\ [x_1, x_2], \ldots, [x_{n-1}, x_n]$, is the *norm* of $P$, written as $\|P\|$; thus,

$$\|P\| = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

**Refinement of a partition**: If $P$ and $P'$ are partitions of $[a, b]$, then $P'$ is a *refinement of* $P$ if every partition point of $P$ is also a partition point of $P'$; that is, if $P'$ is obtained by inserting additional points between those of $P$.

**Riemann sums**: If $f$ is defined on $[a, b]$, then a sum

$$\sigma = \sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}),$$

where

$$x_{j-1} \le c_j \le x_j, \quad 1 \le j \le n,$$

is a *Riemann sum of $f$ over the partition $P = \{x_0, x_1, \ldots, x_n\}$*. (Occasionally we will say more simply that $\sigma$ is a Riemann sum of $f$ over $[a, b]$.)

Since $c_j$ can be chosen arbitrarily in $[x_j, x_{j-1}]$, there are infinitely many Riemann sums for a given function $f$ over a given partition $P$.

## 5.2 Riemann Integral

Let $f$ be defined on $[a, b]$. We say that $f$ is *Riemann integrable on* $[a, b]$ if there is a number $L$ with the following property:

For every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|\sigma - L| < \varepsilon.$$

If $\sigma$ is any Riemann sum of $f$ over a partition $P$ of $[a, b]$ such that $\|P\| < \delta$. In this case, we say that $L$ is *the Riemann integral of $f$ over* $[a, b]$, and write

$$\int_a^b f(x)\, dx = L.$$

**Theorem**: Prove that the Riemann integral $\int_a^b f(x)dx$, if it exists, is unique.

**Proof**: See Lecture.

- Prove: If $\int_a^b f(x)\, dx$ exists, then for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|\sigma_1 - \sigma_2| < \varepsilon$ if $\sigma_1$ and $\sigma_2$ are Riemann sums of $f$ over partitions $P_1$ and $P_2$ of $[a, b]$ with norms less than $\delta$.

**Example**: Determine $f$ is Riemann integrable over the given interval or not? Find $\int_a^b f(x)dx$, where

$$f(x) = 1, \quad a \le x \le b.$$

**Solution**: For the given function, we have

$$\sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}) = \sum_{j=1}^{n} (x_j - x_{j-1}).$$

Most of the terms in the sum on the right cancel in pairs; that is,

$$\begin{aligned} & \sum_{j=1}^{n}(x_j - x_{j-1}) = (x_1 - x_0) + (x_2 - x_1) + \cdots + (x_n - x_{n-1}) \\ = \;& -x_0 + (x_1 - x_1) + (x_2 - x_2) + \cdots + (x_{n-1} - x_{n-1}) + x_n \\ = \;& x_n - x_0 = b - a. \end{aligned}$$

Thus, every Riemann sum of $f$ over any partition of $[a, b]$ equals $b - a$, so

$$\int_a^b dx = b - a.$$

**Example**: For the function

$$f(x) = x, \quad a \le x \le b,$$

Riemann sums are of the form

$$\sigma = \sum_{j=1}^{n} c_j(x_j - x_{j-1}). \tag{5.1}$$

Since $x_{j-1} \le c_j \le x_j$ and $(x_j + x_{j-1})/2$ is the midpoint of $[x_{j-1}, x_j]$, we can write

$$c_j = \frac{x_j + x_{j-1}}{2} + d_j, \tag{5.2}$$

where

$$|d_j| \le \frac{x_j - x_{j-1}}{2} \le \frac{\|P\|}{2}. \tag{5.3}$$

**Example**: Substituting (5.2) into (5.1) yields

$$\begin{aligned} \sigma &= \textstyle\sum_{j=1}^{n} \frac{x_j + x_{j-1}}{2}(x_j - x_{j-1}) + \sum_{j=1}^{n} d_j(x_j - x_{j-1}) \\ &= \tfrac{1}{2}\textstyle\sum_{j=1}^{n}(x_j^2 - x_{j-1}^2) + \sum_{j=1}^{n} d_j(x_j - x_{j-1}). \end{aligned} \tag{5.4}$$

Because of cancelations, we have

$$\sum_{j=1}^{n}(x_j^2 - x_{j-1}^2) = b^2 - a^2,$$

so (5.4) can be rewritten as

$$\sigma = \frac{b^2 - a^2}{2} + \sum_{j=1}^{n} d_j(x_j - x_{j-1}).$$

Hence,

$$\begin{aligned}\left|\sigma - \frac{b^2-a^2}{2}\right| &\leq \sum_{j=1}^{n} |d_j|(x_j - x_{j-1}) \leq \frac{\|P\|}{2}\sum_{j=1}^{n}(x_j - x_{j-1}) \\ &= \frac{\|P\|}{2}(b-a).\end{aligned}$$

Therefore, every Riemann sum of $f$ over a partition $P$ of $[a,b]$ satisfies

$$\left|\sigma - \frac{b^2-a^2}{2}\right| < \varepsilon \quad \text{if} \quad \|P\| < \delta = \frac{2\varepsilon}{b-a}.$$

Hence,

$$\int_a^b x\,dx = \frac{b^2-a^2}{2}.$$

**Theorem**: If $f$ is unbounded on $[a,b]$, then $f$ is not integrable on $[a,b]$.

**Proof**: We will show that if $f$ is unbounded on $[a,b]$, $P$ is any partition of $[a,b]$, and $M > 0$, then there are Riemann sums $\sigma$ and $\sigma'$ of $f$ over $P$ such that

$$|\sigma - \sigma'| \geq M. \tag{5.5}$$

Let

$$\sigma = \sum_{j=1}^{n} f(c_j)(x_j - x_{j-1})$$

be a Riemann sum of $f$ over a partition $P$ of $[a,b]$.

There must be an integer $i$ in $\{1,2,\ldots,n\}$ such that

$$|f(c) - f(c_i)| \geq \frac{M}{x_i - x_{i-1}}$$

for some $c$ in $[x_{i-1}x_i]$.

Because if there were not so, we would have

$$|f(x) - f(c_j)| < \frac{M}{x_j - x_{j-1}}, \quad x_{j-1} \leq x \leq x_j, \quad 1 \leq j \leq n.$$

Then

$$\begin{aligned}|f(x)| &= |f(c_j) + f(x) - f(c_j)| \leq |f(c_j)| + |f(x) - f(c_j)| \\ &\leq |f(c_j)| + \frac{M}{x_j - x_{j-1}}, \quad x_{j-1} \leq x \leq x_j, \quad 1 \leq j \leq n.\end{aligned}$$

which implies that

$$|f(x)| \leq \max_{1\leq j\leq n} |f(c_j)| + \frac{M}{x_j - x_{j-1}}, \quad a \leq x \leq b,$$

contradicting the assumption that $f$ is unbounded on $[a, b]$.

Consider the Riemann sum

$$\sigma' = \sum_{j=1}^{n} f(c_j')(x_j - x_{j-1})$$

over the same partition $P$, where

$$c_j' = \begin{cases} c_j, & j \neq i, \\ c, & j = i. \end{cases}$$

Since

$$|\sigma - \sigma'| = |f(c) - f(c_i)|(x_i - x_{i-1}),$$

hence due to

$$|f(c) - f(c_i)| \geq \frac{M}{x_i - x_{i-1}}$$

implies (5.5).

## 5.3 Upper and Lower Integrals

**Upper and lower integrals**: If $f$ is bounded on $[a, b]$ and $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a, b]$, let

$$M_j = \sup_{x_{j-1} \leq x \leq x_j} f(x) \quad \text{and} \quad m_j = \inf_{x_{j-1} \leq x \leq x_j} f(x).$$

The *upper sum of $f$ over $P$* is

$$S(P) = \sum_{j=1}^{n} M_j(x_j - x_{j-1}),$$

and the *upper integral of $f$ over*, $[a, b]$, denoted by

$$\overline{\int_a^b} f(x)\, dx,$$

is the infimum of all upper sums.

**Lower integrals**: The *lower sum of $f$ over $P$* is

$$s(P) = \sum_{j=1}^{n} m_j(x_j - x_{j-1}),$$

and the *lower integral of $f$ over* $[a, b]$, denoted by

$$\underline{\int_a^b} f(x)\, dx,$$

is the supremum of all lower sums.

**Remark**: If $m \le f(x) \le M$ for all $x$ in $[a,b]$, then

$$m(b-a) \le s(P) \le S(P) \le M(b-a)$$

for every partition $P$. Thus, the set of upper sums of $f$ over all partitions $P$ of $[a,b]$ is bounded, as is the set of lower sums.

Therefore, definition of infimum and supremum imply that $\overline{\int_a^b} f(x)\,dx$ and $\underline{\int_a^b} f(x)\,dx$ exist, are unique, and satisfy the inequalities

$$m(b-a) \le \overline{\int_a^b} f(x)\,dx \le M(b-a)$$

and

$$m(b-a) \le \underline{\int_a^b} f(x)\,dx \le M(b-a).$$

**Theorem**: Let $f$ be bounded on $[a,b]$, and let $P$ be a partition of $[a,b]$. Then

1. The upper sum $S(P)$ of $f$ over $P$ is the supremum of the set of all Riemann sums of $f$ over $P$.
2. The lower sum $s(P)$ of $f$ over $P$ is the infimum of the set of all Riemann sums of $f$ over $P$.

**Proof**: If $P = \{x_0, x_1, \ldots, x_n\}$, then

$$S(P) = \sum_{j=1}^{n} M_j(x_j - x_{j-1}),$$

where

$$M_j = \sup_{x_{j-1} \le x \le x_j} f(x).$$

An arbitrary Riemann sum of $f$ over $P$ is of the form

$$\sigma = \sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}),$$

where $x_{j-1} \le c_j \le x_j$.

Since $f(c_j) \le M_j$, it follows that $\sigma \le S(P)$. Now let $\varepsilon > 0$ and choose $\bar{c}_j$ in $[x_{j-1}, x_j]$ so that

$$f(\bar{c}_j) > M_j - \frac{\varepsilon}{n(x_j - x_{j-1})}, \quad 1 \le j \le n.$$

The Riemann sum produced in this way is

$$\begin{aligned}\overline{\sigma} &= \sum_{j=1}^{n} f(\overline{c}_j)(x_j - x_{j-1}) > \sum_{j=1}^{n} \left[ M_j - \frac{\varepsilon}{n(x_j - x_{j-1})}) \right] (x_j - x_{j-1}) \\ &= S(P) - \varepsilon.\end{aligned}$$

Now from definition $S(P)$ is the supremum of the set of Riemann sums of $f$ over $P$.

**Example**: Let

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ 1 & \text{if } x \text{ is rational,} \end{cases}$$

Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. Since every interval contains both rational and irrational numbers

$$m_j = 0 \quad \text{and} \quad M_j = 1, \quad 1 \le j \le n.$$

Hence,

$$\begin{aligned}S(P) &= \sum_{j=1}^{n} 1 \cdot (x_j - x_{j-1}) = b - a \\ s(P) &= \sum_{j=1}^{n} 0 \cdot (x_j - x_{j-1}) = 0.\end{aligned}$$

Since all upper sums equal $b - a$ and all lower sums equal 0, from definition we have

$$\overline{\int_a^b} f(x)\,dx = b - a \quad \text{and} \quad \underline{\int_a^b} f(x)\,dx = 0.$$

**Example**: Let $f$ be defined on $[1, 2]$ by $f(x) = 0$ if $x$ is irrational and $f(p/q) = 1/q$ if $p$ and $q$ are positive integers with no common factors.

If $P = \{x_0, x_1, \ldots, x_n\}$ is any partition of $[1, 2]$, then $m_j = 0$, $1 \le j \le n$, so $s(P) = 0$; hence,

$$\underline{\int_1^2} f(x)\,dx = 0.$$

We now show that

$$\overline{\int_1^2} f(x)\,dx = 0 \tag{5.6}$$

also.

**Example**: Since $S(P) > 0$ for every $P$, from definition we have

$$\overline{\int_1^2} f(x)\,dx \geq 0.$$

We need only show that

$$\overline{\int_1^2} f(x)\,dx \leq 0,$$

which will follow if we show that no positive number is less than every upper sum.

To this end, we observe that if $0 < \varepsilon < 2$, then $f(x) \geq \varepsilon/2$ for only finitely many values of $x$ in $[1, 2]$.

Let $k$ be the number of such points and let $P_0$ be a partition of $[1, 2]$ such that

$$\|P_0\| < \frac{\varepsilon}{2k}. \tag{5.7}$$

**Example**: Consider the upper sum

$$S(P_0) = \sum_{j=1}^{n} M_j(x_j - x_{j-1}).$$

There are at most $k$ values of $j$ in this sum for which $M_j \geq \varepsilon/2$, and $M_j \leq 1$ even for these. The contribution of these terms to the sum is less than $k(\varepsilon/2k) = \varepsilon/2$, because of (5.7).

Since $M_j < \varepsilon/2$ for all other values of $j$, the sum of the other terms is less than

$$\frac{\varepsilon}{2}\sum_{j=1}^{n}(x_j - x_{j-1}) = \frac{\varepsilon}{2}(x_n - x_0) = \frac{\varepsilon}{2}(2-1) = \frac{\varepsilon}{2}.$$

Therefore, $S(P_0) < \varepsilon$ and, since $\varepsilon$ can be chosen as small as we wish, no positive number is less than all upper sums.

**Lemma**: Suppose that

$$|f(x)| \leq M, \quad a \leq x \leq b,$$

and let $P'$ be a partition of $[a, b]$ obtained by adding $r$ points to a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$.

Then

$$\begin{aligned} S(P) \geq S(P') &\geq S(P) - 2Mr\|P\| \\ &\text{and} \\ s(P) \leq s(P') &\leq s(P) + 2Mr\|P\|. \end{aligned}$$

**Proof**: First suppose that $r = 1$, so $P'$ is obtained by adding one point $c$ to the partition $P = \{x_0, x_1, \ldots, x_n\}$; then $x_{i-1} < c < x_i$ for some $i$ in $\{1, 2, \ldots, n\}$.

If $j \neq i$, the product $M_j(x_j - x_{j-1})$ appears in both $S(P)$ and $S(P')$ and cancels out of the difference $S(P) - S(P')$.

Therefore, if

$$M_{i1} = \sup_{x_{i-1} \le x \le c} f(x) \quad \text{and} \quad M_{i2} = \sup_{c \le x \le x_i} f(x),$$

then

$$\begin{aligned} S(P) - S(P') &= M_i(x_i - x_{i-1}) - M_{i1}(c - x_{i-1}) - M_{i2}(x_i - c) \\ &= (M_i - M_{i1})(c - x_{i-1}) + (M_i - M_{i2})(x_i - c). \end{aligned} \tag{5.8}$$

Since $f$ is bounded implies that

$$0 \le M_i - M_{ir} \le 2M, \quad r = 1, 2,$$

(5.8) implies that

$$0 \le S(P) - S(P') \le 2M(x_i - x_{i-1}) \le 2M\|P\|.$$

This proves (5.11) for $r = 1$.

Now suppose that $r > 1$ and $P'$ is obtained by adding points $c_1$, $c_2$, ..., $c_r$ to $P$.

Let $P^{(0)} = P$ and, for $j \ge 1$, let $P^{(j)}$ be the partition of $[a, b]$ obtained by adding $c_j$ to $P^{(j-1)}$. Then the result just proved implies that

$$0 \le S(P^{(j-1)}) - S(P^{(j)}) \le 2M\|P^{(j-1)}\|, \quad 1 \le j \le r.$$

Adding these inequalities and taking account of cancellations yields

$$0 \le S(P^{(0)}) - S(P^{(r)}) \le 2M(\|P^{(0)}\| + \|P^{(1)}\| + \cdots + \|P^{(r-1)}\|). \tag{5.9}$$

Since $P^{(0)} = P$, $P^{(r)} = P'$, and $\|P^{(k)}\| \le \|P^{(k-1)}\|$ for $1 \le k \le r - 1$, (5.9) implies that

$$0 \le S(P) - S(P') \le 2Mr\|P\|,$$

which is equivalent to (5.11).

**Theorem**: If $f$ is bounded on $[a, b]$, then

$$\underline{\int_a^b} f(x)\,dx \le \overline{\int_a^b} f(x)\,dx. \tag{5.10}$$

**Proof**: Suppose that $P_1$ and $P_2$ are partitions of $[a, b]$ and $P'$ is a refinement of both.

We have proved

$$\begin{aligned} S(P) \geq S(P') &\geq S(P) - 2Mr\|P\| \quad (*) \\ &\text{and} \\ s(P) \leq s(P') &\leq s(P) + 2Mr\|P\| \quad (**). \end{aligned}$$

Letting $P = P_1$ in (**) and $P = P_2$ in (*) shows that

$$s(P_1) \leq s(P') \quad \text{and} \quad S(P') \leq S(P_2).$$

Since $s(P') \leq S(P')$, this implies that $s(P_1) \leq S(P_2)$. Thus, every lower sum is a lower bound for the set of all upper sums.

Since $\overline{\int_a^b} f(x)\,dx$ is the infimum of this set, it follows that

$$s(P_1) \leq \overline{\int_a^b} f(x)\,dx$$

for every partition $P_1$ of $[a, b]$.

This means that $\overline{\int_a^b} f(x)\,dx$ is an upper bound for the set of all lower sums.

Since $\underline{\int_a^b} f(x)\,dx$ is the supremum of this set, this implies (5.10).

**Theorem**: If $f$ is integrable on $[a, b]$, then

$$\underline{\int_a^b} f(x)\,dx = \overline{\int_a^b} f(x)\,dx = \int_a^b f(x)\,dx.$$

**Proof**: Suppose that $P$ is a partition of $[a, b]$ and $\sigma$ is a Riemann sum of $f$ over $P$. Since

$$\begin{aligned} \overline{\int_a^b} f(x)\,dx - \int_a^b f(x)\,dx &= \left(\overline{\int_a^b} f(x)\,dx - S(P)\right) + (S(P) - \sigma) \\ &\quad + \left(\sigma - \int_a^b f(x)\ dx\right). \end{aligned}$$

The triangle inequality implies that

$$\begin{aligned} \left|\overline{\int_a^b} f(x)\,dx - \int_a^b f(x)\,dx\right| &\leq \left|\overline{\int_a^b} f(x)\,dx - S(P)\right| + |S(P) - \sigma| \\ &\quad + \left|\sigma - \int_a^b f(x)\ dx\right|. \end{aligned} \tag{5.11}$$

Now suppose that $\varepsilon > 0$. From definition of Riemann upper sum, there is a partition $P_0$ of $[a, b]$ such that

$$\overline{\int_a^b} f(x)\,dx \leq S(P_0) < \overline{\int_a^b} f(x)\,dx + \frac{\varepsilon}{3}. \tag{5.12}$$

From definition of Riemann integral, there is a $\delta > 0$ such that

$$\left|\sigma - \int_a^b f(x)\,dx\right| < \frac{\varepsilon}{3}. \tag{5.13}$$

If $\|P\| < \delta$ and $P$ is a refinement of $P_0$.
Since $S(P) \le S(P_0)$ by Lemma we have proved, (5.12) implies that

$$\overline{\int_a^b} f(x)\,dx \le S(P) < \overline{\int_a^b} f(x)\,dx + \frac{\varepsilon}{3},$$

so

$$\left|S(P) - \overline{\int_a^b} f(x)\,dx\right| < \frac{\varepsilon}{3} \tag{5.14}$$

in addition to (5.13). Now (5.11), (5.13), and (5.14) imply that

$$\left|\overline{\int_a^b} f(x)\,dx - \int_a^b f(x)\,dx\right| < \frac{2\varepsilon}{3} + |S(P) - \sigma| \tag{5.15}$$

for every Riemann sum $\sigma$ of $f$ over $P$.
Since $S(P)$ is the supremum of these Riemann sums, we may choose $\sigma$ so that

$$|S(P) - \sigma| < \frac{\varepsilon}{3}.$$

Now (5.15) implies that

$$\left|\overline{\int_a^b} f(x)\,dx - \int_a^b f(x)\,dx\right| < \varepsilon.$$

Since $\varepsilon$ is an arbitrary positive number, it follows that

$$\overline{\int_a^b} f(x)\,dx = \int_a^b f(x)\,dx.$$

**Lemma**: If $f$ is bounded on $[a, b]$ and $\varepsilon > 0$, there is a $\delta > 0$ such that

$$\overline{\int_a^b} f(x)\,dx \le S(P) < \overline{\int_a^b} f(x)\,dx + \varepsilon \tag{5.16}$$

and

$$\underline{\int_a^b} f(x)\,dx \ge s(P) > \underline{\int_a^b} f(x)\,dx - \varepsilon$$

if $\|P\| < \delta$.

**Proof**: The first inequality follows immediately from definition of Riemann upper integral.

To establish the second inequality, suppose that $|f(x)| \le K$ if $a \le x \le b$. From definition of the Riemann upper integral, there is a partition $P_0 = \{x_0, x_1, \ldots, x_{r+1}\}$ of $[a, b]$ such that

$$S(P_0) < \overline{\int_a^b} f(x)\,dx + \frac{\varepsilon}{2}. \tag{5.17}$$

If $P$ is any partition of $[a, b]$, let $P'$ be constructed from the partition points of $P_0$ and $P$. Then we know that

$$S(P') \le S(P_0). \tag{5.18}$$

Since $P'$ is obtained by adding at most $r$ points to $P$, we have

$$S(P') \ge S(P) - 2Kr\|P\|. \tag{5.19}$$

Now (5.17), (5.18), and (5.19) imply that

$$\begin{aligned} S(P) &\le S(P') + 2Kr\|P\| \\ &\le S(P_0) + 2Kr\|P\| \\ &< \overline{\int_a^b} f(x)\,dx + \frac{\varepsilon}{2} + 2Kr\|P\|. \end{aligned}$$

Therefore, (5.16) holds if

$$\|P\| < \delta = \frac{\varepsilon}{4Kr}.$$

**Theorem**: If $f$ is bounded on $[a, b]$ and

$$\underline{\int_a^b} f(x)\,dx = \overline{\int_a^b} f(x)\,dx = L,$$

then $f$ is integrable on $[a, b]$ and

$$\int_a^b f(x)\,dx = L.$$

**Proof**: If $\varepsilon > 0$, there is a $\delta > 0$ such that

$$\underline{\int_a^b} f(x)\,dx - \varepsilon < s(P) \le S(P) < \overline{\int_a^b} f(x)\,dx + \varepsilon, \tag{5.20}$$

If $\|P\| < \delta$.

If $\sigma$ is a Riemann sum of $f$ over $P$, then

$$s(P) \le \sigma \le S(P).$$

So (5.21) and (5.20) imply that

$$L - \varepsilon < \sigma < L + \varepsilon$$

if $\|P\| < \delta$.

**Recall the following theorems**:
**Theorem**: If $f$ is integrable on $[a, b]$, then

$$\underline{\int_a^b} f(x)\,dx = \overline{\int_a^b} f(x)\,dx = \int_a^b f(x)\,dx.$$

**Theorem**: If $f$ is bounded on $[a, b]$ and

$$\underline{\int_a^b} f(x)\,dx = \overline{\int_a^b} f(x)\,dx = L, \tag{5.21}$$

then $f$ is integrable on $[a, b]$ and

$$\int_a^b f(x)\,dx = L. \tag{5.22}$$

**Theorem**: A bounded function $f$ is integrable on $[a, b]$ if and only if

$$\underline{\int_a^b} f(x)\,dx = \overline{\int_a^b} f(x)\,dx.$$

**Theorem**: If $f$ is bounded on $[a, b]$, then $f$ is integrable on $[a, b]$ if and only if for each $\varepsilon > 0$ there is a partition $P$ of $[a, b]$ for which

$$S(P) - s(P) < \varepsilon.$$

**Proof**: Since

$$s(P) \le \underline{\int_a^b} f(x)\,dx \le \overline{\int_a^b} f(x)\,dx \le S(P)$$

for all $P$.

The inequality $S(P) - s(P) < \varepsilon$, implies that

$$0 \le \overline{\int_a^b} f(x)\,dx - \underline{\int_a^b} f(x)\,dx < \varepsilon.$$

Since $\varepsilon$ can be any positive number, this implies that

$$\overline{\int_a^b} f(x)\,dx = \underline{\int_a^b} f(x)\,dx.$$

Therefore, $\int_a^b f(x)\,dx$ exists.

Since $\varepsilon$ can be any positive number, this implies that

$$\overline{\int_a^b} f(x)\,dx = \underline{\int_a^b} f(x)\,dx.$$

For the converse of the theorem try yourself

**Theorem**: If $f$ is bounded on $[a,b]$, then $f$ is integrable on $[a,b]$ if and only if for each $\varepsilon > 0$ there is a partition $P$ of $[a,b]$ for which

$$S(P) - s(P) < \varepsilon.$$

**Theorem**: If $f$ is continuous on $[a,b]$, then $f$ is integrable on $[a,b]$.

**Proof**: Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a,b]$.

Since $f$ is continuous on $[a,b]$, there are points $c_j$ and $c_j'$ in $[x_{j-1}, x_j]$ such that

$$f(c_j) = M_j = \sup_{x_{j-1} \le x \le x_j} f(x)$$

and

$$f(c_j') = m_j = \inf_{x_{j-1} \le x \le x_j} f(x).$$

Therefore,

$$S(P) - s(P) = \sum_{j=1}^{n} \left[f(c_j) - f(c_j')\right](x_j - x_{j-1}). \tag{5.23}$$

Since $f$ is uniformly continuous on $[a,b]$, there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$|f(x') - f(x)| < \frac{\varepsilon}{b-a}.$$

If $x$ and $x'$ are in $[a,b]$ and $|x - x'| < \delta$. If $\|P\| < \delta$, then $|c_j - c_j'| < \delta$ and, from (5.23),

$$S(P) - s(P) < \frac{\varepsilon}{b-a} \sum_{j=1}^{n} (x_j - x_{j-1}) = \varepsilon.$$

Hence, $f$ is integrable on $[a,b]$.

**Theorem**: If $f$ is bounded on $[a,b]$, then $f$ is integrable on $[a,b]$ if and only if for each $\varepsilon > 0$ there is a partition $P$ of $[a,b]$ for which

$$S(P) - s(P) < \varepsilon.$$

**Theorem**: If $f$ is monotonic on $[a,b]$, then $f$ is integrable on $[a,b]$.

**Proof**: Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a,b]$. Since $f$ is nondecreasing,

$$\begin{aligned} f(x_j) &= M_j = \sup_{x_{j-1} \le x \le x_j} f(x) \\ f(x_{j-1}) &= m_j = \inf_{x_{j-1} \le x \le x_j} f(x). \end{aligned}$$

Hence,

$$S(P) - s(P) = \sum_{j=1}^{n} (f(x_j) - f(x_{j-1}))(x_j - x_{j-1}).$$

Since $0 < x_j - x_{j-1} \le \|P\|$ and $f(x_j) - f(x_{j-1}) \ge 0$,

$$\begin{aligned} S(P) - s(P) &\le \|P\| \sum_{j=1}^{n} (f(x_j) - f(x_{j-1})) \\ &= \|P\|(f(b) - f(a)). \end{aligned}$$

Therefore,

$$S(P) - s(P) < \varepsilon \quad \text{if} \quad \|P\|(f(b) - f(a)) < \varepsilon,$$

so $f$ is integrable on $[a,b]$.

**Theorem**: If $f$ and $g$ are integrable on $[a,b]$, then so is $f+g$, and

$$\int_a^b (f+g)(x)\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx.$$

**Proof**: Any Riemann sum of $f+g$ over a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a,b]$ can be written as

$$\begin{aligned} \sigma_{f+g} &= \textstyle\sum_{j=1}^{n} [f(c_j) + g(c_j)](x_j - x_{j-1}) \\ &= \textstyle\sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}) + \sum_{j=1}^{n} g(c_j)(x_j - x_{j-1}) \\ &= \sigma_f + \sigma_g, \end{aligned}$$

where $\sigma_f$ and $\sigma_g$ are Riemann sums for $f$ and $g$.

Definition of Riemann integral implies that if $\varepsilon > 0$ there are positive numbers $\delta_1$ and $\delta_2$ such that

$$\left|\sigma_f - \int_a^b f(x)\,dx\right| < \frac{\varepsilon}{2} \quad \text{if} \quad \|P\| < \delta_1$$

and

$$\left|\sigma_g - \int_a^b g(x)\,dx\right| < \frac{\varepsilon}{2} \quad \text{if} \quad \|P\| < \delta_2.$$

If $\|P\| < \delta = \min(\delta_1, \delta_2)$, then

$$\begin{aligned}\left|\sigma_{f+g} - \int_a^b f(x)\,dx - \int_a^b g(x)\,dx\right| &= \left|\left(\sigma_f - \int_a^b f(x)\,dx\right) + \left(\sigma_g - \int_a^b g(x)\,dx\right)\right| \\ &\le \left|\sigma_f - \int_a^b f(x)\,dx\right| + \left|\sigma_g - \int_a^b g(x)\,dx\right| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,\end{aligned}$$

so the conclusion follows from Definition.

**Example**: Determine whether the function $f(x) = 1 + x$ is integrable on $[a, b]$?

**Solution**: If

$$f(x) = \sin x \quad \text{and} \quad g(x) = \frac{1}{x}, \quad x \neq 0,$$

then

$$h(x) = f(g(x)) = \sin\frac{1}{x}, \quad x \neq 0,$$

and

$$h'(x) = f'(g(x))g(x) = \left(\cos\frac{1}{x}\right)\left(-\frac{1}{x^2}\right), \quad x \neq 0.$$

**Theorem**: If $f$ is integrable on $[a, b]$ and $c$ is a constant, then $cf$ is integrable on $[a, b]$ and

$$\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx.$$

**Proof**: See Lecture.

**Theorem**: If $f_1$, $f_2$, ..., $f_n$ are integrable on $[a, b]$ and $c_1$, $c_2$, ..., $c_n$ are constants, then $c_1f_1 + c_2f_2 + \cdots + c_nf_n$ is integrable on $[a, b]$ and

$$\begin{aligned}\int_a^b (c_1f_1 + c_2f_2 + \cdots + c_nf_n)(x)\,dx &= c_1\int_a^b f_1(x)\,dx \\ &\quad + \cdots + c_n\int_a^b f_n(x)\,dx.\end{aligned}$$

**Proof**: See Lecture.

**Theorem**: If $f$ and $g$ are integrable on $[a,b]$ and $f(x) \le g(x)$ for $a \le x \le b$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

**Proof**: Since $g(x) - f(x) \ge 0$, every lower sum of $g - f$ over any partition of $[a,b]$ is nonnegative.

Therefore,

$$\underline{\int_a^b} (g(x) - f(x))\,dx \ge 0.$$

Hence,

$$\begin{array}{rcl} \int_a^b g(x)\,dx - \int_a^b f(x)\,dx & = & \int_a^b (g(x) - f(x))\,dx \\ & = & \underline{\int_a^b} (g(x) - f(x))\,dx \ge 0. \end{array} \tag{5.24}$$

which yields . (The first equality in (5.24) follows from Theorems **??** and **??**; the second, from Theorem **??**.)

**Theorem**: If $f$ is integrable on $[a,b]$, then so is $|f|$, and

$$\left| \int_a^b f(x)\,dx \right| \le \int_a^b |f(x)|\,dx.$$

**Proof**: Let $P$ be a partition of $[a,b]$ and define

$$\begin{array}{rcl} M_j & = & \sup\{f(x) : x_{j-1} \le x \le x_j\}, \\ m_j & = & \inf\{f(x) : x_{j-1} \le x \le x_j\}, \\ \overline{M}_j & = & \sup\{|f(x)| : x_{j-1} \le x \le x_j\}, \\ \overline{m}_j & = & \inf\{|f(x)| : x_{j-1} \le x \le x_j\}. \end{array}$$

Then

$$\begin{array}{rcl} \overline{M}_j - \overline{m}_j & = & \sup\{|f(x)| - |f(x')| : x_{j-1} \le x, x' \le x_j\} \\ & \le & \sup\{|f(x) - f(x')| : x_{j-1} \le x, x' \le x_j\} \\ & = & M_j - m_j. \end{array} \tag{5.25}$$

Therefore,

$$\overline{S}(P) - \overline{s}(P) \le S(P) - s(P),$$

where the upper and lower sums on the left are associated with $|f|$ and those on the right are associated with $f$.

Suppose that $\varepsilon > 0$. Since $f$ is integrable on $[a, b]$, there is a partition $P$ of $[a, b]$ such that $S(P) - s(P) < \varepsilon$.

This inequality and (5.25) imply that $\overline{S}(P) - \overline{s}(P) < \varepsilon$. Therefore, $|f|$ is integrable on $[a, b]$.

Since

$$f(x) \le |f(x)| \quad \text{and} \quad -f(x) \le |f(x)|, \quad a \le x \le b,$$

We have

$$\int_a^b f(x)\,dx \le \int_a^b |f(x)|\,dx \quad \text{and} \quad -\int_a^b f(x)\,dx \le \int_a^b |f(x)|\,dx,$$

hence the result.

**Theorem**: If $f$ and $g$ are integrable on $[a, b]$, then so is the product $fg$.

**Proof**: We consider the case where $f$ and $g$ are nonnegative, and leave the rest of the proof to you. The subscripts $f$, $g$, and $fg$ in the following argument identify the functions with which the various quantities are associated. We assume that neither $f$ nor $g$ is identically zero on $[a, b]$, since the conclusion is obvious if one of them is.

If $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a, b]$, then

$$S_{fg}(P) - s_{fg}(p) = \sum_{j=1}^{n} (M_{fg,j} - m_{fg,j})(x_j - x_{j-1}). \tag{5.26}$$

Since $f$ and $g$ are nonnegative, $M_{fg,j} \le M_{f,j}M_{g,j}$ and $m_{fg,j} \ge m_{f,j}m_{g,j}$. Hence,

$$\begin{aligned} M_{fg,j} - m_{fg,j} &\le M_{f,j}M_{g,j} - m_{f,j}m_{g,j} \\ &= (M_{f,j} - m_{f,j})M_{g,j} + m_{f,j}(M_{g,j} - m_{g,j}) \\ &\le M_g(M_{f,j} - m_{f,j}) + M_f(M_{g,j} - m_{g,j}), \end{aligned}$$

where $M_f$ and $M_g$ are upper bounds for $f$ and $g$ on $[a, b]$.

From (5.26) and the last inequality,

$$S_{fg}(P) - s_{fg}(P) \le M_g[S_f(P) - s_f(P)] + M_f[S_g(P) - s_g(P)]. \tag{5.27}$$

Now suppose that $\varepsilon > 0$. There are partitions $P_1$ and $P_2$ of $[a, b]$ such that

$$S_f(P_1) - s_f(P_1) < \frac{\varepsilon}{2M_g} \quad \text{and} \quad S_g(P_2) - s_g(P_2) < \frac{\varepsilon}{2M_f}. \tag{5.28}$$

If $P$ is a refinement of both $P_1$ and $P_2$, then (5.28) and we can write

$$S_f(P) - s_f(P) < \frac{\varepsilon}{2M_g} \quad \text{and} \quad S_g(P) - s_g(P) < \frac{\varepsilon}{2M_f}.$$

This and (5.27) yield

$$S_{fg}(P) - s_{fg}(P) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore, $fg$ is integrable on $[a, b]$, by Theorem **??**.

**Theorem**: Suppose that $u$ is continuous and $v$ is integrable and nonnegative on $[a, b]$. Then

$$\int_a^b u(x)v(x)\,dx = u(c)\int_a^b v(x)\,dx \tag{5.29}$$

for some $c$ in $[a, b]$.

**Proof**: Since $u$ is continuous implies $u$ is integrable on $[a, b]$. The function $uv$ is integrable on $[a, b]$.

If $m = \min\{u(x) : a \le x \le b\}$ and $M = \max\{u(x) : a \le x \le b\}$, then

$$m \le u(x) \le M$$

and, since $v(x) \ge 0$,

$$mv(x) \le u(x)v(x) \le Mv(x).$$

Therefore, we can write

$$m\int_a^b v(x)\,dx \le \int_a^b u(x)v(x)\,dx \le M\int_a^b v(x)\,dx. \tag{5.30}$$

This implies that (5.32) holds for any $c$ in $[a, b]$ if $\int_a^b v(x)\,dx = 0$.

If $\int_a^b v(x)\,dx \ne 0$, let

$$\overline{u} = \frac{\int_a^b u(x)v(x)\,dx}{\int_a^b v(x)\,dx} \tag{5.31}$$

Since $\int_a^b v(x)\,dx > 0$ in this case (why?), (5.33) implies that $m \le \overline{u} \le M$. The intermediate value theorem implies that $\overline{u} = u(c)$ for some $c$ in $[a, b]$. This implies (5.32).

**Theorem**: Suppose that $u$ is continuous and $v$ is integrable and nonnegative on $[a, b]$. Then

$$\int_a^b u(x)v(x)\,dx = u(c)\int_a^b v(x)\,dx \tag{5.32}$$

for some $c$ in $[a, b]$.

**Proof**: Since $u$ is continuous implies $u$ is integrable on $[a, b]$. The function $uv$ is integrable on $[a, b]$.

If $m = \min\{u(x) : a \le x \le b\}$ and $M = \max\{u(x) : a \le x \le b\}$, then

$$m \le u(x) \le M$$

and, since $v(x) \geq 0$,

$$mv(x) \leq u(x)v(x) \leq Mv(x).$$

**Proof**: Therefore, we can write

$$m \int_a^b v(x)\,dx \leq \int_a^b u(x)v(x)\,dx \leq M \int_a^b v(x)\,dx. \tag{5.33}$$

This implies that (5.32) holds for any $c$ in $[a, b]$ if $\int_a^b v(x)\,dx = 0$.

If $\int_a^b v(x)\,dx \neq 0$, let

$$\overline{u} = \frac{\int_a^b u(x)v(x)\,dx}{\int_a^b v(x)\,dx} \tag{5.34}$$

Since $\int_a^b v(x)\,dx > 0$ in this case (why?), (5.33) implies that $m \leq \overline{u} \leq M$. The intermediate value theorem implies that $\overline{u} = u(c)$ for some $c$ in $[a, b]$. This implies (5.32).

**Average of $u(x)$ on $[a, b]$ and weighted average** : Recall the following

$$\overline{u} = \frac{\int_a^b u(x)v(x)\,dx}{\int_a^b v(x)dx}$$

If $v(x) = 1$, we have

$$\overline{u} = \frac{1}{b-a} \int_a^b u(x)dx,$$

is known as average of $u(x)$ over $[a, b]$.

**Theorem**: If $f$ is integrable on $[a, b]$ and $a \leq a_1 < b_1 \leq b$, then $f$ is integrable on $[a_1, b_1]$.

**Proof**: Suppose that $\varepsilon > 0$, there is a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that

$$S(P) - s(P) = \sum_{j=1}^{n} (M_j - m_j)(x_j - x_{j-1}) < \varepsilon. \tag{5.35}$$

We may assume that $a_1$ and $b_1$ are partition points of $P$, because if not they can be inserted to obtain a refinement $P'$ such that $S(P') - s(P') \leq S(P) - s(P)$.

Let $a_1 = x_r$ and $b_1 = x_s$. Since every term in (5.35) is nonnegative,

$$\sum_{j=r+1}^{s} (M_j - m_j)(x_j - x_{j-1}) < \varepsilon.$$

**Theorem**: If $f$ is integrable on $[a, b]$ and $a \leq a_1 < b_1 \leq b$, then $f$ is integrable on $[a_1, b_1]$.

**Proof**: Thus, $\overline{P} = \{x_r, x_{r+1}, \ldots, x_s\}$ is a partition of $[a_1, b_1]$ over which the upper and lower sums of $f$ satisfy

$$S(\overline{P}) - s(\overline{P}) < \varepsilon.$$

Therefore, $f$ is integrable on $[a_1, b_1]$.

If $f$ is integrable on $[a, b]$ and $a \leq a_1 < b_1 \leq b$, then $f$ is integrable on $[a_1, b_1]$.

**Theorem**: If $f$ is integrable on $[a, b]$ and $[b, c]$, then $f$ is integrable on $[a, c]$.

$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx. \tag{5.36}$$

So far we have defined $\int_\alpha^\beta f(x)\,dx$ only for the case where $\alpha < \beta$.

If $\alpha < \beta$, we define

$$\int_\beta^\alpha f(x)\,dx = -\int_\alpha^\beta f(x)\,dx.$$

$$\int_\alpha^\alpha f(x)\,dx = 0.$$

**Theorem**: If $f$ is integrable on $[a, b]$ and $[b, c]$, then $f$ is integrable on $[a, c]$.

$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx. \tag{5.37}$$

**Proof**: See Lecture.

## 5.4 Fundamental Theorem of Calculus

**Theorem**: If $f$ is integrable on $[a, b]$ and $a \leq c \leq b$, then the function $F$ defined by

$$F(x) = \int_c^x f(t)\,dt$$

satisfies a Lipschitz condition on $[a, b]$, and is therefore continuous on $[a, b]$.

**Proof**: If $x$ and $x'$ are in $[a, b]$, then

$$F(x) - F(x') = \int_c^x f(t)\,dt - \int_c^{x'} f(t)\,dt = \int_{x'}^x f(t)\,dt.$$

Since $|f(t)| \leq K$ $(a \leq t \leq b)$ for some constant $K$,

$$\left|\int_{x'}^x f(t)\,dt\right| \leq K|x - x'|, \quad a \leq x,\, x' \leq b.$$

**Theorem**: If $f$ is integrable on $[a,b]$ and $a \le c \le b$, then the function $F$ defined by

$$F(x) = \int_c^x f(t)\,dt$$

satisfies a Lipschitz condition on $[a,b]$, and is therefore continuous on $[a,b]$.

**Proof**: Consequently,

$$|F(x) - F(x')| \le K|x - x'|, \quad a \le x,\, x' \le b.$$

**Theorem**: If $f$ is integrable on $[a,b]$ and $a \le c \le b$, then $F(x) = \int_c^x f(t)\,dt$ is differentiable at any point $x_0$ in $(a,b)$ where $f$ is continuous, with $F'(x_0) = f(x_0)$.

If $f$ is continuous from the right at $a$, then $F'_+(a) = f(a)$. If $f$ is continuous from the left at $b$, then $F'_-(b) = f(b)$.

**Proof**: Since

$$\frac{1}{x - x_0}\int_{x_0}^x f(x_0)\,dt = f(x_0).$$

We can write

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x_0) = \frac{1}{x - x_0}\int_{x_0}^x [f(t) - f(x_0)]\,dt.$$

From this

$$\left|\frac{F(x) - F(x_0)}{x - x_0} - f(x_0)\right| \le \frac{1}{|x - x_0|}\left|\int_{x_0}^x |f(t) - f(x_0)|\,dt\right|.$$

Since $f$ is continuous at $x_0$, there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$|f(t) - f(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta$$

and $t$ is between $x$ and $x_0$.

Therefore, we have

$$\left|\frac{F(x) - F(x_0)}{x - x_0} - f(x_0)\right| < \varepsilon\frac{|x - x_0|}{|x - x_0|} = \varepsilon \quad \text{if} \quad 0 < |x - x_0| < \delta.$$

Hence, $F'(x_0) = f(x_0)$.

**Theorem**: If $f$ is integrable on $[a,b]$ and $a \le c \le b$, then $F(x) = \int_c^x f(t)\,dt$ is differentiable at any point $x_0$ in $(a,b)$ where $f$ is continuous, with $F'(x_0) = f(x_0)$.

If $f$ is continuous from the right at $a$, then $F'_+(a) = f(a)$. If $f$ is continuous from the left at $b$, then $F'_-(b) = f(b)$.

**Example**: If

$$f(x) = \begin{cases} x, & 0 \le x \le 1, \\ x+1, & 1 < x \le 2. \end{cases}$$

Then the function

$$F(x) = \int_0^x f(t)\,dt = \begin{cases} \frac{x^2}{2}, & 0 < x \le 1, \\ \frac{x^2}{2} + x - 1, & 1 < x \le 2, \end{cases}$$

is continuous on $[0, 2]$.

We can conclude

$$F'(x) = \begin{cases} x = f(x), & 0 < x < 1, \\ x + 1 = f(x), & 1 < x < 2. \end{cases}$$

$$\begin{aligned} F'_+(0) &= \lim_{x\to 0+} \frac{F(x) - F(0)}{x} = \lim_{x\to 0+} \frac{(x^2/2) - 0}{x} = 0 = f(0), \\ F'_-(2) &= \lim_{x\to 2-} \frac{F(x) - F(2)}{x - 2} = \lim_{x\to 2-} \frac{(x^2/2) + x - 1 - 3}{x - 2} \\ &= \lim_{x\to 2-} \frac{x+4}{2} = 3 = f(2). \end{aligned}$$

$F$ does not have a derivative at $x = 1$, where $f$ is discontinuous, since

$$F'_-(1) = 1 \quad \text{and} \quad F'_+(1) = 2.$$

**Theorem**: Suppose that $F$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval $(a, b)$, and $f$ is integrable on $[a, b]$.

Suppose also that

$$F'(x) = f(x), \quad a < x < b.$$

Then

$$\int_a^b f(x)\,dx = F(b) - F(a). \tag{5.38}$$

**Proof**: If $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a, b]$, then

$$F(b) - F(a) = \sum_{j=1}^{n} (F(x_j) - F(x_{j-1})). \tag{5.39}$$

From mean value theorem, there is in each open interval $(x_{j-1}, x_j)$ a point $c_j$ such that

$$F(x_j) - F(x_{j-1}) = f(c_j)(x_j - x_{j-1}).$$

Hence, (5.39) can be written as

$$F(b) - F(a) = \sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}) = \sigma,$$

where $\sigma$ is a Riemann sum for $f$ over $P$.

Since $f$ is integrable on $[a, b]$, there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$\left| \sigma - \int_a^b f(x)\, dx \right| < \varepsilon \quad \text{if} \quad \|P\| < \delta.$$

Therefore,

$$\left| F(b) - F(a) - \int_a^b f(x)\, dx \right| < \varepsilon$$

for every $\varepsilon > 0$.

**Lemma**: If $f'$ is integrable on $[a, b]$, then

$$\int_a^b f'(x)\, dx = f(b) - f(a).$$

**Proof**: See Lecture.

### 5.4.1 Anti-derivative of a Function

**Anti-derivative of a function**: A function $F$ is an *antiderivative* of $f$ on $[a, b]$ if $F$ is continuous on $[a, b]$ and differentiable on $(a, b)$, with

$$F'(x) = f(x), \quad a < x < b.$$

If $F$ is an antiderivative of $f$ on $[a, b]$, then so is $F + c$ for any constant $c$.

Conversely, if $F_1$ and $F_2$ are antiderivatives of $f$ on $[a, b]$, then $F_1 - F_2$ is constant on $[a, b]$.

**Theorem**: If $f$ is continuous on $[a, b]$, then $f$ has an antiderivative on $[a, b]$. Moreover, if $F$ is any antiderivative of $f$ on $[a, b]$, then

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

**Proof**: The function $F_0(x) = \int_a^x f(t)\, dt$ is continuous on $[a, b]$.

Furthermore, $F_0'(x) = f(x)$ on $(a, b)$ by previous theorem.

Therefore, $F_0$ is an antiderivative of $f$ on $[a, b]$.

Now let $F = F_0 + c$ ($c =$ constant) be an arbitrary antiderivative of $f$ on $[a, b]$. Then

$$F(b) - F(a) = \int_a^b f(x)\,dx + c - \int_a^a f(x)\,dx - c = \int_a^b f(x)\,dx.$$

When applying this theorem, we will use the familiar notation

$$F(b) - F(a) = F(x)\Big|_a^b.$$

## 5.5 Integration by Parts

**Theorem**: If $u'$ and $v'$ are integrable on $[a, b]$, then

$$\int_a^b u(x)v'(x)\,dx = u(x)v(x)\Big|_a^b - \int_a^b v(x)u'(x)\,dx. \tag{5.40}$$

**Proof**: Since $u$ and $v$ are continuous on $[a, b]$, they are integrable on $[a, b]$.

Therefore, using the product of two integrable function is integrable and sum of two integrable functions is integrable imply

$$(uv)' = u'v + uv'$$

is integrable on $[a, b]$.

We have

$$\int_a^b [u(x)v'(x) + u'(x)v(x)]\,dx = u(x)v(x)\Big|_a^b,$$

**Theorem**: Suppose that $f'$ is nonnegative and integrable and $g$ is continuous on $[a, b]$. Then

$$\int_a^b f(x)g(x)\,dx = f(a)\int_a^c g(x)\,dx + f(b)\int_c^b g(x)\,dx \tag{5.41}$$

for some $c$ in $[a, b]$.

**Proof**: Since $f$ is differentiable on $[a, b]$, it is continuous on $[a, b]$. Since $g$ is continuous on $[a, b]$, so is $fg$.

Since we know if $f$ is continuous then $f$ is integrable, thus the integrals in (5.41) exist. If

$$G(x) = \int_a^x g(t)\,dt, \tag{5.42}$$

then $G'(x) = g(x),\ a < x < b$.

Therefore, by using integration by parts with $u = f$ and $v = G$ yields

$$\int_a^b f(x)g(x)\,dx = f(x)G(x)\Big|_a^b - \int_a^b f'(x)G(x)\,dx. \tag{5.43}$$

Since $f'$ is nonnegative and $G$ is continuous, the first mean value theorem of integral implies that

$$\int_a^b f'(x)G(x)\,dx = G(c)\int_a^b f'(x)\,dx \tag{5.44}$$

for some $c$ in $[a,b]$.

But we know that

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

From this and (5.42), (5.44) can be rewritten as

$$\int_a^b f'(x)G(x)\,dx = (f(b) - f(a))\int_a^c g(x)\,dx.$$

Substituting this into (5.43) and noting that $G(a) = 0$ yields

$$\begin{aligned}\int_a^b f(x)g(x)\,dx &= f(b)\int_a^b g(x)\,dx - (f(b) - f(a))\int_a^c g(x)\,dx,\\ &= f(a)\int_a^c g(x)\,dx + f(b)\Big(\int_a^b g(x)\,dx\\ &\quad -\int_c^a g(x)\,dx\Big)\\ &= f(a)\int_a^c g(x)\,dx + f(b)\int_c^b g(x)\,dx.\end{aligned}$$

## 5.6 Integration by Substitution

**Theorem**: Suppose that the transformation $x = \phi(t)$ maps the interval $c \le t \le d$ into the interval $a \le x \le b$, with $\phi(c) = \alpha$ and $\phi(d) = \beta$, and let $f$ be continuous on $[a,b]$.

Let $\phi'$ be integrable on $[c,d]$. Then

$$\int_\alpha^\beta f(x)\,dx = \int_c^d f(\phi(t))\phi'(t)\,dt. \tag{5.45}$$

**Proof**: Both integrals in (5.45) exist: the one on the left by the fact that if $f$ is continuous on $[a,b]$, then $f$ is integrable on $[a,b]$, the one on the right by the continuity of $f(\phi(t))$.

The function

$$F(x) = \int_a^x f(y)\,dy$$

is an antiderivative of $f$ on $[a,b]$ and, therefore, also on the closed interval with endpoints $\alpha$ and $\beta$.

Hence, by fundamental theorem of calculus,

$$\int_{\alpha}^{\beta} f(x)\,dx = F(\beta) - F(\alpha). \tag{5.46}$$

By the chain rule, the function

$$G(t) = F(\phi(t))$$

is an antiderivative of $f(\phi(t))\phi'(t)$ on $[c, d]$. Therefore, we have

$$\begin{aligned}\int_{c}^{d} f(\phi(t))\phi'(t)\,dt &= G(d) - G(c) = F(\phi(d)) - F(\phi(c)) \\ &= F(\beta) - F(\alpha).\end{aligned}$$

Comparing this with (5.46) yields (5.45).

**Example**: Evaluate the integral

$$I = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} (1 - 2x^2)(1 - x^2)^{-1/2} dx.$$

**Solution**: We let

$$f(x) = (1 - 2x^2)(1 - x^2)^{-1/2}, \quad -1/\sqrt{2} \le x \le 1/\sqrt{2},$$

and

$$x = \phi(t) = \sin t, \quad -\pi/4 \le t \le \pi/4.$$

Then $\phi'(t) = \cos t$ and

$$\begin{aligned}I = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} f(x)\,dx &= \int_{-\pi/4}^{\pi/4} f(\sin t)\cos t\,dt \\ &= \int_{-\pi/4}^{\pi/4} (1 - 2\sin^2 t)(1 - \sin^2 t)^{-1/2}\cos t\,dt.\end{aligned} \tag{5.47}$$

$$\begin{aligned}(1 - \sin^2 t)^{-1/2} &= (\cos t)^{-1}, -\pi/4 \le t \le \pi/4 \\ &\text{and} \\ 1 - 2\sin^2 t &= \cos 2t,\end{aligned}$$

(5.47) yields

$$I = \int_{-\pi/4}^{\pi/4} \cos 2t\,dt = \frac{\sin 2t}{2}\Big|_{-\pi/4}^{\pi/4} = 1.$$

**Example**: Evaluate the integral

$$I = \int_{0}^{5\pi} \frac{\sin t}{2 + \cos t}\,dt.$$